Additional files

for the thesis

Noise, uncertainty, and feature selection in molecular property prediction

Adelaide Punt

Contents

S Additional files **2**

S.1 QSAR modelling and label noise 2

S.2 Noise robustness across molecular representations and model architectures 2

S.3 Limitations of the similar property principle 37

S.4 Building adaptive hybrid representations 40

S.5 Similarity in an adaptive hybrid representation 75

S.6 Detecting and handling label noise 80

S.7 Datasets and curation 89

S.7.1 Small molecules 89

S.7.2 The Chapter 3 versions of LogD, Caco-2, hERG K_i, and QM9 . 92

S.7.3 Chemical reactions 93

S.7.4 Antibodies 93

S.7.5 Nucleotide sequences 94

S.8 Data and code availability 95

Chapter S

Additional files

This document holds the tables and figures that support the main chapters but are too long to include in them. They are grouped by the chapter they support and numbered in order as Additional files.

S.1 QSAR modelling and label noise

Additional file 1: Tolerance band for the delivered noise

The delivered noise is checked against a band of three standard errors,

$$\left|\frac{\hat{\tau}}{\tau} - 1\right| \le \max\left(3\sqrt{\frac{\kappa - 1}{4\,n_{\text{eff}}}},\ 0.005\right), \qquad n_{\text{eff}} = \frac{\left(\sum_i s_i^2\right)^2}{\sum_i s_i^4}, \tag{S.1}$$

where $\hat{\tau}$ is the delivered root-mean-square noise and κ is the sample kurtosis of the unscaled draws z, bounded between 3 and 60. The effective count n_{eff} is the number of independent contributions the delivered noise averages over, and falls below n when the noise is concentrated on a few molecules. Under grouped-shifted it is $n_{\text{eff}} = (\rho^2/g_{\text{eff}} + (1-\rho)^2/n)^{-1}$, with $g_{\text{eff}} = \left(\sum_g n_g\right)^2 / \sum_g n_g^2$ for family sizes n_g. At $\nu \le 4$ the Student-t distribution has no fourth moment, so the standard error is instead $1.3\,n_{\text{eff}}^{2/\nu-1}$, with the constant calibrated on the measured spread. A draw outside the band raises a warning, which a correct injector does on about 1% of draws; the delivered amount is recorded either way.

S.2 Noise robustness across molecular representations and model architectures

Additional file 2: Hyperparameters for every model configuration, and the descriptors that make up PDV

Table S.1 lists the default settings, which both the QM9 and the assay pipelines read from one file. Table S.2 lists the tuned settings that replace a default, per model and per dataset, on QM9, LogD, Caco-2, and hERG K_i. Table S.3 lists the 200 descriptor names that make up the physicochemical descriptor vector (PDV), in the order they are computed.

Table S.1: **Default hyperparameters for every model configuration in the study.** Every value is read from `models/model_defaults.py` at spec version 1.7.0, spec hash `9249926c38dc`, which is the file both the QM9 pipeline and the assay pipeline load their settings from. Every QM9 results row carries the spec version and the spec hash as two of its columns, so a QM9 table can be traced back to the settings that produced it; the three assay datasets are written by a second pipeline whose results files do not carry those two columns. The configuration list is the model roster the job generator submits. One setting is used for every representation and for every dataset, except where Table B gives a tuned replacement. The support vector machine (SVM) uses an RBF kernel on every representation, so no result in this study carries a kernel that changes with the representation. Values not held in the shared spec name their source file in the third column.

Configuration	**Hyperparameter**	**Default value**
Shared by every neural network below		
Training, all networks	optimizer	adam
	lr	0.001
	weight_decay	0.0
	batch_size	32
	epochs	100 cap; early stopping usually reaches it first
	patience	10 epochs without a strictly better mean validation loss
	restore_best_weights	True the reported fit is the best epoch, not the last
	val_loss_reduction	mean
	improvement_tolerance	0.0
	mc_passes	100 stochastic forward passes at inference
	standardise_targets	True fitted on training labels alone

Configuration	Hyperparameter	Default value
Tree and boosting models		
RF `rf`	n_estimators	100
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
RF300 `rf300`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
XGBoost `xgboost`	n_estimators	100
	max_depth	6
	learning_rate	0.1
	subsample	1.0
	colsample_bytree	1.0
	colsample_bylevel	1.0
	min_child_weight	1
	gamma	0.0
	reg_alpha	0.0
	reg_lambda	1.0
LightGBM `lgb`	n_estimators	100
	learning_rate	0.1
	num_leaves	31
	max_depth	-1 no limit
	subsample	1.0
	colsample_bytree	1.0
	min_child_samples	20
	reg_alpha	0.0
	reg_lambda	0.0
NGBoost `ngboost`	n_estimators	500
	learning_rate	0.01
	natural_gradient	True
	dist	Normal ngboost.distns.Normal
	score	MLE ngboost.scores.MLE, the log scoring rule

Configuration	Hyperparameter	Default value
	minibatch_frac	1.0
	col_sample	1.0
	base_max_depth	3 depth of one boosting stage
	base_criterion	friedman_mse
	early_stopping_rounds	50 stages without a better validation log likelihood
	use_best_iteration	True predictions use the best stage, not every stage fitted
	ngboost_ensemble_seeds	1 1 = one fit, no seed ensemble
QRF `qrf`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
Kernel models		
SVM `svm`	C	1.0
	gamma	scale
	kernel	rbf
GP `gauche_rbf`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model

Configuration	Hyperparameter	Default value
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Tanimoto `gauche` run on ecfp4 only	kernel	tanimoto set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model

Configuration	Hyperparameter	Default value
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Hetero `heteroscedastic_gp`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime

Configuration	Hyperparameter	Default value
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
	noise network hidden sizes	[64, 64] models/models.py, not in the shared spec
	noise network output	Softplus, plus 10^{-4} models/models.py, not in the shared spec
	Adam learning rate, process	0.1 models/models.py, not in the shared spec
	Adam learning rate, noise network	0.001 models/models.py, not in the shared spec
	noise network epochs	100 the shared epoch cap
Neural networks		
NN-α `dnn`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
NN-β `mlp`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
BNN-α `dnn_bnn_full`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules

Configuration	Hyperparameter	Default value
BNN-β `mlp_bnn_full`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
VBLL-α `dnn_bnn_full_variational`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-β `mlp_bnn_full_variational`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-α (noise head) `dnn_bnn_full_variational_hetero`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
VBLL-β (noise head) `mlp_bnn_full_variational_hetero`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0

Configuration	Hyperparameter	Default value
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
BNN-α (variance head) `dnn_bnn_full_mve`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec
BNN-β (variance head) `mlp_bnn_full_mve`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec

Table S.2: **The tuned settings that replace a default, and the four datasets they apply to.** A setting is chosen once per model per dataset, not per representation, and is then used for all six representations: a two-way comparison across model and representation cannot be read if each cell carries its own hyperparameters. Only the Bayesian and variational networks are tuned; every other configuration in Table S.1 trains at the default. A model with no row for a dataset kept the default on that dataset. Learning rates are printed to three significant figures; the source files hold the full value. Source files: `results/master_tuned_hyperparameters.json` for QM9 and `results/master_tuned_hyperparameters_lab.json` for the three assay datasets.

Dataset	Configuration	Hyperparameter	Tuned value
QM9	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	VBLL-β	dropout_rate	0.357
		hidden_size	64
		lr	0.00119
		num_hidden_layers	1
Caco-2	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	VBLL-α	activation	relu
		hidden_size1	64
		hidden_size2	32
Caco-2	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
Caco-2	BNN-β	dropout_rate	0.401

Dataset	Configuration	Hyperparameter	Tuned value
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	VBLL-α	activation	tanh
		hidden_size1	256
		hidden_size2	32
LogD	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	VBLL-β	dropout_rate	0.079
		hidden_size	128
		lr	0.00011
		num_hidden_layers	2

Table S.3: **The 200 descriptors that make up the PDV representation.** One entry is one feature of the vector, in the order the pipeline computes them. The list is fixed in the code rather than taken from RDKit's current default set, which grows between releases, so the same 200 features are built on every dataset and in every run. They were computed with RDKit 2022.09.5. Descriptors that return a non-finite value on a molecule have that value replaced by zero before the standardization constants are computed. Source: `DEFAULT_DESCRIPTOR_LIST` in `scripts/process_and_train.py`.

BalabanJ	NOCount	SlogP_VSA7	fr_bicyclic
BertzCT	NumAliphaticCarbocycles	SlogP_VSA8	fr_diazo
Chi0	NumAliphaticHeterocycles	SlogP_VSA9	fr_dihydropyridine
Chi0n	NumAliphaticRings	TPSA	fr_epoxide
Chi0v	NumAromaticCarbocycles	VSA_EState1	fr_ester
Chi1	NumAromaticHeterocycles	VSA_EState10	fr_ether
Chi1n	NumAromaticRings	VSA_EState2	fr_furan
Chi1v	NumHAcceptors	VSA_EState3	fr_guanido
Chi2n	NumHDonors	VSA_EState4	fr_halogen
Chi2v	NumHeteroatoms	VSA_EState5	fr_hdrzine
Chi3n	NumRadicalElectrons	VSA_EState6	fr_hdrzone
Chi3v	NumRotatableBonds	VSA_EState7	fr_imidazole
Chi4n	NumSaturatedCarbocycles	VSA_EState8	fr_imide
Chi4v	NumSaturatedHeterocycles	VSA_EState9	fr_isocyan
EState_VSA1	NumSaturatedRings	fr_Al_COO	fr_isothiocyan
EState_VSA10	NumValenceElectrons	fr_Al_OH	fr_ketone
EState_VSA11	PEOE_VSA1	fr_Al_OH_noTert	fr_ketone_Topliss
EState_VSA2	PEOE_VSA10	fr_ArN	fr_lactam
EState_VSA3	PEOE_VSA11	fr_Ar_COO	fr_lactone
EState_VSA4	PEOE_VSA12	fr_Ar_N	fr_methoxy
EState_VSA5	PEOE_VSA13	fr_Ar_NH	fr_morpholine
EState_VSA6	PEOE_VSA14	fr_Ar_OH	fr_nitrile
EState_VSA7	PEOE_VSA2	fr_COO	fr_nitro
EState_VSA8	PEOE_VSA3	fr_COO2	fr_nitro_arom
EState_VSA9	PEOE_VSA4	fr_C_O	fr_nitro_arom_nonortho
ExactMolWt	PEOE_VSA5	fr_C_O_noCOO	fr_nitroso
FpDensityMorgan1	PEOE_VSA6	fr_C_S	fr_oxazole
FpDensityMorgan2	PEOE_VSA7	fr_HOCCN	fr_oxime
FpDensityMorgan3	PEOE_VSA8	fr_Imine	fr_para_hydroxylation
FractionCSP3	PEOE_VSA9	fr_NH0	fr_phenol
HallKierAlpha	RingCount	fr_NH1	fr_phenol_noOrthoHbond
HeavyAtomCount	SMR_VSA1	fr_NH2	fr_phos_acid
HeavyAtomMolWt	SMR_VSA10	fr_N_O	fr_phos_ester
Ipc	SMR_VSA2	fr_Ndealkylation1	fr_piperdine
Kappa1	SMR_VSA3	fr_Ndealkylation2	fr_piperzine
Kappa2	SMR_VSA4	fr_Nhpyrrole	fr_priamide
Kappa3	SMR_VSA5	fr_SH	fr_prisulfonamd
LabuteASA	SMR_VSA6	fr_aldehyde	fr_pyridine
MaxAbsEStateIndex	SMR_VSA7	fr_alkyl_carbamate	fr_quatN
MaxAbsPartialCharge	SMR_VSA8	fr_alkyl_halide	fr_sulfide
MaxEStateIndex	SMR_VSA9	fr_allylic_oxid	fr_sulfonamd
MaxPartialCharge	SlogP_VSA1	fr_amide	fr_sulfone
MinAbsEStateIndex	SlogP_VSA10	fr_amidine	fr_term_acetylene
MinAbsPartialCharge	SlogP_VSA11	fr_aniline	fr_tetrazole
MinEStateIndex	SlogP_VSA12	fr_aryl_methyl	fr_thiazole
MinPartialCharge	SlogP_VSA2	fr_azide	fr_thiocyan
MolLogP	SlogP_VSA3	fr_azo	fr_thiophene
MolMR	SlogP_VSA4	fr_barbitur	fr_unbrch_alkane
MolWt	SlogP_VSA5	fr_benzene	fr_urea
NHOHCount	SlogP_VSA6	fr_benzodiazepine	qed

Additional file 3: Scaffold folds excluded from the robustness analysis

Every scaffold fold left out of the AUC_{norm} analysis, by dataset, model, representation, and noise condition, with its clean R^2. Every exclusion is a clean R^2 below 0.3; no QM9 replicate fell below it.

Table S.4: **Scaffold folds excluded from the robustness analysis.** One row is one model on one representation under one noise condition, in one scaffold fold, that has no AUC_{norm}. Every exclusion is a clean R^2 below 0.3, the gate below which a ratio to the clean baseline is unstable; its clean R^2 is given. All excluded folds are on the three assay datasets: no QM9 replicate fell below the gate.

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
LogD	BNN-α (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.077
LogD	BNN-α (var. head)	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.289
Caco-2	RF	ChemBERTa	Gaussian	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Laplace	1	clean R2 below the gate	0.292
Caco-2	QRF	ChemBERTa	Gaussian	1	clean R2 below the gate	0.282
Caco-2	QRF	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.282
Caco-2	QRF	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.282
Caco-2	XGBoost	ChemBERTa	Gaussian	1	clean R2 below the gate	0.276
Caco-2	XGBoost	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.276
Caco-2	XGBoost	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.276
Caco-2	NGBoost	Avalon	Gaussian	1	clean R2 below the gate	0.249
Caco-2	NGBoost	Avalon	Gaussian	3	clean R2 below the gate	0.281
Caco-2	NGBoost	Avalon	Gaussian	4	clean R2 below the gate	0.287
Caco-2	NGBoost	Avalon	Grouped, wider	1	clean R2 below the gate	0.249
Caco-2	NGBoost	Avalon	Grouped, wider	3	clean R2 below the gate	0.281
Caco-2	NGBoost	Avalon	Grouped, wider	4	clean R2 below the gate	0.287
Caco-2	NGBoost	Avalon	Grouped, shifted	1	clean R2 below the gate	0.249
Caco-2	NGBoost	Avalon	Grouped, shifted	3	clean R2 below the gate	0.281
Caco-2	NGBoost	Avalon	Grouped, shifted	4	clean R2 below the gate	0.287
Caco-2	NGBoost	ChemBERTa	Gaussian	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Gaussian	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Gaussian	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Laplace	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Laplace	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Laplace	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ECFP4	Gaussian	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Gaussian	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Grouped, wider	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Grouped, wider	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Grouped, shifted	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Student-t (ν=5)	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Student-t (ν=5)	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Outlier (10%)	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Outlier (10%)	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Laplace	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Laplace	4	clean R2 below the gate	0.270
Caco-2	NGBoost	MHG-GNN	Gaussian	1	clean R2 below the gate	0.294
Caco-2	NGBoost	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.294
Caco-2	NGBoost	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	NGBoost	Sort & Slice	Gaussian	4	clean R2 below the gate	0.296

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	NGBoost	Sort & Slice	Grouped, wider	4	clean R2 below the gate	0.296
Caco-2	NGBoost	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.296
Caco-2	BNN-α	Avalon	Gaussian	1	clean R2 below the gate	0.274
Caco-2	BNN-α	Avalon	Gaussian	3	clean R2 below the gate	0.214
Caco-2	BNN-α	Avalon	Grouped, wider	3	clean R2 below the gate	0.251
Caco-2	BNN-α	Avalon	Grouped, wider	4	clean R2 below the gate	0.247
Caco-2	BNN-α	Avalon	Grouped, shifted	4	clean R2 below the gate	0.248
Caco-2	BNN-α	ChemBERTa	Gaussian	0	clean R2 below the gate	0.297
Caco-2	BNN-α	ChemBERTa	Gaussian	1	clean R2 below the gate	0.175
Caco-2	BNN-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.202
Caco-2	BNN-α	ChemBERTa	Gaussian	3	clean R2 below the gate	0.205
Caco-2	BNN-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.144
Caco-2	BNN-α	ChemBERTa	Grouped, wider	0	clean R2 below the gate	0.230
Caco-2	BNN-α	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.207
Caco-2	BNN-α	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.181
Caco-2	BNN-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.144
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.241
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.250
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.245
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.233
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.143
Caco-2	BNN-α	MHG-GNN	Gaussian	0	clean R2 below the gate	0.130
Caco-2	BNN-α	MHG-GNN	Gaussian	1	clean R2 below the gate	0.226
Caco-2	BNN-α	MHG-GNN	Gaussian	2	clean R2 below the gate	0.273
Caco-2	BNN-α	MHG-GNN	Gaussian	3	clean R2 below the gate	0.128
Caco-2	BNN-α	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.228
Caco-2	BNN-α	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.191
Caco-2	BNN-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.188
Caco-2	BNN-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.232
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	0	clean R2 below the gate	0.240
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.091
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.266
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.223
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.132
Caco-2	BNN-α	PDV	Gaussian	3	clean R2 below the gate	0.282
Caco-2	BNN-α	PDV	Gaussian	4	clean R2 below the gate	0.290
Caco-2	BNN-α	Sort & Slice	Grouped, wider	1	clean R2 below the gate	0.292
Caco-2	BNN-α (var. head)	Avalon	Gaussian	0	clean R2 below the gate	0.296
Caco-2	BNN-α (var. head)	Avalon	Gaussian	1	clean R2 below the gate	0.258
Caco-2	BNN-α (var. head)	Avalon	Gaussian	3	clean R2 below the gate	0.258
Caco-2	BNN-α (var. head)	Avalon	Gaussian	4	clean R2 below the gate	0.132
Caco-2	BNN-α (var. head)	Avalon	Grouped, wider	1	clean R2 below the gate	0.289
Caco-2	BNN-α (var. head)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.170
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	0.229
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.103
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.112
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.085
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.144
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.143
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.293
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.203
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.225
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.128
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.114
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.106
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.246
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	2	clean R2 below the gate	0.177
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.115
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.240
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.154
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	2	clean R2 below the gate	0.212
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.262
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.113
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	0	clean R2 below the gate	0.204
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	1	clean R2 below the gate	0.224
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	2	clean R2 below the gate	0.157
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	3	clean R2 below the gate	0.074
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	4	clean R2 below the gate	0.093
Caco-2	BNN-α (var. head)	ECFP4	Laplace	3	clean R2 below the gate	0.256
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	0	clean R2 below the gate	-0.065
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.057
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.126

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.088
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.103
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.060
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.092
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.249
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.043
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, shifted	0	clean R2 below the gate	0.120
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.019
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.069
Caco-2	BNN-α (var. head)	PDV	Gaussian	4	clean R2 below the gate	0.289
Caco-2	BNN-α (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.226
Caco-2	BNN-α (var. head)	PDV	Grouped, shifted	4	clean R2 below the gate	0.210
Caco-2	BNN-α (var. head)	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.275
Caco-2	BNN-α (var. head)	PDV	Outlier (10%)	1	clean R2 below the gate	0.251
Caco-2	BNN-α (var. head)	PDV	Outlier (10%)	3	clean R2 below the gate	0.297
Caco-2	BNN-α (var. head)	PDV	Laplace	0	clean R2 below the gate	0.273
Caco-2	VBLL-α	Avalon	Gaussian	0	clean R2 below the gate	0.292
Caco-2	VBLL-α	Avalon	Gaussian	1	clean R2 below the gate	0.212
Caco-2	VBLL-α	Avalon	Gaussian	3	clean R2 below the gate	0.191
Caco-2	VBLL-α	Avalon	Gaussian	4	clean R2 below the gate	0.212
Caco-2	VBLL-α	Avalon	Grouped, wider	0	clean R2 below the gate	0.268
Caco-2	VBLL-α	Avalon	Grouped, wider	1	clean R2 below the gate	0.177
Caco-2	VBLL-α	Avalon	Grouped, wider	3	clean R2 below the gate	0.194
Caco-2	VBLL-α	Avalon	Grouped, wider	4	clean R2 below the gate	0.201
Caco-2	VBLL-α	Avalon	Grouped, shifted	0	clean R2 below the gate	0.231
Caco-2	VBLL-α	Avalon	Grouped, shifted	1	clean R2 below the gate	0.239
Caco-2	VBLL-α	Avalon	Grouped, shifted	3	clean R2 below the gate	0.146
Caco-2	VBLL-α	Avalon	Grouped, shifted	4	clean R2 below the gate	0.182
Caco-2	VBLL-α	ChemBERTa	Gaussian	0	clean R2 below the gate	0.193
Caco-2	VBLL-α	ChemBERTa	Gaussian	1	clean R2 below the gate	0.249
Caco-2	VBLL-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.278
Caco-2	VBLL-α	ChemBERTa	Gaussian	3	clean R2 below the gate	0.214
Caco-2	VBLL-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.238
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	0	clean R2 below the gate	0.247
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.237
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.214
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.208
Caco-2	VBLL-α	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.258
Caco-2	VBLL-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.210
Caco-2	VBLL-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.213
Caco-2	VBLL-α	ECFP4	Gaussian	3	clean R2 below the gate	0.267
Caco-2	VBLL-α	ECFP4	Gaussian	4	clean R2 below the gate	0.299
Caco-2	VBLL-α	ECFP4	Grouped, wider	1	clean R2 below the gate	0.254
Caco-2	VBLL-α	ECFP4	Grouped, wider	3	clean R2 below the gate	0.290
Caco-2	VBLL-α	ECFP4	Grouped, shifted	3	clean R2 below the gate	0.260
Caco-2	VBLL-α	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.272
Caco-2	VBLL-α	MHG-GNN	Gaussian	0	clean R2 below the gate	0.242
Caco-2	VBLL-α	MHG-GNN	Gaussian	1	clean R2 below the gate	0.211
Caco-2	VBLL-α	MHG-GNN	Gaussian	2	clean R2 below the gate	0.285
Caco-2	VBLL-α	MHG-GNN	Gaussian	3	clean R2 below the gate	0.219
Caco-2	VBLL-α	MHG-GNN	Gaussian	4	clean R2 below the gate	0.159
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	0	clean R2 below the gate	0.267
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.268
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.189
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.186
Caco-2	VBLL-α	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.242
Caco-2	VBLL-α	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.203
Caco-2	VBLL-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.227
Caco-2	VBLL-α	PDV	Gaussian	4	clean R2 below the gate	0.299
Caco-2	VBLL-α	PDV	Grouped, wider	4	clean R2 below the gate	0.269
Caco-2	VBLL-α	PDV	Grouped, shifted	3	clean R2 below the gate	0.295
Caco-2	VBLL-α	PDV	Grouped, shifted	4	clean R2 below the gate	0.292
Caco-2	VBLL-α	Sort & Slice	Gaussian	3	clean R2 below the gate	0.297
Caco-2	VBLL-α	Sort & Slice	Grouped, wider	3	clean R2 below the gate	0.237
Caco-2	VBLL-α	Sort & Slice	Grouped, shifted	3	clean R2 below the gate	0.277
Caco-2	VBLL-α	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.297
Caco-2	VBLL-α (het.)	Avalon	Gaussian	1	clean R2 below the gate	0.246
Caco-2	VBLL-α (het.)	Avalon	Gaussian	3	clean R2 below the gate	0.214
Caco-2	VBLL-α (het.)	Avalon	Gaussian	4	clean R2 below the gate	0.210
Caco-2	VBLL-α (het.)	Avalon	Grouped, wider	1	clean R2 below the gate	0.259
Caco-2	VBLL-α (het.)	Avalon	Grouped, wider	3	clean R2 below the gate	0.166
Caco-2	VBLL-α (het.)	Avalon	Grouped, wider	4	clean R2 below the gate	0.231

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	VBLL-α (het.)	Avalon	Grouped, shifted	1	clean R2 below the gate	0.259
Caco-2	VBLL-α (het.)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.235
Caco-2	VBLL-α (het.)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.203
Caco-2	VBLL-α (het.)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.249
Caco-2	VBLL-α (het.)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.243
Caco-2	VBLL-α (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.253
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.281
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.192
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.282
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.250
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.265
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.272
Caco-2	VBLL-α (het.)	ECFP4	Gaussian	3	clean R2 below the gate	0.242
Caco-2	VBLL-α (het.)	ECFP4	Grouped, wider	3	clean R2 below the gate	0.282
Caco-2	VBLL-α (het.)	ECFP4	Grouped, shifted	3	clean R2 below the gate	0.294
Caco-2	VBLL-α (het.)	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.293
Caco-2	VBLL-α (het.)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.236
Caco-2	VBLL-α (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.261
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.221
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.280
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.220
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.252
Caco-2	VBLL-α (het.)	PDV	Grouped, shifted	4	clean R2 below the gate	0.282
Caco-2	VBLL-α (het.)	Sort & Slice	Gaussian	3	clean R2 below the gate	0.293
Caco-2	BNN-β	Avalon	Gaussian	3	clean R2 below the gate	0.121
Caco-2	BNN-β	Avalon	Gaussian	4	clean R2 below the gate	0.298
Caco-2	BNN-β	Avalon	Grouped, wider	3	clean R2 below the gate	0.271
Caco-2	BNN-β	Avalon	Grouped, wider	4	clean R2 below the gate	0.276
Caco-2	BNN-β	Avalon	Grouped, shifted	0	clean R2 below the gate	0.256
Caco-2	BNN-β	Avalon	Grouped, shifted	3	clean R2 below the gate	0.186
Caco-2	BNN-β	ChemBERTa	Gaussian	1	clean R2 below the gate	0.187
Caco-2	BNN-β	ChemBERTa	Gaussian	3	clean R2 below the gate	0.126
Caco-2	BNN-β	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.190
Caco-2	BNN-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.209
Caco-2	BNN-β	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.285
Caco-2	BNN-β	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.238
Caco-2	BNN-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.192
Caco-2	BNN-β	MHG-GNN	Gaussian	1	clean R2 below the gate	0.137
Caco-2	BNN-β	MHG-GNN	Gaussian	3	clean R2 below the gate	0.256
Caco-2	BNN-β	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.157
Caco-2	BNN-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.240
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.152
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.274
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.239
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.211
Caco-2	BNN-β (var. head)	Avalon	Gaussian	2	clean R2 below the gate	0.265
Caco-2	BNN-β (var. head)	Avalon	Gaussian	3	clean R2 below the gate	0.239
Caco-2	BNN-β (var. head)	Avalon	Gaussian	4	clean R2 below the gate	0.298
Caco-2	BNN-β (var. head)	Avalon	Grouped, wider	3	clean R2 below the gate	0.271
Caco-2	BNN-β (var. head)	Avalon	Grouped, wider	4	clean R2 below the gate	0.291
Caco-2	BNN-β (var. head)	Avalon	Grouped, shifted	1	clean R2 below the gate	0.244
Caco-2	BNN-β (var. head)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.220
Caco-2	BNN-β (var. head)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.249
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	0.209
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.173
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.279
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.124
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.255
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.129
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.121
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.088
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.229
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	0	clean R2 below the gate	-0.260
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.145
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.195
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.132
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, wider	0	clean R2 below the gate	-0.280
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.056
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.095
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	0	clean R2 below the gate	-0.057
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.194

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.201
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.083
Caco-2	BNN-β (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.290
Caco-2	BNN-β (var. head)	PDV	Grouped, shifted	3	clean R2 below the gate	0.270
Caco-2	VBLL-β	Avalon	Gaussian	3	clean R2 below the gate	0.259
Caco-2	VBLL-β	Avalon	Gaussian	4	clean R2 below the gate	0.299
Caco-2	VBLL-β	Avalon	Grouped, wider	3	clean R2 below the gate	0.270
Caco-2	VBLL-β	Avalon	Grouped, wider	4	clean R2 below the gate	0.284
Caco-2	VBLL-β	Avalon	Grouped, shifted	3	clean R2 below the gate	0.159
Caco-2	VBLL-β	Avalon	Grouped, shifted	4	clean R2 below the gate	0.296
Caco-2	VBLL-β	ChemBERTa	Gaussian	3	clean R2 below the gate	0.272
Caco-2	VBLL-β	ChemBERTa	Gaussian	4	clean R2 below the gate	0.272
Caco-2	VBLL-β	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.291
Caco-2	VBLL-β	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.291
Caco-2	VBLL-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.283
Caco-2	VBLL-β	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.299
Caco-2	VBLL-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.265
Caco-2	VBLL-β	MHG-GNN	Gaussian	1	clean R2 below the gate	0.260
Caco-2	VBLL-β	MHG-GNN	Gaussian	3	clean R2 below the gate	0.272
Caco-2	VBLL-β	MHG-GNN	Gaussian	4	clean R2 below the gate	0.253
Caco-2	VBLL-β	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.256
Caco-2	VBLL-β	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.262
Caco-2	VBLL-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.280
Caco-2	VBLL-β	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	VBLL-β	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.239
Caco-2	VBLL-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.253
Caco-2	VBLL-β	PDV	Gaussian	4	clean R2 below the gate	0.272
Caco-2	VBLL-β (het.)	Avalon	Gaussian	3	clean R2 below the gate	0.259
Caco-2	VBLL-β (het.)	Avalon	Gaussian	4	clean R2 below the gate	0.257
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	0	clean R2 below the gate	0.268
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	1	clean R2 below the gate	0.281
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	3	clean R2 below the gate	0.265
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	4	clean R2 below the gate	0.281
Caco-2	VBLL-β (het.)	Avalon	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	VBLL-β (het.)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.262
Caco-2	VBLL-β (het.)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.228
Caco-2	VBLL-β (het.)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.253
Caco-2	VBLL-β (het.)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.262
Caco-2	VBLL-β (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.227
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.260
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.294
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.261
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.299
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.287
Caco-2	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.247
Caco-2	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.268
Caco-2	VBLL-β (het.)	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.266
Caco-2	VBLL-β (het.)	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.269
Caco-2	VBLL-β (het.)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.296
Caco-2	VBLL-β (het.)	ChemBERTa	Laplace	3	clean R2 below the gate	0.271
Caco-2	VBLL-β (het.)	ChemBERTa	Laplace	4	clean R2 below the gate	0.296
Caco-2	VBLL-β (het.)	ECFP4	Gaussian	3	clean R2 below the gate	0.297
Caco-2	VBLL-β (het.)	ECFP4	Outlier (10%)	3	clean R2 below the gate	0.292
Caco-2	VBLL-β (het.)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.270
Caco-2	VBLL-β (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.282
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.202
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.273
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.149
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.265
hERG K_i	NGBoost	Avalon	Gaussian	4	clean R2 below the gate	0.283
hERG K_i	NGBoost	Avalon	Grouped, wider	4	clean R2 below the gate	0.283
hERG K_i	NGBoost	Avalon	Grouped, shifted	4	clean R2 below the gate	0.283
hERG K_i	NGBoost	ChemBERTa	Gaussian	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Laplace	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	PDV	Gaussian	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Grouped, wider	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Grouped, shifted	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Censoring	4	clean R2 below the gate	0.297

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	NGBoost	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Outlier (10%)	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Laplace	4	clean R2 below the gate	0.297
hERG K_i	NN-α	PDV	Grouped, shifted	2	clean R2 below the gate	0.233
hERG K_i	NN-α	Sort & Slice	Grouped, wider	3	clean R2 below the gate	0.265
hERG K_i	BNN-α	Avalon	Grouped, wider	3	clean R2 below the gate	0.261
hERG K_i	BNN-α	ChemBERTa	Gaussian	0	clean R2 below the gate	0.072
hERG K_i	BNN-α	ChemBERTa	Gaussian	1	clean R2 below the gate	0.231
hERG K_i	BNN-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.067
hERG K_i	BNN-α	ChemBERTa	Gaussian	3	clean R2 below the gate	0.243
hERG K_i	BNN-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.194
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.137
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.152
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.257
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.050
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.061
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.112
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.109
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.258
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.218
hERG K_i	BNN-α	MHG-GNN	Gaussian	0	clean R2 below the gate	0.242
hERG K_i	BNN-α	MHG-GNN	Gaussian	2	clean R2 below the gate	0.031
hERG K_i	BNN-α	MHG-GNN	Gaussian	3	clean R2 below the gate	0.127
hERG K_i	BNN-α	MHG-GNN	Gaussian	4	clean R2 below the gate	0.032
hERG K_i	BNN-α	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.051
hERG K_i	BNN-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.109
hERG K_i	BNN-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.054
hERG K_i	BNN-α	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.079
hERG K_i	BNN-α	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.181
hERG K_i	BNN-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.138
hERG K_i	BNN-α	PDV	Gaussian	2	clean R2 below the gate	0.154
hERG K_i	BNN-α	PDV	Gaussian	4	clean R2 below the gate	0.173
hERG K_i	BNN-α	PDV	Grouped, wider	2	clean R2 below the gate	0.132
hERG K_i	BNN-α	PDV	Grouped, wider	4	clean R2 below the gate	0.153
hERG K_i	BNN-α	PDV	Grouped, shifted	4	clean R2 below the gate	0.097
hERG K_i	BNN-α	Sort & Slice	Gaussian	2	clean R2 below the gate	0.037
hERG K_i	BNN-α (var. head)	Avalon	Gaussian	0	clean R2 below the gate	0.141
hERG K_i	BNN-α (var. head)	Avalon	Gaussian	4	clean R2 below the gate	0.223
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	-0.011
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.060
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	2	clean R2 below the gate	0.011
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.279
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.152
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	0	clean R2 below the gate	-0.003
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.031
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.027
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.152
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.061
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.055
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.011
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.007
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.093
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.015
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	0	clean R2 below the gate	-0.066
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.119
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	2	clean R2 below the gate	0.240
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.066
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.036
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	0	clean R2 below the gate	0.016
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.054
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	2	clean R2 below the gate	0.032
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.181
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.036
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	0	clean R2 below the gate	0.283
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	1	clean R2 below the gate	0.053
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	2	clean R2 below the gate	0.076
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	3	clean R2 below the gate	0.031
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	4	clean R2 below the gate	0.012
hERG K_i	BNN-α (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.093
hERG K_i	BNN-α (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	-0.047
hERG K_i	BNN-α (var. head)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.023
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	0	clean R2 below the gate	-0.001

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.007
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	-0.010
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.041
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.033
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.083
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.033
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	-0.426
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	-0.035
hERG K_i	BNN-α (var. head)	PDV	Gaussian	2	clean R2 below the gate	0.075
hERG K_i	BNN-α (var. head)	PDV	Gaussian	3	clean R2 below the gate	0.169
hERG K_i	BNN-α (var. head)	PDV	Gaussian	4	clean R2 below the gate	0.259
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	1	clean R2 below the gate	0.077
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	2	clean R2 below the gate	0.210
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	3	clean R2 below the gate	0.098
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.217
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	0	clean R2 below the gate	0.051
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	2	clean R2 below the gate	0.036
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	3	clean R2 below the gate	0.296
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	4	clean R2 below the gate	0.153
hERG K_i	BNN-α (var. head)	PDV	Censoring	0	clean R2 below the gate	0.016
hERG K_i	BNN-α (var. head)	PDV	Censoring	1	clean R2 below the gate	0.110
hERG K_i	BNN-α (var. head)	PDV	Censoring	2	clean R2 below the gate	0.019
hERG K_i	BNN-α (var. head)	PDV	Censoring	3	clean R2 below the gate	0.219
hERG K_i	BNN-α (var. head)	PDV	Censoring	4	clean R2 below the gate	0.194
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	1	clean R2 below the gate	0.074
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	2	clean R2 below the gate	0.031
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	3	clean R2 below the gate	0.281
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.084
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	0	clean R2 below the gate	0.158
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	1	clean R2 below the gate	0.152
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	2	clean R2 below the gate	0.213
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	3	clean R2 below the gate	0.110
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	4	clean R2 below the gate	0.015
hERG K_i	BNN-α (var. head)	PDV	Laplace	0	clean R2 below the gate	0.289
hERG K_i	BNN-α (var. head)	PDV	Laplace	2	clean R2 below the gate	0.113
hERG K_i	BNN-α (var. head)	PDV	Laplace	3	clean R2 below the gate	0.088
hERG K_i	BNN-α (var. head)	PDV	Laplace	4	clean R2 below the gate	0.213
hERG K_i	BNN-α (var. head)	Sort & Slice	Gaussian	1	clean R2 below the gate	0.070
hERG K_i	BNN-α (var. head)	Sort & Slice	Gaussian	2	clean R2 below the gate	0.033
hERG K_i	BNN-α (var. head)	Sort & Slice	Gaussian	3	clean R2 below the gate	0.288
hERG K_i	BNN-α (var. head)	Sort & Slice	Grouped, wider	4	clean R2 below the gate	0.201
hERG K_i	BNN-α (var. head)	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.222
hERG K_i	VBLL-α	Avalon	Gaussian	4	clean R2 below the gate	0.275
hERG K_i	VBLL-α	Avalon	Grouped, shifted	4	clean R2 below the gate	0.299
hERG K_i	VBLL-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.293
hERG K_i	VBLL-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.221
hERG K_i	VBLL-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.190
hERG K_i	VBLL-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.287
hERG K_i	VBLL-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.224
hERG K_i	VBLL-α	ECFP4	Gaussian	4	clean R2 below the gate	0.288
hERG K_i	VBLL-α	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.290
hERG K_i	VBLL-α	MHG-GNN	Gaussian	4	clean R2 below the gate	0.198
hERG K_i	VBLL-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.289
hERG K_i	VBLL-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.222
hERG K_i	VBLL-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.224
hERG K_i	VBLL-α	PDV	Gaussian	4	clean R2 below the gate	0.234
hERG K_i	VBLL-α	PDV	Grouped, wider	4	clean R2 below the gate	0.267
hERG K_i	VBLL-α	PDV	Grouped, shifted	4	clean R2 below the gate	0.243
hERG K_i	VBLL-α (het.)	Avalon	Grouped, wider	4	clean R2 below the gate	0.300
hERG K_i	VBLL-α (het.)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.249
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	0	clean R2 below the gate	0.246
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	2	clean R2 below the gate	0.227
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.176
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.184
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	0	clean R2 below the gate	0.277
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.230
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.253
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.191
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.241
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.208
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.271
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.202

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	VBLL-α (het.)	ECFP4	Gaussian	4	clean R2 below the gate	0.245
hERG K_i	VBLL-α (het.)	ECFP4	Grouped, wider	4	clean R2 below the gate	0.298
hERG K_i	VBLL-α (het.)	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.261
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	0	clean R2 below the gate	0.297
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.289
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.260
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.281
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.165
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, wider	0	clean R2 below the gate	0.286
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.219
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.202
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.197
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.285
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.175
hERG K_i	VBLL-α (het.)	PDV	Gaussian	2	clean R2 below the gate	0.280
hERG K_i	VBLL-α (het.)	PDV	Gaussian	3	clean R2 below the gate	0.276
hERG K_i	VBLL-α (het.)	PDV	Gaussian	4	clean R2 below the gate	0.189
hERG K_i	VBLL-α (het.)	PDV	Grouped, wider	2	clean R2 below the gate	0.264
hERG K_i	VBLL-α (het.)	PDV	Grouped, wider	4	clean R2 below the gate	0.188
hERG K_i	VBLL-α (het.)	PDV	Grouped, shifted	2	clean R2 below the gate	0.235
hERG K_i	VBLL-α (het.)	PDV	Grouped, shifted	3	clean R2 below the gate	0.209
hERG K_i	VBLL-α (het.)	PDV	Grouped, shifted	4	clean R2 below the gate	0.139
hERG K_i	VBLL-α (het.)	Sort & Slice	Gaussian	4	clean R2 below the gate	0.282
hERG K_i	VBLL-α (het.)	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.289
hERG K_i	NN-β	PDV	Grouped, wider	3	clean R2 below the gate	0.221
hERG K_i	NN-β	PDV	Grouped, shifted	3	clean R2 below the gate	-0.092
hERG K_i	BNN-β	Avalon	Gaussian	2	clean R2 below the gate	0.271
hERG K_i	BNN-β	Avalon	Gaussian	3	clean R2 below the gate	0.209
hERG K_i	BNN-β	Avalon	Grouped, shifted	4	clean R2 below the gate	0.298
hERG K_i	BNN-β	ChemBERTa	Gaussian	3	clean R2 below the gate	0.221
hERG K_i	BNN-β	ChemBERTa	Gaussian	4	clean R2 below the gate	0.210
hERG K_i	BNN-β	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.124
hERG K_i	BNN-β	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.296
hERG K_i	BNN-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.217
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.063
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.148
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.272
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.180
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.201
hERG K_i	BNN-β	MHG-GNN	Gaussian	0	clean R2 below the gate	0.202
hERG K_i	BNN-β	MHG-GNN	Gaussian	2	clean R2 below the gate	0.084
hERG K_i	BNN-β	MHG-GNN	Gaussian	3	clean R2 below the gate	0.285
hERG K_i	BNN-β	MHG-GNN	Gaussian	4	clean R2 below the gate	0.181
hERG K_i	BNN-β	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.196
hERG K_i	BNN-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.074
hERG K_i	BNN-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.062
hERG K_i	BNN-β	PDV	Gaussian	2	clean R2 below the gate	0.257
hERG K_i	BNN-β	PDV	Gaussian	4	clean R2 below the gate	0.207
hERG K_i	BNN-β	PDV	Grouped, wider	2	clean R2 below the gate	0.156
hERG K_i	BNN-β	PDV	Grouped, wider	4	clean R2 below the gate	0.249
hERG K_i	BNN-β	PDV	Grouped, shifted	1	clean R2 below the gate	0.269
hERG K_i	BNN-β	PDV	Grouped, shifted	2	clean R2 below the gate	0.181
hERG K_i	BNN-β	PDV	Grouped, shifted	4	clean R2 below the gate	0.197
hERG K_i	BNN-β	Sort & Slice	Gaussian	4	clean R2 below the gate	0.063
hERG K_i	BNN-β	Sort & Slice	Grouped, shifted	1	clean R2 below the gate	0.198
hERG K_i	BNN-β	Sort & Slice	Grouped, shifted	3	clean R2 below the gate	0.074
hERG K_i	BNN-β (var. head)	Avalon	Grouped, wider	4	clean R2 below the gate	0.134
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	-0.015
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.191
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	2	clean R2 below the gate	0.248
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.057
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	-0.022
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	0	clean R2 below the gate	-0.061
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.138
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.039
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.296
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	-0.038
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	-0.060
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.056
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.026
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.039
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	0	clean R2 below the gate	-0.388

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.036
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	3	clean R2 below the gate	-0.024
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	4	clean R2 below the gate	-0.147
hERG K_i	BNN-β (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	-0.075
hERG K_i	BNN-β (var. head)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.024
hERG K_i	BNN-β (var. head)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	-0.116
hERG K_i	BNN-β (var. head)	PDV	Gaussian	0	clean R2 below the gate	0.039
hERG K_i	BNN-β (var. head)	PDV	Gaussian	2	clean R2 below the gate	0.072
hERG K_i	BNN-β (var. head)	PDV	Gaussian	3	clean R2 below the gate	0.083
hERG K_i	BNN-β (var. head)	PDV	Gaussian	4	clean R2 below the gate	0.023
hERG K_i	BNN-β (var. head)	PDV	Grouped, wider	2	clean R2 below the gate	0.111
hERG K_i	BNN-β (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.151
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	0	clean R2 below the gate	0.107
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	1	clean R2 below the gate	0.120
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	2	clean R2 below the gate	0.190
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	4	clean R2 below the gate	0.197
hERG K_i	BNN-β (var. head)	Sort & Slice	Gaussian	4	clean R2 below the gate	0.047
hERG K_i	BNN-β (var. head)	Sort & Slice	Grouped, wider	1	clean R2 below the gate	0.114
hERG K_i	BNN-β (var. head)	Sort & Slice	Grouped, wider	2	clean R2 below the gate	0.007
hERG K_i	BNN-β (var. head)	Sort & Slice	Grouped, shifted	0	clean R2 below the gate	0.062
hERG K_i	VBLL-β	ChemBERTa	Gaussian	4	clean R2 below the gate	0.282
hERG K_i	VBLL-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.287
hERG K_i	VBLL-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.277
hERG K_i	VBLL-β	MHG-GNN	Gaussian	4	clean R2 below the gate	0.258
hERG K_i	VBLL-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.260
hERG K_i	VBLL-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.295
hERG K_i	VBLL-β	PDV	Gaussian	4	clean R2 below the gate	0.264
hERG K_i	VBLL-β	PDV	Grouped, wider	4	clean R2 below the gate	0.265
hERG K_i	VBLL-β	PDV	Grouped, shifted	4	clean R2 below the gate	0.271
hERG K_i	VBLL-β (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.198
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.281
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.291
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.184
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.295
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.198
hERG K_i	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	0	clean R2 below the gate	0.286
hERG K_i	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.218
hERG K_i	VBLL-β (het.)	ChemBERTa	Outlier (10%)	2	clean R2 below the gate	0.286
hERG K_i	VBLL-β (het.)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.198
hERG K_i	VBLL-β (het.)	ChemBERTa	Laplace	2	clean R2 below the gate	0.286
hERG K_i	VBLL-β (het.)	ChemBERTa	Laplace	3	clean R2 below the gate	0.272
hERG K_i	VBLL-β (het.)	ChemBERTa	Laplace	4	clean R2 below the gate	0.191
hERG K_i	VBLL-β (het.)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.297
hERG K_i	VBLL-β (het.)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.260
hERG K_i	VBLL-β (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.220
hERG K_i	VBLL-β (het.)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.272
hERG K_i	VBLL-β (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.258
hERG K_i	VBLL-β (het.)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.254
hERG K_i	VBLL-β (het.)	PDV	Gaussian	2	clean R2 below the gate	0.234
hERG K_i	VBLL-β (het.)	PDV	Gaussian	3	clean R2 below the gate	0.266
hERG K_i	VBLL-β (het.)	PDV	Gaussian	4	clean R2 below the gate	0.176
hERG K_i	VBLL-β (het.)	PDV	Grouped, wider	3	clean R2 below the gate	0.299
hERG K_i	VBLL-β (het.)	PDV	Grouped, wider	4	clean R2 below the gate	0.183
hERG K_i	VBLL-β (het.)	PDV	Grouped, shifted	2	clean R2 below the gate	0.264
hERG K_i	VBLL-β (het.)	PDV	Grouped, shifted	3	clean R2 below the gate	0.049
hERG K_i	VBLL-β (het.)	PDV	Grouped, shifted	4	clean R2 below the gate	0.226
hERG K_i	VBLL-β (het.)	PDV	Student-t (ν=5)	2	clean R2 below the gate	0.270
hERG K_i	VBLL-β (het.)	PDV	Student-t (ν=5)	3	clean R2 below the gate	-0.087
hERG K_i	VBLL-β (het.)	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.220
hERG K_i	VBLL-β (het.)	PDV	Outlier (10%)	2	clean R2 below the gate	0.285
hERG K_i	VBLL-β (het.)	PDV	Outlier (10%)	3	clean R2 below the gate	0.274
hERG K_i	VBLL-β (het.)	PDV	Outlier (10%)	4	clean R2 below the gate	0.224
hERG K_i	VBLL-β (het.)	PDV	Laplace	2	clean R2 below the gate	0.293
hERG K_i	VBLL-β (het.)	PDV	Laplace	3	clean R2 below the gate	0.279
hERG K_i	VBLL-β (het.)	PDV	Laplace	4	clean R2 below the gate	0.179

Additional file 4: Candidate model families

For each pair of models considered for grouping, on how many of the six representations the two differ by less than their standard deviation (SD) across replicates, for clean R^2 and for $\mathrm{AUC}_{\mathrm{norm}}$ under each noise condition run on every representation.

Table S.5: **Candidate model families: how often two models differ by less than their split-to-split variation.** Each row is one pair of base models considered for a family. Each cell counts the representations, of six, on which the two differ by less than their split-to-split SD, followed by the range of the absolute difference. Grouped says whether the pair shares a family in the ANOVA. Outcomes are clean R^2 and $\mathrm{AUC}_{\mathrm{norm}}$ under the three noise conditions every representation ran, on QM9. The difference is the mean over the ten splits of one member minus the other, paired on the split; the split-to-split SD is that of the two members' values over the splits.

Model A	Model B	Grouped	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
RF	QRF	yes	6 of 6 (0.000–0.008)	2 of 6 (0.005–0.010)	6 of 6 (0.011–0.016)	2 of 6 (0.001–0.009)
XGBoost	LightGBM	yes	5 of 6 (0.001–0.022)	6 of 6 (0.000–0.006)	6 of 6 (0.001–0.016)	6 of 6 (0.000–0.004)
XGBoost	NGBoost	no	0 of 6 (0.044–0.125)	0 of 6 (0.015–0.043)	3 of 6 (0.007–0.047)	0 of 6 (0.008–0.028)
LightGBM	NGBoost	no	0 of 6 (0.045–0.138)	0 of 6 (0.021–0.040)	2 of 6 (0.018–0.041)	0 of 6 (0.011–0.028)
NN-α	BNN-α	yes	5 of 6 (0.001–0.023)	6 of 6 (0.004–0.016)	6 of 6 (0.002–0.026)	5 of 6 (0.002–0.025)
NN-α	VBLL-α	no	0 of 6 (0.024–0.073)	2 of 6 (0.008–0.023)	0 of 6 (0.026–0.079)	2 of 6 (0.008–0.019)
BNN-α	VBLL-α	no	0 of 6 (0.042–0.066)	2 of 6 (0.004–0.039)	2 of 6 (0.006–0.059)	2 of 6 (0.006–0.043)
NN-β	BNN-β	yes	5 of 6 (0.000–0.054)	5 of 6 (0.003–0.021)	5 of 6 (0.002–0.033)	4 of 6 (0.001–0.017)
NN-β	VBLL-β	no	0 of 6 (0.044–0.089)	2 of 6 (0.004–0.048)	0 of 6 (0.045–0.067)	3 of 6 (0.005–0.040)
BNN-β	VBLL-β	no	1 of 6 (0.035–0.076)	4 of 6 (0.003–0.056)	1 of 6 (0.017–0.074)	5 of 6 (0.003–0.058)
VBLL-α	VBLL-β	yes	5 of 6 (0.001–0.037)	5 of 6 (0.003–0.017)	6 of 6 (0.001–0.010)	4 of 6 (0.001–0.022)
NN-α	NN-β	no	4 of 6 (0.003–0.045)	4 of 6 (0.004–0.035)	5 of 6 (0.007–0.042)	4 of 6 (0.002–0.031)
BNN-α	BNN-β	no	4 of 6 (0.005–0.063)	4 of 6 (0.005–0.032)	6 of 6 (0.002–0.031)	4 of 6 (0.004–0.025)

Additional file 5: Tukey's honestly significant difference test within each model family

For each pair of models in a family, on how many representations the two differ at adjusted $p < 0.05$, with the size of the difference.

Table S.6: **Tukey HSD between the members of each model family.** One test per representation and outcome, after removing each split's mean. Each cell counts the representations, of six, on which the pair differs at adjusted $p < 0.05$, followed by the range of the absolute difference. With ten splits the test detects differences smaller than the split-to-split variation, so the size of each difference is given beside it.

Family	Model A	Model B	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
Forest	QRF	RF	3 of 6 (0.000–0.008)	6 of 6 (0.005–0.010)	6 of 6 (0.011–0.016)	5 of 6 (0.001–0.009)
Boosting	LightGBM	XGBoost	4 of 6 (0.001–0.022)	3 of 6 (0.000–0.006)	4 of 6 (0.001–0.017)	1 of 6 (0.000–0.004)
NN-α	NN-α	BNN-α	5 of 6 (0.001–0.023)	3 of 6 (0.004–0.016)	4 of 6 (0.002–0.026)	2 of 6 (0.002–0.025)
NN-β	NN-β	BNN-β	1 of 6 (0.000–0.054)	2 of 6 (0.003–0.021)	2 of 6 (0.002–0.033)	2 of 6 (0.001–0.018)
VBLL	VBLL-α	VBLL-β	5 of 6 (0.001–0.037)	4 of 6 (0.003–0.017)	2 of 6 (0.001–0.010)	4 of 6 (0.001–0.022)

Additional file 6: Candidate representation groups

The same comparison as Additional file 4, for pairs of representations over the models.

Table S.7: **Candidate representation groups: how often two representations differ by less than their split-to-split variation.** Each row is one pair of representations considered for a group. Each cell counts the base models on which the two differ by less than their split-to-split SD, followed by the range of the absolute difference. No pair was grouped. Outcomes are clean R^2 and $\mathrm{AUC}_{\mathrm{norm}}$ under the three noise conditions every representation ran, on QM9. The difference is the mean over the ten splits of one member minus the other, paired on the split; the split-to-split SD is that of the two members' values over the splits.

Representation A	Representation B	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
ECFP4	Sort & Slice	0 of 13 (0.035–0.105)	5 of 13 (0.001–0.041)	12 of 13 (0.005–0.025)	5 of 13 (0.001–0.047)

Additional file 7: Intervals for the shares of variance

Every share in the variance decomposition tables, with the lower and upper ends of its 95% interval from 1,000 bootstrap resamples of whole replicates, on all four datasets.

Table S.8: **Intervals for the shares of variance in the decomposition tables.** One row is one term of one decomposition: a dataset, a noise condition, and an outcome. The share is the one printed in Chapter 3; the lower and upper ends are its 95% interval from 1,000 bootstrap resamples of whole replicates. QM9 has 10 replicates; on each assay dataset a replicate is one of five scaffold folds, so its interval rests on five. Predictive performance under noise is R^2 at each dataset's reporting noise level: 1.0 on QM9, LogD, and hERG K_i, and 0.75 on Caco-2.

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	10.6	2.0	16.9
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	35.4	30.5	42.1
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	0.6	0.5	0.8
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	31.1	28.6	36.1
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	12.1	9.7	16.8
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	10.2	5.7	11.9
QM9	Gaussian	Robustness (AUC_{norm})	Replicate	1.7	0.5	2.8
QM9	Gaussian	Robustness (AUC_{norm})	Model family	47.2	41.3	54.1
QM9	Gaussian	Robustness (AUC_{norm})	Model within family	0.6	0.4	1.3
QM9	Gaussian	Robustness (AUC_{norm})	Representation	8.1	6.0	11.2
QM9	Gaussian	Robustness (AUC_{norm})	Interaction	19.9	17.6	27.8
QM9	Gaussian	Robustness (AUC_{norm})	Residual	22.6	14.3	24.8
QM9	Grouped, wider	Robustness (AUC_{norm})	Replicate	1.7	0.6	2.6
QM9	Grouped, wider	Robustness (AUC_{norm})	Model family	46.3	40.0	54.0
QM9	Grouped, wider	Robustness (AUC_{norm})	Model within family	0.3	0.1	0.7
QM9	Grouped, wider	Robustness (AUC_{norm})	Representation	7.9	6.4	10.2
QM9	Grouped, wider	Robustness (AUC_{norm})	Interaction	18.5	16.2	27.9
QM9	Grouped, wider	Robustness (AUC_{norm})	Residual	25.3	16.4	27.7
QM9	Grouped, shifted	Robustness (AUC_{norm})	Replicate	28.4	2.1	38.1
QM9	Grouped, shifted	Robustness (AUC_{norm})	Model family	32.7	25.2	49.5
QM9	Grouped, shifted	Robustness (AUC_{norm})	Model within family	1.0	0.7	1.9
QM9	Grouped, shifted	Robustness (AUC_{norm})	Representation	7.0	5.0	12.1
QM9	Grouped, shifted	Robustness (AUC_{norm})	Interaction	7.2	6.8	11.8
QM9	Grouped, shifted	Robustness (AUC_{norm})	Residual	23.7	17.0	27.5
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Replicate	13.4	1.8	19.7
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model family	26.6	22.1	33.7
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model within family	0.8	0.6	1.3
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Representation	29.0	27.3	33.8
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Interaction	14.2	12.2	18.7
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Residual	16.0	11.7	16.6
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Replicate	12.7	1.5	19.4

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model family	33.0	28.8	40.4
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model within family	0.4	0.2	1.0
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Representation	25.7	22.3	31.0
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Interaction	13.1	12.0	17.0
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Residual	15.1	10.3	16.5
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Replicate	51.0	6.5	69.0
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model family	8.4	2.4	26.4
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model within family	0.4	0.2	1.8
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Representation	10.3	5.0	27.0
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Interaction	6.2	4.7	15.8
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Residual	23.7	16.3	32.9
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	3.3	0.2	4.7
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	60.2	58.9	63.2
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	6.7	5.7	8.1
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	16.8	14.3	20.1
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	9.5	9.1	11.2
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	3.6	1.4	4.0
LogD	Gaussian	Robustness (AUC_{norm})	Replicate	1.3	0.1	2.4
LogD	Gaussian	Robustness (AUC_{norm})	Model family	58.0	49.7	67.1
LogD	Gaussian	Robustness (AUC_{norm})	Model within family	5.2	3.9	7.5
LogD	Gaussian	Robustness (AUC_{norm})	Representation	7.1	5.7	9.9
LogD	Gaussian	Robustness (AUC_{norm})	Interaction	12.2	10.9	19.2
LogD	Gaussian	Robustness (AUC_{norm})	Residual	16.3	5.6	18.5
LogD	Grouped, wider	Robustness (AUC_{norm})	Replicate	2.5	0.1	4.2
LogD	Grouped, wider	Robustness (AUC_{norm})	Model family	56.1	47.3	65.6
LogD	Grouped, wider	Robustness (AUC_{norm})	Model within family	3.6	2.0	6.3
LogD	Grouped, wider	Robustness (AUC_{norm})	Representation	7.5	6.9	9.6
LogD	Grouped, wider	Robustness (AUC_{norm})	Interaction	13.4	13.4	22.9
LogD	Grouped, wider	Robustness (AUC_{norm})	Residual	16.8	6.5	17.9
LogD	Grouped, shifted	Robustness (AUC_{norm})	Replicate	13.9	0.6	26.3
LogD	Grouped, shifted	Robustness (AUC_{norm})	Model family	26.2	22.2	35.7
LogD	Grouped, shifted	Robustness (AUC_{norm})	Model within family	7.5	4.7	14.5
LogD	Grouped, shifted	Robustness (AUC_{norm})	Representation	11.2	10.2	15.4
LogD	Grouped, shifted	Robustness (AUC_{norm})	Interaction	13.3	10.3	26.8
LogD	Grouped, shifted	Robustness (AUC_{norm})	Residual	27.9	13.7	30.2
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Replicate	4.5	0.6	7.4
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model family	59.6	56.2	66.0
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model within family	5.6	3.4	8.8
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Representation	14.8	11.4	19.3
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Interaction	9.3	9.1	12.0
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Residual	6.3	2.6	6.9
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Replicate	4.6	0.4	6.7
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model family	62.4	59.9	69.6
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model within family	4.1	3.1	5.9

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Representation	15.3	10.9	20.2
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Interaction	7.7	7.4	9.9
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Residual	5.9	2.8	5.9
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Replicate	12.1	1.0	15.9
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model family	34.7	30.8	48.0
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model within family	4.7	3.4	7.3
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Representation	25.0	19.3	33.9
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Interaction	11.2	10.2	16.7
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Residual	12.4	5.2	13.1
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	38.7	0.6	47.6
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	22.8	19.3	41.6
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	13.4	10.8	22.0
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	9.9	8.4	16.4
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	8.9	7.9	17.2
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	6.2	2.7	8.9
Caco-2	Gaussian	Robustness (AUC_{norm})	Replicate	18.6	0.1	25.7
Caco-2	Gaussian	Robustness (AUC_{norm})	Model family	43.2	37.5	60.6
Caco-2	Gaussian	Robustness (AUC_{norm})	Model within family	4.8	3.3	9.6
Caco-2	Gaussian	Robustness (AUC_{norm})	Representation	3.1	2.3	5.0
Caco-2	Gaussian	Robustness (AUC_{norm})	Interaction	9.4	9.2	20.9
Caco-2	Gaussian	Robustness (AUC_{norm})	Residual	20.8	7.3	23.6
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Replicate	17.8	1.4	27.7
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Model family	46.7	39.2	66.9
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Model within family	1.8	1.3	5.7
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Representation	3.2	1.2	7.2
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Interaction	8.8	8.0	22.7
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Residual	21.7	9.0	22.8
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Replicate	39.7	3.3	48.9
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Model family	29.1	23.6	49.7
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Model within family	3.6	1.7	12.4
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Representation	4.4	3.5	9.6
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Interaction	5.2	4.7	17.3
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Residual	18.1	8.5	20.0
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Replicate	56.0	2.2	66.2
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Model family	11.4	8.2	31.9
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Model within family	5.8	4.0	13.8
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Representation	9.7	6.9	23.3
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Interaction	6.6	5.4	18.8
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Residual	10.5	5.8	20.2
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Replicate	53.4	2.7	65.6
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Model family	14.0	10.6	33.8
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Model within family	6.0	4.2	12.9
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Representation	8.7	6.1	18.2
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Interaction	8.2	6.7	22.3

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Residual	9.7	5.2	15.5
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Replicate	67.9	11.7	78.6
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Model family	4.4	3.1	18.6
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Model within family	2.3	1.4	10.3
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Representation	7.3	5.1	20.2
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Interaction	3.6	3.0	17.7
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Residual	14.4	6.7	32.3
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	28.1	2.5	35.6
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	18.4	15.9	29.0
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	15.1	9.5	27.6
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	6.7	4.2	14.3
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	14.3	13.2	23.6
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	17.5	7.4	21.2
hERG K_i	Gaussian	Robustness (AUC_{norm})	Replicate	5.0	0.7	10.0
hERG K_i	Gaussian	Robustness (AUC_{norm})	Model family	47.9	45.8	55.6
hERG K_i	Gaussian	Robustness (AUC_{norm})	Model within family	5.3	3.9	8.5
hERG K_i	Gaussian	Robustness (AUC_{norm})	Representation	0.8	0.4	3.6
hERG K_i	Gaussian	Robustness (AUC_{norm})	Interaction	13.0	13.0	23.1
hERG K_i	Gaussian	Robustness (AUC_{norm})	Residual	28.0	8.7	31.2
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Replicate	13.2	0.2	16.9
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Model family	40.5	37.1	55.2
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Model within family	4.4	3.0	7.7
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Representation	2.9	2.3	4.9
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Interaction	6.3	6.3	18.2
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Residual	32.7	12.2	34.9
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Replicate	17.3	2.1	29.3
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Model family	34.6	29.6	51.1
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Model within family	4.0	2.3	8.7
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Representation	0.9	0.4	3.8
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Interaction	14.2	10.9	31.4
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Residual	29.0	9.4	31.4
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Replicate	39.4	6.0	53.8
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model family	22.5	14.8	45.6
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model within family	2.1	0.9	6.6
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Representation	5.6	3.8	14.0
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Interaction	7.0	6.6	20.9
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Residual	23.5	10.1	29.6
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Replicate	29.4	3.0	43.3
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model family	29.3	25.0	39.5
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model within family	2.5	0.6	7.7
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Representation	5.6	2.9	12.2
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Interaction	11.6	10.1	26.2
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Residual	21.6	10.4	22.5
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Replicate	32.6	5.4	47.3

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model family	15.6	12.8	29.0
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model within family	2.5	1.6	7.5
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Representation	1.9	0.7	8.5
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Interaction	12.5	10.1	29.8
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Residual	35.0	17.4	39.4

Additional file 8: Robustness grid under grouped-wider noise

AUC_{norm} of every base model on every representation under grouped-wider noise on QM9, the condition Figure 3.5 does not show.

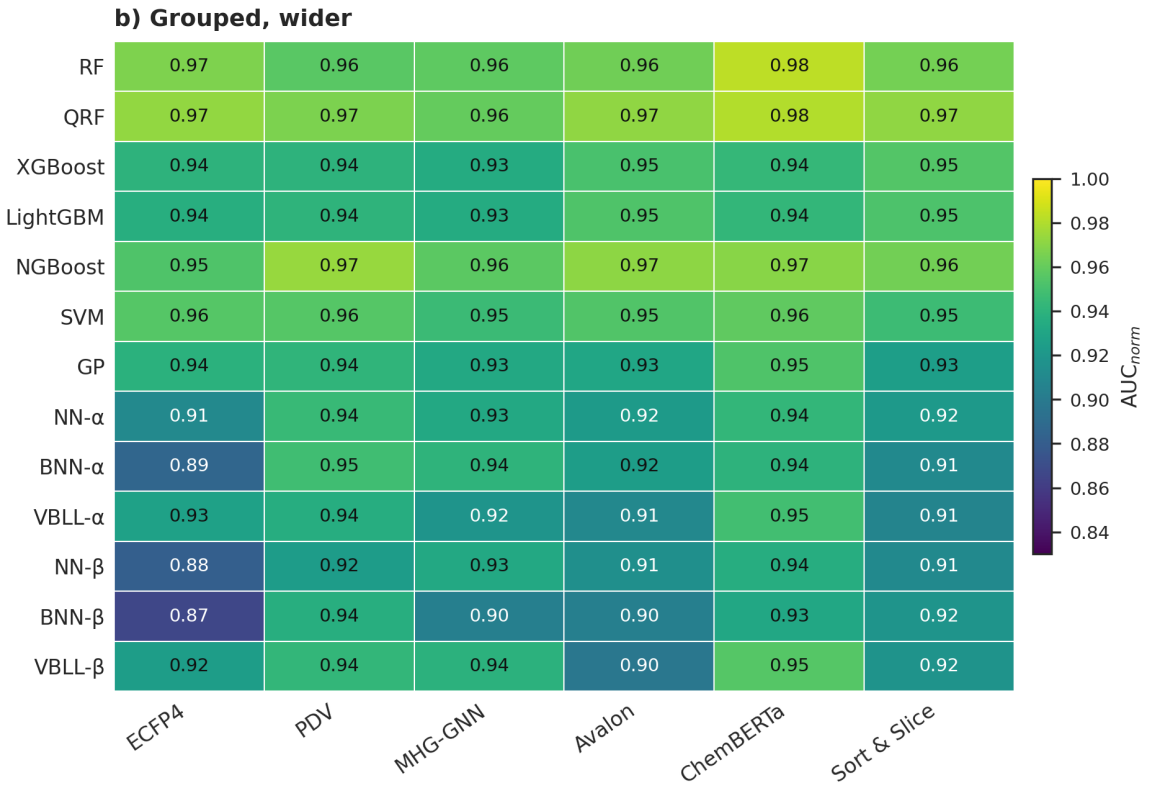


Figure S.1: **Robustness of every base model on every representation under grouped-wider noise, on the QM9 HOMO–LUMO gap.** Rows are the 13 base models, ordered by family, and columns the six representations. One cell is one model on one representation, the median AUC_{norm} over 10 replicates, with its value printed on it. Color runs on the same fixed range as Figure 3.5, which shows Gaussian and grouped-shifted noise.

Additional file 9: Robustness of the variant models on the three assay datasets

Clean R^2 and $\mathrm{AUC}_{\mathrm{norm}}$ of the six variant models on LogD, Caco-2, and hERG K_i, on ECFP4, beside Figure 3.10.

Table S.9: **Robustness of the six variant models on the three assay datasets, on ECFP4.** One row is one variant model on one dataset. The first column is clean R^2; the others are $\mathrm{AUC}_{\mathrm{norm}}$ under each noise condition run on every model. Values are medians over five scaffold folds.

Dataset	Model	Clean R^2	Gaussian	Grouped, wider	Grouped, shifted
LogD	RF (300 trees)	0.643	0.947	0.958	0.897
LogD	GP (het.)	0.804	0.896	0.893	0.798
LogD	BNN-α (var. head)	0.748	0.881	0.852	0.773
LogD	VBLL-α (het.)	0.556	0.693	0.731	0.768
LogD	BNN-β (var. head)	0.752	0.884	0.867	0.797
LogD	VBLL-β (het.)	0.659	0.796	0.864	0.837
Caco-2	RF (300 trees)	0.412	0.876	0.878	0.774
Caco-2	GP (het.)	0.517	0.815	0.834	0.706
Caco-2	BNN-α (var. head)	0.394	0.787	0.769	0.721
Caco-2	VBLL-α (het.)	0.329	0.653	0.733	0.781
Caco-2	BNN-β (var. head)	0.454	0.739	0.739	0.766
Caco-2	VBLL-β (het.)	0.414	0.707	0.739	0.806
hERG K_i	RF (300 trees)	0.553	0.910	0.917	0.800
hERG K_i	GP (het.)	0.578	0.843	0.864	0.775
hERG K_i	BNN-α (var. head)	0.491	0.809	0.800	0.679
hERG K_i	VBLL-α (het.)	0.420	0.730	0.845	0.749
hERG K_i	BNN-β (var. head)	0.539	0.812	0.753	0.728
hERG K_i	VBLL-β (het.)	0.410	0.836	0.834	0.811

Additional file 10: Base models against their probabilistic counterparts on the three assay datasets

The change in $\mathrm{AUC}_{\mathrm{norm}}$ and in clean R^2, counterpart minus base, for each pair, noise condition, and representation, on LogD, Caco-2, and hERG K_i. These tables give the numbers behind Figure 3.12 of the thesis for these three datasets.

Table S.10: **LogD: what replacing a model with its own probabilistic counterpart changes.** Two rows per pair and noise condition, one column per representation: the change in $\mathrm{AUC}_{\mathrm{norm}}$, then the change in clean R^2, both counterpart minus base, the median of the paired differences over five scaffold folds. † marks an $\mathrm{AUC}_{\mathrm{norm}}$ change with the same sign in all five. "Settings" says whether the two models share one setting.

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF (300 trees) → QRF	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.017	-0.020†	-0.021†	-0.021†	-0.019†	-0.021†
RF (300 trees) → QRF	shared	Gaussian	Clean R^2	-0.005	+0.009	+0.008	-0.001	+0.014	-0.004
RF (300 trees) → QRF	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.037†	-0.027†	-0.032†	-0.032†	-0.023†	-0.045
RF (300 trees) → QRF	shared	Grouped, shifted	Clean R^2	-0.005	+0.009	+0.008	-0.001	+0.014	-0.004
RF (300 trees) → QRF	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.005†	-0.004†	-0.017	-0.003	-0.015†	-0.010
RF (300 trees) → QRF	shared	Grouped, wider	Clean R^2	-0.005	+0.009	+0.008	-0.001	+0.014	-0.004
RF → QRF	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.008†	-0.015†	-0.022†	-0.010	-0.005	-0.017†
RF → QRF	differ	Gaussian	Clean R^2	-0.005	+0.011	+0.012	+0.002	+0.020	-0.004
RF → QRF	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.034†	-0.015	-0.028†	-0.018†	-0.014	-0.033†
RF → QRF	differ	Grouped, shifted	Clean R^2	-0.005	+0.011	+0.012	+0.002	+0.020	-0.004
RF → QRF	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.003	-0.001	-0.006	+0.001	+0.003	-0.007
RF → QRF	differ	Grouped, wider	Clean R^2	-0.005	+0.011	+0.012	+0.002	+0.020	-0.004
NN-α → BNN-α	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.037†	-0.023	-0.058	+0.003	-0.028	+0.040†
NN-α → BNN-α	shared	Gaussian	Clean R^2	-0.018	-0.094	-0.157	-0.084	-0.056	-0.060
NN-α → BNN-α	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.054†	+0.112	-0.087	+0.138†	+0.011	+0.029†
NN-α → BNN-α	shared	Grouped, shifted	Clean R^2	-0.028	-0.128	-0.049	-0.139	-0.045	-0.049
NN-α → BNN-α	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.027†	-0.009	-0.075	+0.067†	-0.010	+0.038†
NN-α → BNN-α	shared	Grouped, wider	Clean R^2	-0.026	-0.062	-0.053	-0.144	-0.111	-0.053
BNN-α → VBLL-α	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.159†	-0.117	-0.075	-0.334†	-0.208†	-0.106†
BNN-α → VBLL-α	differ	Gaussian	Clean R^2	-0.234	-0.053	-0.136	-0.258	-0.270	-0.170
BNN-α → VBLL-α	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.112†	-0.115	+0.030	-0.183	-0.114	-0.061
BNN-α → VBLL-α	differ	Grouped, shifted	Clean R^2	-0.210	-0.012	-0.221	-0.261	-0.290	-0.135
BNN-α → VBLL-α	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	-0.124†	-0.041	-0.094†	-0.357†	-0.200†	-0.141†
BNN-α → VBLL-α	differ	Grouped, wider	Clean R^2	-0.215	-0.125	-0.200	-0.227	-0.235	-0.114
BNN-α → BNN-α (var. head)	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.007	-0.013	-0.251	+0.004	-0.060	-0.053
BNN-α → BNN-α (var. head)	differ	Gaussian	Clean R^2	-0.018	-0.027	-0.036	-0.013	-0.038	+0.016
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.060†	-0.119	-0.047	-0.133†	-0.085†	-0.027
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	Clean R^2	+0.005	+0.085	-0.123	+0.030	-0.064	+0.016
BNN-α → BNN-α (var. head)	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	-0.027	-0.025	-0.000	-0.105	-0.122†	-0.050†
BNN-α → BNN-α (var. head)	differ	Grouped, wider	Clean R^2	-0.000	+0.005	-0.175	-0.026	-0.016	+0.030
NN-β → BNN-β	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.035†	+0.074†	+0.009	+0.022	+0.021	+0.002
NN-β → BNN-β	shared	Gaussian	Clean R^2	-0.030	-0.103	-0.091	-0.093	-0.062	-0.016

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
NN-β → BNN-β	shared	Grouped, shifted	AUC_{norm}	+0.043†	+0.026	+0.058†	+0.036	+0.037	+0.046†
NN-β → BNN-β	shared	Grouped, shifted	Clean R^2	-0.026	-0.036	-0.139	-0.061	-0.053	-0.051
NN-β → BNN-β	shared	Grouped, wider	AUC_{norm}	+0.039†	-0.009	-0.047	+0.013	-0.004	+0.040
NN-β → BNN-β	shared	Grouped, wider	Clean R^2	-0.024	-0.028	-0.040	-0.060	-0.034	-0.046
BNN-β → VBLL-β	differ	Gaussian	AUC_{norm}	-0.098†	-0.081†	-0.118†	-0.095†	-0.153†	-0.046
BNN-β → VBLL-β	differ	Gaussian	Clean R^2	-0.144	-0.029	-0.111	-0.130	-0.167	-0.095
BNN-β → VBLL-β	differ	Grouped, shifted	AUC_{norm}	-0.042	-0.010	-0.061†	-0.051	-0.053	+0.015
BNN-β → VBLL-β	differ	Grouped, shifted	Clean R^2	-0.118	-0.062	-0.103	-0.109	-0.166	-0.079
BNN-β → VBLL-β	differ	Grouped, wider	AUC_{norm}	-0.077†	-0.045†	-0.079	-0.096	-0.086†	-0.060†
BNN-β → VBLL-β	differ	Grouped, wider	Clean R^2	-0.133	-0.064	-0.145	-0.132	-0.178	-0.044
BNN-β → BNN-β (var. head)	differ	Gaussian	AUC_{norm}	-0.008	+0.013	-0.041	+0.018	-0.169	+0.011
BNN-β → BNN-β (var. head)	differ	Gaussian	Clean R^2	-0.013	-0.006	-0.045	-0.014	-0.042	-0.031
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.038†	-0.052	-0.153	+0.063	-0.035	-0.043
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	Clean R^2	-0.011	+0.012	+0.016	-0.060	-0.048	+0.007
BNN-β → BNN-β (var. head)	differ	Grouped, wider	AUC_{norm}	-0.009	-0.025	-0.176†	-0.008	-0.102	-0.007
BNN-β → BNN-β (var. head)	differ	Grouped, wider	Clean R^2	-0.014	-0.000	-0.085	-0.039	-0.000	+0.007
GP → GP (het.)	shared	Gaussian	AUC_{norm}	+0.006	-0.001	-0.006	+0.000	-0.004	-0.005
GP → GP (het.)	shared	Gaussian	Clean R^2	+0.000	+0.000	+0.000	+0.000	-0.000	-0.000
GP → GP (het.)	shared	Grouped, shifted	AUC_{norm}	-0.003	+0.010	+0.022	+0.020†	+0.024†	-0.003
GP → GP (het.)	shared	Grouped, shifted	Clean R^2	+0.000	-0.000	+0.000	+0.000	-0.001	-0.000
GP → GP (het.)	shared	Grouped, wider	AUC_{norm}	-0.009	-0.002	+0.002	-0.001	-0.002	-0.006
GP → GP (het.)	shared	Grouped, wider	Clean R^2	+0.000	-0.000	+0.000	+0.000	-0.001	-0.000

Table S.11: **Caco-2: what replacing a model with its own probabilistic counterpart changes.** Two rows per pair and noise condition, one column per representation: the change in AUC_{norm}, then the change in clean R^2, both counterpart minus base, the median of the paired differences over five scaffold folds. † marks an AUC_{norm} change with the same sign in all five. "Settings" says whether the two models share one setting.

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF (300 trees) → QRF	shared	Gaussian	AUC_{norm}	+0.022	-0.004	-0.012	-0.001	+0.016†	+0.025
RF (300 trees) → QRF	shared	Gaussian	Clean R^2	-0.027	-0.033	-0.023	-0.020	-0.022	-0.042
RF (300 trees) → QRF	shared	Grouped, shifted	AUC_{norm}	-0.029†	-0.025†	-0.033†	-0.040†	-0.017	-0.028
RF (300 trees) → QRF	shared	Grouped, shifted	Clean R^2	-0.027	-0.033	-0.023	-0.020	-0.022	-0.042
RF (300 trees) → QRF	shared	Grouped, wider	AUC_{norm}	+0.048†	+0.015†	+0.009	+0.033	+0.027†	+0.040
RF (300 trees) → QRF	shared	Grouped, wider	Clean R^2	-0.027	-0.033	-0.023	-0.020	-0.022	-0.042
RF → QRF	differ	Gaussian	AUC_{norm}	-0.031	+0.029	+0.018	+0.033	+0.016	+0.010
RF → QRF	differ	Gaussian	Clean R^2	-0.021	-0.031	-0.019	-0.019	-0.024	-0.036
RF → QRF	differ	Grouped, shifted	AUC_{norm}	-0.054	-0.016	-0.056	-0.059	+0.025†	-0.043
RF → QRF	differ	Grouped, shifted	Clean R^2	-0.021	-0.031	-0.019	-0.019	-0.024	-0.036
RF → QRF	differ	Grouped, wider	AUC_{norm}	+0.038	+0.055†	+0.013	+0.046	+0.062†	+0.042

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF → QRF	differ	Grouped, wider	Clean R^2	-0.021	-0.031	-0.019	-0.019	-0.024	-0.036
NN-α → BNN-α	shared	Gaussian	AUC_{norm}	+0.091	+0.016	-0.267†	+0.158	—	+0.095†
NN-α → BNN-α	shared	Gaussian	Clean R^2	-0.011	-0.102	-0.095	-0.127	—	-0.120
NN-α → BNN-α	shared	Grouped, shifted	AUC_{norm}	+0.136	+0.094	—	+0.004	—	+0.119
NN-α → BNN-α	shared	Grouped, shifted	Clean R^2	-0.030	-0.106	—	-0.110	—	-0.088
NN-α → BNN-α	shared	Grouped, wider	AUC_{norm}	-0.010	-0.061	+0.024†	-0.019	+0.111†	+0.117
NN-α → BNN-α	shared	Grouped, wider	Clean R^2	-0.027	-0.115	-0.237	-0.074	-0.269	-0.125
BNN-α → VBLL-α	differ	Gaussian	AUC_{norm}	+0.145†	-0.063	—	-0.137†	—	+0.064
BNN-α → VBLL-α	differ	Gaussian	Clean R^2	-0.146	+0.070	—	+0.024	—	-0.065
BNN-α → VBLL-α	differ	Grouped, shifted	AUC_{norm}	+0.184†	+0.185†	—	+0.274†	—	+0.191
BNN-α → VBLL-α	differ	Grouped, shifted	Clean R^2	-0.142	-0.039	—	-0.085	—	-0.032
BNN-α → VBLL-α	differ	Grouped, wider	AUC_{norm}	+0.112	+0.154	—	+0.378†	+0.002†	-0.140
BNN-α → VBLL-α	differ	Grouped, wider	Clean R^2	-0.152	+0.002	—	-0.167	+0.062	+0.026
BNN-α → BNN-α (var. head)	differ	Gaussian	AUC_{norm}	+0.014	-0.068	—	-0.276†	—	-0.139†
BNN-α → BNN-α (var. head)	differ	Gaussian	Clean R^2	-0.046	+0.015	—	+0.007	—	+0.011
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.073	-0.118	—	-0.160	—	+0.031
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	Clean R^2	+0.005	+0.023	—	-0.016	—	-0.008
BNN-α → BNN-α (var. head)	differ	Grouped, wider	AUC_{norm}	+0.018	-0.068	-0.686†	-0.268†	-0.487†	-0.220
BNN-α → BNN-α (var. head)	differ	Grouped, wider	Clean R^2	-0.043	+0.023	+0.125	+0.022	+0.020	-0.018
NN-β → BNN-β	shared	Gaussian	AUC_{norm}	+0.075†	-0.043	+0.119	-0.000	+0.096	-0.004
NN-β → BNN-β	shared	Gaussian	Clean R^2	-0.018	-0.035	-0.247	-0.091	-0.193	-0.035
NN-β → BNN-β	shared	Grouped, shifted	AUC_{norm}	+0.075	+0.021	+0.016†	+0.070†	+0.029	+0.138
NN-β → BNN-β	shared	Grouped, shifted	Clean R^2	-0.043	-0.021	-0.201	-0.140	-0.134	-0.075
NN-β → BNN-β	shared	Grouped, wider	AUC_{norm}	+0.048	+0.059	-0.032	+0.102	-0.125†	+0.055
NN-β → BNN-β	shared	Grouped, wider	Clean R^2	-0.016	-0.102	-0.137	-0.074	-0.070	-0.082
BNN-β → VBLL-β	differ	Gaussian	AUC_{norm}	+0.081†	+0.124†	+0.037	+0.161	+0.085†	+0.117
BNN-β → VBLL-β	differ	Gaussian	Clean R^2	-0.091	-0.017	+0.075	-0.036	+0.055	-0.051
BNN-β → VBLL-β	differ	Grouped, shifted	AUC_{norm}	+0.119†	-0.044	+0.171†	+0.209†	+0.051†	+0.132†
BNN-β → VBLL-β	differ	Grouped, shifted	Clean R^2	-0.091	+0.006	+0.022	-0.001	+0.046	-0.046
BNN-β → VBLL-β	differ	Grouped, wider	AUC_{norm}	+0.042	+0.073	+0.334†	+0.188†	+0.223†	+0.082†
BNN-β → VBLL-β	differ	Grouped, wider	Clean R^2	-0.057	-0.002	-0.063	-0.048	-0.079	-0.021
BNN-β → BNN-β (var. head)	differ	Gaussian	AUC_{norm}	-0.073	-0.082†	—	-0.135†	-0.455†	-0.052
BNN-β → BNN-β (var. head)	differ	Gaussian	Clean R^2	-0.016	-0.035	—	+0.012	+0.067	+0.007
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.006	-0.088	—	-0.119†	-0.079†	-0.022
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	Clean R^2	-0.060	-0.011	—	-0.085	-0.015	-0.018
BNN-β → BNN-β (var. head)	differ	Grouped, wider	AUC_{norm}	-0.095	-0.241	—	-0.190†	-0.122	-0.161†
BNN-β → BNN-β (var. head)	differ	Grouped, wider	Clean R^2	-0.028	-0.003	—	-0.061	-0.076	+0.035
GP → GP (het.)	shared	Gaussian	AUC_{norm}	+0.001	+0.002	+0.002	+0.004	+0.011	+0.005†
GP → GP (het.)	shared	Gaussian	Clean R^2	-0.000	+0.001	+0.002	-0.000	+0.002	+0.000
GP → GP (het.)	shared	Grouped, shifted	AUC_{norm}	+0.010†	+0.013†	+0.025	+0.021†	+0.055†	+0.025
GP → GP (het.)	shared	Grouped, shifted	Clean R^2	-0.000	+0.000	+0.002	-0.000	+0.002	+0.000
GP → GP (het.)	shared	Grouped, wider	AUC_{norm}	+0.002	+0.002	-0.001	+0.004	+0.003	+0.008†
GP → GP (het.)	shared	Grouped, wider	Clean R^2	-0.000	+0.001	+0.002	-0.001	+0.002	+0.000

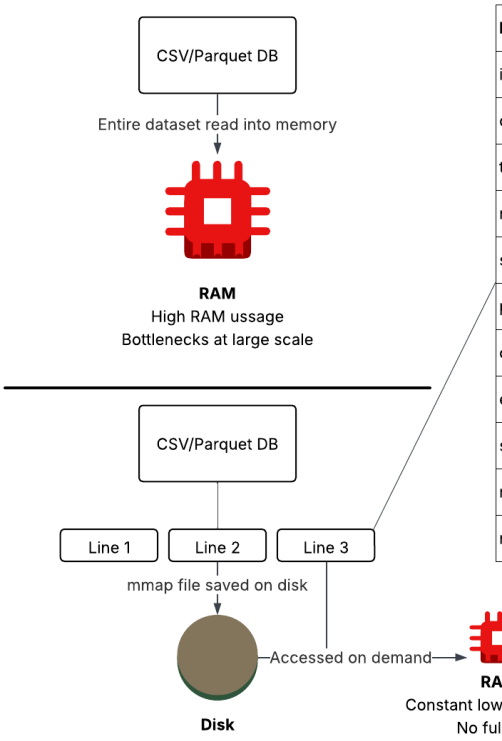
Table S.12: **hERG K_i: what replacing a model with its own probabilistic counterpart changes.** Two rows per pair and noise condition, one column per representation: the change in $\mathrm{AUC}_{\mathrm{norm}}$, then the change in clean R^2, both counterpart minus base, the median of the paired differences over five scaffold folds. † marks an $\mathrm{AUC}_{\mathrm{norm}}$ change with the same sign in all five. "Settings" says whether the two models share one setting.

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF (300 trees) → QRF	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.000	-0.043†	-0.029	-0.032	-0.023	-0.017
RF (300 trees) → QRF	shared	Gaussian	Clean R^2	-0.010	+0.015	-0.007	-0.001	+0.005	-0.023
RF (300 trees) → QRF	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.044†	-0.073†	-0.034†	-0.058†	-0.033†	-0.047†
RF (300 trees) → QRF	shared	Grouped, shifted	Clean R^2	-0.010	+0.015	-0.007	-0.001	+0.005	-0.023
RF (300 trees) → QRF	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.013†	-0.016†	+0.010	+0.012	-0.018	+0.005
RF (300 trees) → QRF	shared	Grouped, wider	Clean R^2	-0.010	+0.015	-0.007	-0.001	+0.005	-0.023
RF → QRF	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.004	-0.035†	-0.015	-0.037	-0.033	+0.015
RF → QRF	differ	Gaussian	Clean R^2	-0.009	+0.020	+0.001	+0.005	+0.019	-0.016
RF → QRF	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.070†	-0.054†	-0.018†	-0.056†	-0.030†	-0.026
RF → QRF	differ	Grouped, shifted	Clean R^2	-0.009	+0.020	+0.001	+0.005	+0.019	-0.016
RF → QRF	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.031	-0.001	+0.015	+0.009	-0.006	+0.026†
RF → QRF	differ	Grouped, wider	Clean R^2	-0.009	+0.020	+0.001	+0.005	+0.019	-0.016
NN-α → BNN-α	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.004	+0.215†	-0.145†	+0.114	—	+0.006
NN-α → BNN-α	shared	Gaussian	Clean R^2	+0.011	-0.145	-0.266	-0.155	—	-0.057
NN-α → BNN-α	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.052	+0.022	-0.359†	+0.105	—	-0.030
NN-α → BNN-α	shared	Grouped, shifted	Clean R^2	+0.015	-0.158	-0.253	-0.126	—	-0.071
NN-α → BNN-α	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.049	+0.048	+0.095†	+0.059	+0.178†	+0.126
NN-α → BNN-α	shared	Grouped, wider	Clean R^2	+0.029	-0.124	-0.334	-0.076	-0.248	-0.103
BNN-α → VBLL-α	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.146†	+0.142†	+0.358†	+0.088	—	+0.169
BNN-α → VBLL-α	differ	Gaussian	Clean R^2	-0.094	-0.009	+0.065	-0.007	—	+0.009
BNN-α → VBLL-α	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.074	+0.059	+0.590†	+0.027	—	+0.211†
BNN-α → VBLL-α	differ	Grouped, shifted	Clean R^2	-0.076	+0.025	-0.027	-0.012	—	-0.025
BNN-α → VBLL-α	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.069†	+0.024†	+0.017	+0.135	-0.035†	-0.034
BNN-α → VBLL-α	differ	Grouped, wider	Clean R^2	-0.114	+0.029	+0.077	-0.072	+0.065	+0.005
BNN-α → BNN-α (var. head)	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.040	-0.625†	—	-0.384†	—	-0.324†
BNN-α → BNN-α (var. head)	differ	Gaussian	Clean R^2	-0.057	-0.052	—	-0.040	—	+0.060
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.010	-0.506†	—	-0.335†	—	-0.377†
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	Clean R^2	-0.029	-0.108	—	-0.029	—	+0.007
BNN-α → BNN-α (var. head)	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	-0.060	-0.448†	—	-0.251†	—	-0.412†
BNN-α → BNN-α (var. head)	differ	Grouped, wider	Clean R^2	-0.011	-0.055	—	-0.034	—	-0.011
NN-β → BNN-β	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.008	+0.135†	+0.123†	+0.124	-0.142†	+0.003
NN-β → BNN-β	shared	Gaussian	Clean R^2	+0.076	-0.118	-0.190	-0.163	-0.215	-0.080
NN-β → BNN-β	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.077†	-0.133†	-0.052	+0.184	—	-0.217†
NN-β → BNN-β	shared	Grouped, shifted	Clean R^2	+0.015	-0.053	-0.212	-0.049	—	-0.086
NN-β → BNN-β	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.022	+0.055	+0.120†	+0.035	+0.030	+0.052
NN-β → BNN-β	shared	Grouped, wider	Clean R^2	+0.010	-0.157	-0.230	-0.059	-0.185	-0.093
BNN-β → VBLL-β	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.009	+0.037	+0.162†	-0.103	+0.408†	+0.065
BNN-β → VBLL-β	differ	Gaussian	Clean R^2	-0.091	+0.053	+0.015	+0.078	+0.089	+0.011
BNN-β → VBLL-β	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.025	-0.034†	+0.250	-0.040	—	+0.268†
BNN-β → VBLL-β	differ	Grouped, shifted	Clean R^2	-0.071	+0.039	+0.039	+0.016	—	+0.035

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
BNN-β → VBLL-β	differ	Grouped, wider	AUC_{norm}	+0.020	-0.232	-0.049†	+0.032	+0.097†	-0.165
BNN-β → VBLL-β	differ	Grouped, wider	Clean R^2	-0.063	+0.045	+0.132	+0.034	+0.061	+0.055
BNN-β → BNN-β (var. head)	differ	Gaussian	AUC_{norm}	-0.009	-0.300†	—	-0.418†	—	-0.351
BNN-β → BNN-β (var. head)	differ	Gaussian	Clean R^2	-0.007	-0.004	—	+0.082	—	-0.054
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.003	-0.156†	—	-0.444†	—	-0.143
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	Clean R^2	-0.037	-0.065	—	-0.004	—	+0.003
BNN-β → BNN-β (var. head)	differ	Grouped, wider	AUC_{norm}	-0.084	-0.514†	—	-0.235†	—	-0.308†
BNN-β → BNN-β (var. head)	differ	Grouped, wider	Clean R^2	-0.014	+0.033	—	-0.071	—	+0.059
GP → GP (het.)	shared	Gaussian	AUC_{norm}	+0.001	-0.008	+0.001	+0.007	+0.010	+0.009
GP → GP (het.)	shared	Gaussian	Clean R^2	+0.000	-0.002	-0.005	-0.000	-0.010	+0.000
GP → GP (het.)	shared	Grouped, shifted	AUC_{norm}	+0.007	+0.020	+0.029	+0.030	+0.025	+0.011
GP → GP (het.)	shared	Grouped, shifted	Clean R^2	+0.000	-0.000	-0.005	-0.000	-0.011	+0.000
GP → GP (het.)	shared	Grouped, wider	AUC_{norm}	-0.002	+0.001	+0.006	+0.008	+0.017†	+0.004†
GP → GP (het.)	shared	Grouped, wider	Clean R^2	+0.000	-0.001	-0.005	-0.000	-0.012	+0.000

Additional file 11: Memory-mapped data files of the QM9 pipeline

Rather than parse a CSV or Parquet file into memory for every replicate, our pipeline writes the preprocessed data to memory-mapped binary files (`.mmap`), which stay on disk and are read record by record (Figure S.2). Python writes one file per split, and the Rust program writes a copy of it in which the processed labels replace the raw ones; Python reads that copy for training. Each record holds the isomeric and canonical SMILES, the raw and processed target values, and the requested representations as fixed-width blocks. Python and Rust read the same pages on disk without deserialization.



Field Name	Required?	Data Type & Size	Size
isomeric_smiles	Yes	UTF-8	Variable length
canonical_smiles	Yes	UTF-8	Variable length
target_value	Yes	float32	4 bytes
randomised_smiles	Optional	UTF-8	Variable length
sns	Optional	uint8[16]	16 bytes
processed_target	Yes	float32	4 bytes
domain_label	Optional	uint8	1 byte
ecfp4	Optional	uint8[32]	32 bytes
smiles_ohe	Optional	bit-packed uint8[]	Variable size
randomised_smiles_ohe	Optional	bit-packed uint8[]	Variable size
newline separator	Yes	ASCII '\n' character	1 byte

Figure S.2: In-memory (top left) and memory-mapped (bottom left) data loading. Memory-mapped loading keeps each molecule as one record of a binary file on disk and reads records into memory on demand. The table on the right lists the fields of one record.

S.3 Limitations of the similar property principle

One row of Table S.13 is one representation on one dataset. Representations are listed in each dataset's pool order.

Table S.13: All-pairs correlation, five-nearest-neighbor prediction, and activity cliffs (ACs) for every representation in each regression dataset's pool. All pairs: Spearman correlation between pair distance and absolute label difference, on a fixed sample of 2,000 molecules (seed 0) where a dataset has more. Five neighbors: Spearman correlation and RMSE (in the label's units) of the mean label of the five nearest molecules from other scaffold groups. Cliff molecules: molecules in at least one of this representation's closest pairs whose labels differ by more than the dataset's gap (Table 4.2). The number of closest pairs is the number ECFP4 has at Tanimoto ≥ 0.55, a few more where distances tie at the cut-off. Shared: percentage of ECFP4's similar pairs that are also among this representation's closest pairs. HLM clearance's pool has no ECFP4.

Dataset	Representation	All pairs Spearman	Five neighbors Spearman	Five neighbors RMSE	Cliff molecules	Shared (%)
hERG K_i	MHG-GNN	0.07	0.71	0.65	208	67
	Mordred	0.07	0.72	0.62	165	62
	WL graph kernel	0.04	0.71	0.64	267	74
	Avalon	0.02	0.69	0.65	238	63
	GROVER	0.05	0.70	0.63	192	63
	PDV	0.08	0.73	0.62	208	61
	ECFP4	0.06	0.72	0.63	211	100
	Atom pairs	0.04	0.74	0.60	138	68
	ChemBERT	0.07	0.68	0.66	166	57
	SMI-TED	0.09	0.69	0.65	187	58
LogD	MHG-GNN	0.09	0.81	0.66	286	42
	Mordred	0.06	0.80	0.68	308	35
	GROVER	0.09	0.81	0.67	399	51
	ECFP4	0.12	0.76	0.74	731	100
	SMI-TED	0.10	0.76	0.74	758	42
	MoLFormer	0.13	0.77	0.73	768	46
	Avalon	0.04	0.73	0.77	801	49
	Topological torsions	0.10	0.74	0.76	808	71
	ChemBERT	0.09	0.75	0.75	808	42
	GraphMVP	0.11	0.76	0.74	415	38
Caco-2 efflux	WL graph kernel	0.13	0.58	0.35	398	54
	MHG-GNN	0.16	0.60	0.34	347	53
	ECFP4	0.08	0.59	0.34	324	100
	PDV	0.18	0.62	0.34	237	43
	GROVER	0.06	0.61	0.34	302	55
	Avalon	−0.00	0.56	0.35	335	46
	Topological torsions	0.15	0.61	0.34	348	74
	SMI-TED	0.14	0.57	0.35	400	44
	Atom pairs	0.08	0.59	0.34	315	57
	MoLFormer	0.09	0.59	0.34	311	51
RLM clearance	PDV	0.06	0.56	0.62	433	34
	MHG-GNN	0.02	0.47	0.67	487	46
	SMI-TED	0.04	0.46	0.67	483	48
	Mordred	0.07	0.49	0.66	445	42
	MoLFormer	0.08	0.44	0.68	412	60
	GROVER	0.05	0.43	0.69	436	46

Dataset	Representation	All pairs Spearman	Five neighbors Spearman	Five neighbors RMSE	Cliff molecules	Shared (%)
	WL graph kernel	0.00	0.40	0.71	554	36
	Atom pairs	0.07	0.47	0.67	437	60
	Avalon	0.04	0.29	0.74	389	55
	ECFP4	0.02	0.36	0.71	458	100
MDR1–MDCK efflux	Mordred	0.16	0.58	0.57	363	45
	PDV	0.23	0.64	0.54	302	39
	MHG-GNN	0.20	0.52	0.62	344	48
	Avalon	−0.04	0.50	0.61	378	61
	GROVER	−0.03	0.47	0.63	356	49
	SMI-TED	0.10	0.59	0.56	389	51
	Atom pairs	0.00	0.58	0.57	374	64
	WL graph kernel	0.17	0.47	0.64	413	40
	MACCS keys	−0.04	0.54	0.59	379	46
	ECFP4	0.02	0.51	0.61	397	100
HLM clearance	Mordred	0.07	0.49	0.55	429	43
	PDV	0.10	0.55	0.52	390	34
	MHG-GNN	0.08	0.46	0.56	435	46
	SMI-TED	0.03	0.45	0.56	475	48
	MoLFormer	0.00	0.43	0.56	422	60
	GROVER	−0.03	0.41	0.59	413	46
	Atom pairs	−0.04	0.47	0.55	426	60
	ChemBERT	−0.02	0.39	0.58	452	45
	WL graph kernel	0.12	0.38	0.59	490	37
	Avalon	0.01	0.32	0.61	416	56
QM9	Mordred	0.18	0.84	0.68	1,443	3
	Avalon	0.19	0.83	0.70	1,148	9
	SMI-TED	0.13	0.85	0.65	1,576	13
	ECFP4	0.19	0.88	0.64	1,495	100
	SELFormer	0.20	0.82	0.73	300	2
	ChemBERTa-2	0.19	0.89	0.61	1,108	1
	Mol2vec	0.14	0.86	0.66	1,274	24
	ChemBERT	0.19	0.83	0.72	1,295	7
	GraphMVP	0.20	0.83	0.71	1,231	9
	MoLFormer	0.21	0.84	0.70	1,423	3

Additional file 12: Constant and near-constant features per representation

Representation	Features	hERG K_i (1,412)	Caco-2 (2,158)	LogD (5,028)	QM9 (9,978)
ECFP4	2,048	105 / 1,815	194 / 1,826	39 / 1,834	0 / 1,997
Atom pairs	2,048	201 / 977	563 / 1,239	–	–
Topological torsions	2,048	–	1,179 / 1,822	1,059 / 1,818	–
Avalon	2,048	50 / 985	98 / 1,128	30 / 1,076	4 / 1,380
MACCS keys	167	24 / 60	36 / 86	31 / 87	n/a
WL graph kernel	140–317	0 / 0 / 276	0 / 0 / 317	–	–
PDV	200	21 / 55	35 / 66	–	–
Mordred	1,613	242 / 336	–	247 / 340	307 / 394
GROVER	4,800	0 / 0	0 / 0	0 / 0	–
MHG-GNN	1,024	292 / 292	292 / 310	292 / 304	–
GraphMVP	300	–	–	0 / 0	0 / 0
MoLFormer	768	–	0 / 0	0 / 0	0 / 0
SMI-TED	768	0 / 0	0 / 0	0 / 0	0 / 0
ChemBERT	256	0 / 0	–	0 / 0	0 / 0
ChemBERTa-2	384	–	–	–	0 / 0
SELFormer	768	–	–	–	0 / 0
Mol2vec	300	–	–	–	0 / 0

Table S.14: Constant and near-constant features of each representation on hERG K_i, Caco-2, LogD, and QM9, with the number of molecules under each dataset name. Each entry gives the number of constant features and the number of near-constant features (one value in at least 95% of molecules); WL graph kernel entries add its number of features, which varies by dataset. A dash marks a representation outside the dataset's single representations, and n/a one that was not computed.

S.4 Building adaptive hybrid representations

Representation	Features	MDR1–MDCK (2,641)	HLM clearance (3,086)	RLM clearance (3,053)	hERG blockers (7,772)
ECFP4	2,048	1 / 1,890	–	1 / 1,887	0 / 1,840
Atom pairs	2,048	308 / 1,233	260 / 1,240	262 / 1,238	28 / 929
Topological torsions	2,048	–	–	–	625 / 1,754
Avalon	2,048	5 / 1,032	8 / 1,044	8 / 1,045	0 / 866
MACCS keys	167	22 / 58	20 / 59	20 / 58	9 / 50
WL graph kernel	140–317	0 / 0 / 143	0 / 0 / 141	0 / 0 / 140	0 / 0 / 162
PDV	200	21 / 59	14 / 59	16 / 59	–
Mordred	1,613	226 / 331	222 / 336	222 / 331	197 / 304
GROVER	4,800	0 / 0	0 / 0	0 / 0	0 / 0
MHG-GNN	1,024	292 / 292	292 / 293	292 / 293	292 / 292
MoLFormer	768	–	0 / 0	0 / 0	0 / 0
SMI-TED	768	0 / 0	0 / 0	0 / 0	–
ChemBERT	256	–	0 / 0	–	–
Mol2vec	300	–	–	–	0 / 0

Table S.15: Constant and near-constant features of each representation on MDR1–MDCK efflux, HLM clearance, RLM clearance, and the Lhasa hERG blocker training set of 7,772 molecules. Entries read as in Table S.14.

Additional file 13: Settings of every model, as downstream model and as internal evaluator

One row is one model. The middle two columns give its settings when it scores a hybrid (downstream model) and when it scores one representation inside the search (internal evaluator). The models fitted only to compute feature importance are described in Chapter 5.

Table S.16: Settings of every model. p is the number of features. "Molecules" is the number of training molecules the internal evaluator is fitted on; above it, a random subset of that size is drawn with a fixed seed. Settings not listed are the library's defaults. Every setting is the same on every dataset. The Gaussian process (GP) runs only on the datasets whose row of Table 5.6 lists it.

Model	As downstream model	As internal evaluator	Molecules	Library; source
Random forest (RF)	100 trees, no depth limit; $\sqrt{p}$ features tried at each split	50 trees, depth at most 10; all p features tried at each split on regression, $\sqrt{p}$ on classification	all	scikit-learn [1]; Breiman [2]
LightGBM	100 trees	50 trees, depth at most 4	all	lightgbm; Ke et al. [3]

Model	As downstream model	As internal evaluator	Molecules	Library; source
XGBoost	100 trees; half of the features sampled for each tree	50 trees, depth at most 4	all	xgboost; Chen and Guestrin [4]
Support vector machine (SVM)	RBF kernel, $C = 1$. Not fitted above 10,000 molecules, or above 5,000 under uncapped augmentation	RBF kernel, $C = 1$	at most 500	scikit-learn; Vapnik [5]
GP	Regression: RBF plus white-noise kernel, labels normalized, 3 optimizer restarts. Classification: RBF kernel. Not fitted above 1,000 molecules	Regression: scaled RBF kernel, fixed noise variance 0.01, no restarts. Classification: scaled RBF kernel, no restarts	at most 500	scikit-learn; Rasmussen and Williams [6]
Deep neural network (DNN)	Hidden layers of 256, 128, and 64 units, ReLU, dropout 0.2 [7]; Adam [8] at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules that share no group with the rest, using the outer split's grouping; labels z-normalized	The same, except that the early-stopping 20% is drawn at random	all	PyTorch [9]

Additional file 14: Size of the search

One row is one downstream model. The counts are for one search over every value of Table 5.5, with adaptive pruning off. The searches in this research use each dataset's smaller grid, of 108 to 3,360 configurations (Table 5.6).

Table S.17: Number of configurations in one search over every value of the search settings. A configuration is one choice of downstream model, internal evaluator, allocation method, importance method, representation count, and budget. Each (downstream model, internal evaluator) pair runs 5 allocation methods × 3 importance methods × 4 representation counts × 8 budgets = 480 configurations (Table 5.5). "Matched" is counted once where it repeats an evaluator already in the list. Each configuration is scored on 5 folds, so five downstream models mean 40,800 fits of a downstream model.

Downstream model	Internal evaluators	Configurations
RF	3 (matched is RF)	1,440
LightGBM	3 (matched is LightGBM)	1,440
DNN	3 (matched is DNN)	1,440
XGBoost	4	1,920
SVM	4	1,920
GP	4	1,920
Total, first five models	17 pairs	8,160

Additional file 15: Design experiments behind the search settings

One row is one setting of the search, or one rule that was tested and then fixed instead of searched. Every number comes from preliminary design runs made before the main experiments; they are not results of the thesis, and several used other budgets than the final search.

Table S.18: Design experiments behind the search settings. Preliminary design runs, not the thesis's full runs.

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Downstream model	Nine models, among them RF, LightGBM, XGBoost, CatBoost, DNN, k-nearest neighbors (kNN), and a GP, each on each of 10 single representations at full width	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, LogD; 5 folds	Best held-out score over the 10 representations, and seconds per fit. The DNN scored best on 3 of the 4 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio)	Full: RF, LightGBM, XGBoost, SVM, DNN, and the GP on humanness / fast: the same five
Budget	None compared	–	–	10, 25, 50, 100, 250, 500, 1,000, and 2,000 features at both tiers
Representation count	2, 3, 5, and all 10	9 datasets, 3 tree-ensemble downstream models, budgets 500 and 1,000 features; 5 folds	The best count for each (downstream model, budget) combination	Full: 2, 3, 5, all / fast: 3, all
Allocation method	8 candidates: the five searched and 3 further pooled candidates, each at every representation count	The same 9 datasets, models, and budgets: 54 (dataset, model, budget) combinations	Each candidate was scored at its best representation count, and the number of combinations in which it was in the top 3 of the 8 was counted. Fixed shares: linear 30, squared 27, rank decay 26, softmax 25; global pool with squared weights 14	Full: the five described in Chapter 5 / fast: linear, squared, rank decay

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Importance method, agreement	8 methods	LogD, hERG K_i, Buchwald–Hartwig yield, nanobody thermostability, antibody FcRn retention; each method fitted once on the whole dataset	Spearman correlation between two methods' scores over every feature.	One method kept from each group whose rankings agree above 0.95
Importance method, score	28 methods, among them three controls: feature variance, Laplacian score, and random ranking	Caco-2 efflux; 3 folds; budget 100 features; RF and LightGBM downstream	RMSE of the best single representation minus RMSE of the hybrid ($\log_{10}$ efflux ratio; positive means the hybrid has the lower error): DeepLIFT +0.0144, LightGBM gain +0.0100, TreeSHAP +0.0067, random ranking −0.0118	Full: LightGBM gain, TreeSHAP, DeepLIFT / fast: LightGBM gain
Internal evaluator settings	20 hyperparameter settings from 5 model families	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression; 3 folds; budget 100 features	Spread of mean scores across all 20 settings: 0.0253 R^2, 0.0179 R^2, and 0.0085 Spearman. Median standard deviation across the 3 folds for one setting, allocation method, and downstream model: 0.0462 R^2, 0.0693 R^2, and 0.0186 Spearman	Matched, RF, LightGBM, and DNN, at fixed settings, both tiers
Rescaling of each representation's score (a rule)	5 rescalings	9 datasets; 5 folds; budgets 50 and 300 features	Whether any rescaling beat the one in use by more than that rule's own spread across folds: in 0 of 90 (dataset, allocation method, budget) combinations	One fixed rule per metric (Chapter 5)

The representation-count and allocation rows come from one design run on 10 datasets, 3 downstream models, and 2 budgets (60 combinations); QM9 is left out, which gives the 54.

Values of the search settings. Before the full runs, we ran a series of preliminary design experiments to determine the values each setting of the search may take. Each used a few datasets and 1 to 5 folds, and each dataset was scored in its own metric, with scores never averaged across datasets. When making a selection, we considered whether a value gave the best score for some dataset, downstream model, and budget, as well as the compute time in hours. A value that won somewhere but cost many hours stayed in the full search and was left out of the fast tier. Each setting's values were then either kept, because each won for some dataset, model, and budget, or fixed at one value. A setting was fixed when the spread between its values was smaller than the spread of one value across folds. As an example of a kept value, nine downstream models were fitted on each of 10 single representations over 5 folds. The datasets were Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, and LogD. The DNN scored best on 3 of the 4 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio), so it was added. As an example of a fixed rule, we tested five ways of rescaling each representation's score before the budget is split. On hERG K_i, with linear allocation at 300 features, the five rescalings gave RMSE values from 0.5870 to 0.5924 (pK_i). A single rescaling, however, ranged from 0.5475 to 0.6294 RMSE across the 5 folds, so rescaling is one fixed rule per metric. Every design experiment is listed in Table S.18.

Importance methods. Two comparisons were run, the first of how the methods rank features and the second of the hybrids those rankings build. The ranking comparison scored 8 importance methods on 5 datasets, namely LogD, hERG K_i, Buchwald–Hartwig yield, nanobody thermostability, and antibody FcRn retention. For each pair of methods, it measured the Spearman correlation between their scores over every feature of every representation. The three retained methods rank features differently. This comparison is preliminary, because each method was fitted once on the whole dataset rather than inside folds.

The hybrid comparison used Caco-2, with 2,158 molecules, 10 representations, 3 scaffold folds, and a budget of 100 features. Hybrids were built from each method's ranking and scored with RF and LightGBM downstream models. Three control rankings were added, namely feature variance, Laplacian score, and a random order. The measure was the RMSE of the best single representation minus the RMSE of the hy-

brid, in $\log_{10}$ efflux ratio units, so that a positive value means the hybrid has the lower error. The preliminary results were +0.0144 for DeepLIFT, +0.0100 for LightGBM gain, +0.0067 for TreeSHAP, and −0.0118 for the random ranking. The gap between DeepLIFT and LightGBM gain, 0.0044 RMSE, is smaller than the fold-to-fold spread. The methods also differ at the top of their rankings. Of each representation's 100 top-ranked features on Caco-2, the retained pairs of methods share a median of 28 to 54 features.

Three rules determined which methods were retained. First, a method must score every feature of every representation for regression, binary, and multiclass labels, which all three retained methods do. Second, of two methods whose rankings agree above a Spearman correlation of 0.95, we keep the faster one. Under this rule, DeepLIFT replaced integrated gradients, which agreed with it above 0.95 on 4 of 6 datasets, and gradient SHAP, which agreed above 0.95 on 3 of 6. Third, we keep methods that rank features differently, because a different ranking builds a different hybrid. TreeSHAP came last of the three on Caco-2 and was kept under this third rule. Only two filters act on features before allocation, the cut at a score of 0 and the pre-screen on representations wider than 2,000 features. On a 2,048-bit fingerprint, the pre-screen drops the 48 lowest-variance bits. On hERG K_i all 48 are among the 105 constant ECFP4 bits, while on QM9, which has no constant ECFP4 bits, they are near-constant ones. Near-constant features otherwise remain candidates and are ranked by importance.

No single importance method suits every dataset and downstream model. A preliminary run used 4 datasets (nanobody thermostability, hERG K_i, humanness, and LogD), 5 folds, 10 representations, 5 downstream models, and budgets of 25, 50, 100, 150, and 300 features. For each of the 25 pairs of downstream model and budget per dataset, we counted which importance method gave the best configuration, and Table S.19 gives the counts. LightGBM gain won the most pairs on two datasets and tied with TreeSHAP on humanness. DeepLIFT won the most on one dataset, LogD, with 17 of its 25 pairs, and outside LogD it gave the best configuration for 1 of 75 pairs. On LogD, the DNN took DeepLIFT at 3 of 5 budgets. On LogD, XGBoost and LightGBM each took DeepLIFT at 5 of 5 budgets, and RF took LightGBM gain at 5 of 5.

Allocation methods. We first tried greedy forward selection, which picks features for how well they work together, something fixed shares cannot do [10, 11]. In each round, it added the next block of features from whichever representation most improved the internal validation score. It stopped when the budget was spent or when the

Dataset	Metric	Wins (of 25)			Best score		
		LightGBM gain	TreeSHAP	DeepLIFT	LightGBM gain	TreeSHAP	DeepLIFT
Nanobody thermostability	Spearman	15	10	0	0.6696	0.6471	0.6440
hERG K_i	RMSE (pK_i)	14	11	0	0.5589	0.5605	0.5728
Humanness	AUROC	12	12	1	0.9723	0.9713	0.9689
LogD	RMSE (log units)	8	0	17	0.4250	0.4345	0.4170

Table S.19: Importance method of the best configuration in a preliminary run on four datasets, with 5 folds, 10 representations, 5 downstream models, and budgets of 25, 50, 100, 150, and 300 features. Wins count the pairs of downstream model and budget, out of 25 per dataset, at which a method gave the best configuration, and Best score is the best score reached with that method.

score had not improved for a set number of rounds. With blocks of s features and K representations, it needs up to $(B/s) \times K$ fits per configuration and fold. With blocks of 10 features, this amounts to 500 fits at 500 features and 10 representations, against 10 for the score-based methods. In preliminary runs, it reached a Spearman correlation of 0.3600 on Jain antibody FcRn retention, against 0.4526 for linear allocation, and placed 16th of 17 methods. It was removed from the search.

The five methods were chosen in a preliminary comparison run before the main experiments. It compared 8 candidate allocation methods, each at 4 representation counts, on 9 regression datasets. It used 3 downstream models (LightGBM, RF, and CatBoost) and budgets of 500 and 1,000 features, giving 54 combinations of dataset, downstream model, and budget. A fixed-share method was the single best method in 38 of the 54 combinations and the global pool in 16. All five were kept because no single method was best in most combinations. This comparison predates the current score rescaling, and it sized the weights with an inner 3-fold group split instead of the 80/20 internal split used now.

Matched internal evaluator. Matching the evaluator to the downstream model is the obvious default. However, the evaluator only has to rank representations, and another model may rank them better for the downstream model. Because the matched evaluator is one of the four searched values, the search already covers this default, and a mismatched evaluator winning is informative rather than a negative result. We first counted wins in the search itself, in a preliminary run on 4 datasets (nanobody thermostability, hERG K_i, humanness, and LogD). These runs used 10 representations, budgets of 25 to 300 features, and 5 folds, and their scores are the search's own fold scores rather than held-out scores. On nanobody thermostability with a DNN

downstream model, the matched evaluator reached a Spearman correlation of 0.6696 against 0.6506 for the LightGBM evaluator, a gap of 0.0190. Its 5 fold scores ranged from 0.5095 to 0.7920. On LogD with an SVM downstream model, the matched evaluator scored 0.4322 RMSE (log units) against 0.4217 for the LightGBM evaluator, where lower is better. This is a gap of 0.0105, the matched evaluator lost at all 5 budgets, and its 5 fold scores ranged from 0.4118 to 0.4634. We then scored the evaluators on held-out folds in a second preliminary run, with 3 outer folds, a budget of 250 features, 5 representations, and LightGBM gain only. The downstream models were RF and SVM. On nanobody thermostability with an RF downstream model, the matched evaluator reached a Spearman correlation of 0.6098, the LightGBM evaluator 0.6220, and the SVM evaluator 0.6172. The matched evaluator's fold scores ranged from 0.5659 to 0.6734. With an SVM downstream model, the matched evaluator reached 0.6189, the GP evaluator 0.6265, and the LightGBM evaluator 0.6317, with fold scores for the matched evaluator from 0.5530 to 0.6703. On hERG K_i with an RF downstream model, the matched evaluator scored 0.6046 RMSE (pK_i), the SVM evaluator 0.6059, and the LightGBM evaluator 0.6075. Its fold scores ranged from 0.5957 to 0.6178, and it beat the SVM evaluator on 2 of 3 folds. With an SVM downstream model, the matched evaluator scored 0.5723 RMSE (pK_i), the LightGBM evaluator 0.5724, and the GP evaluator 0.5765, with fold scores for the matched evaluator from 0.5605 to 0.5862. Every gap above, both held out and within the search, is smaller than the range of the matched evaluator's own fold scores.

Fast tier. The fast lists were chosen by one rule in a preliminary comparison. A value stays in the fast lists if it is inexpensive and improves the best score, whereas an expensive value that wins often stays in the full search only. Importance methods and downstream models were measured on four datasets, with 5 folds each, against the unpruned full search. The four were nanobody thermostability, humanness, hERG K_i, and LogD. DeepLIFT took about half of the fast run's hours on all four datasets, and dropping it changed only LogD's best score, from 0.4905 to 0.4930 RMSE (log units). Adding the DNN as a downstream model moved LogD from 0.4930 to 0.4273 RMSE, against a full-search best of 0.4170. Adding the SVM moved hERG K_i from 0.5776 to 0.5599 RMSE (pK_i), while adding XGBoost changed no dataset's best score. All five downstream models are now searched at both tiers. The global pool gave the best configuration for some combinations of dataset, downstream model, and budget on those four datasets. It is expensive, however, so it is in the full search only. The three fast allocation methods are linear, squared, and rank-decay allocation.

Additional file 16: Worked example of fold counts

When a training fold holds fewer than five groups, both splits use as many folds as there are groups. For example, Koenig expression has five groups, so each outer training fold holds four. The search inside that fold uses 4 folds of three groups each, and the internal split holds out one of the three.

Additional file 17: Fields of a blueprint and their types

```
Blueprint
  # What the search chose. A rebuild reads all of these.
  reps                  : list[str]           # every representation searched, not only those used
  rep_count             : int or None         # how many of them may contribute; None = all
  budget                : int                 # most features the hybrid may have
  allocation_method     : str
  importance_method     : str
  internal_evaluator    : str
  objective             : str                 # 'predictive' or 'similarity'

  # How the search scored configurations.
  metric                : str
  lower_is_better       : bool
  task_type             : str                 # 'regression' or 'classification'
  n_folds               : int
  seed                  : int

  # Augmentation settings. A rebuild reads all of these.
  augmentation_strategy : str
  aug_per_source_budget : int, or {source: int}, or None
  augmentation_margin   : float
  augmentation_fraction : float or None
  candidate_ranking     : str
  augmentation_threshold: float or None
  augmentation_budget   : int
  aug_max_candidates    : int

  # Stored for inspection.
  rep_scores            : {representation: float}
                          # each representation's best score over every molecule the search saw;
                          # the rebuild uses it to choose which representations to keep only when
                          # rep_count is set and no per-fold scores are given
  allocation_per_fold   : {fold: {representation: int}}      # features each representation got
  kept_augmentations    : {fold: {source: list[int]}}         # extra features kept, by index
```

Listing S.1: Fields of a blueprint and their types. Comment lines group the fields, and each field is given with its type and, where needed, a comment on its meaning.

Additional file 18: How augmentation ranks and adds donor features

Donors are set per dataset (Augmentation column of Table 5.12), and on QM9 the topological torsion fingerprint (2,048 features) is the only wide donor. Bahia et al. [12] reported that combining 2D and 3D descriptors gave many more significant models across six protein–ligand series (461 structures), which they attribute to complementary properties. Their 3D descriptors came from bioactive conformations in the Protein Data Bank, whereas our 3D donors use one generated conformer per molecule.

Donor features are ranked by what the base gets wrong. Inside each outer training fold, an 80/20 internal split keeps each scaffold family or sequence group on one side. A small RF (50 trees, depth at most 10) is fitted on the base features. Three-fold group-aware CV gives out-of-fold residuals $r_i = y_i - \hat{y}_i^{\text{oof}}$ for the internal-training samples, since in-sample residuals would shrink toward zero. For binary tasks, $\hat{y}$ is the positive-class probability. All donor features are pooled, each is scored by its absolute Pearson correlation with the residuals, $c_j = |\rho_{\text{P}}(\mathbf{x}_j, \mathbf{r})|$, and the top $M = 50$ go forward. Ranking against $\mathbf{y}$ would favor features that repeat what the base captures. Ranking against $\mathbf{r}$ favors features that explain its errors, as in gradient boosting [13] and least angle regression [14].

Blocks of the top $k \in \{0, 5, 10, 20, 40\}$ ranked features, capped at the number of candidates, are each added to the base. Each is scored with the downstream model on the internal-validation samples. The chosen k^* has the best score, and $k = 0$ wins ties, which makes keeping no extras the default. Five block choices instead of about 50 accept-or-reject decisions leave less room to overfit a small validation set [15]. The chosen donors and feature positions are stored and re-extracted for the outer fold.

In uncapped mode, extras are appended outside the budget and the final width is $|\mathbf{H}|+k^*$, at most $B+40$ features. In capped mode they replace the base features with the lowest RF importance, and thus the width stays at most B. Uncapped augmentation and no augmentation differ in width at the same B, and a comparison between them mixes better features with more features. Capped mode removes this difference, since each extra replaces a base feature.

Augmentation is not a searched setting: a dataset with donors is searched three times, with no, uncapped, and capped augmentation (Table S.20). An earlier search of each whole dataset covered 16 datasets of an earlier list. On none did either augmented condition beat no augmentation by more than 1% (3% on datasets under about 1,000 samples). We therefore ran the searches of Chapter 5 without augmentation on all 20 datasets (Section 5.2.3). Only the nearest-neighbor search on SMAC colloidal stability and HIC retention keeps uncapped augmentation (Chapter 6).

The residual ranking has four limitations. $|\rho_{\mathrm{P}}|$ captures only linear relations, one feature at a time. It ignores redundancy: two near-identical donor features can both rank high. Donors are pooled with no per-donor limit, which gives a 2,048-feature donor more chances than a 49-feature one. Finally, with a zero margin over $k = 0$, a near-tie on one internal split can admit extras.

Condition	Final width (features)	What an extra feature must beat	Datasets that run it
No augmentation	at most B	nothing is offered	all 20
Uncapped	$\lvert\mathbf{H}\rvert + k^*$, at most $B + 40$	keeping no extras	SMAC colloidal stability and HIC retention, nearest-neighbor search only
Capped	at most B	keeping no extras, and the weakest base feature it replaces	none (cut after the earlier search)

Table S.20: The three augmentation conditions. B is the budget and $|\mathbf{H}|$ the width of the base hybrid, both in features, and k^* is the chosen number of extra features, which are drawn from a small or specialized feature set offered to the hybrid.

Additional file 19: What each cut removes from the hERG K_i search

Table S.21 follows one dataset, hERG K_i, through the cuts; its earlier search, without augmentation, covered six downstream models including kNN. The best had an RMSE of 0.547 pK_i (SVM, 1,000 features, LightGBM gain, softmax allocation, RF evaluator, all representations), with an SD of 0.044 pK_i over the five folds. Most of the fitting time removed by keeping only the SVM and the DNN came from XGBoost (35%) and LightGBM (25%). For each later cut, the best configuration it removed was behind the best kept by less than the spread over folds. The final grid still contains the best configuration, which is expected, because the cuts were chosen on this same search.

Step	Configurations kept	Fitting time kept (%)	Best kept	Best removed
Full search	8,220	100.0	0.547	–
Downstream models: SVM and DNN only	3,360	28.7	0.547	0.571
Budgets: 250 to 2,000 features	1,680	20.0	0.547	0.563
Importance: DeepLIFT removed	1,120	16.3	0.547	0.552
Allocation: global pool removed	896	4.6	0.547	0.554
Representation count: all only	224	1.4	0.547	0.5545

Table S.21: The hERG K_i search after each cut, applied in turn. One row is one cut. Fitting time is the share of the full search's recorded fitting time that remains. Best kept and best removed are the lowest mean RMSE over five folds, in pK_i (lower is better), among the configurations kept and among those the cut removed.

Additional file 20: Default and tuned model settings on single representations

We tested whether tuning helps on one outer fold of QM9, hERG K_i, Caco-2 efflux, LogD, and Buchwald–Hartwig yield. Each downstream model was given its two best single representations, chosen on that fold's training molecules. The tuner tried the default and 20 draws from grids that reach stronger regularization than the optional search. In this test the DNN was tuned as well, over dropout, learning rate, and layer sizes. The tuner scored each setting on 3 inner folds grouped by scaffold family of the training molecules, which kept the held-out fold out of the tuning. Table S.22 gives the held-out RMSE at both settings. The SVM gained most, on LogD with MHG-GNN. Tuning made the RF worse on Buchwald–Hartwig yield. At default settings, two repeat fits of the DNN on QM9 with Mol2vec scored 0.487 and 0.495. The DNN rows for Caco-2 and hERG K_i were computed before DNN predictions were capped at the range of the training labels. With one fold, these pairs show direction only.

Table S.22: Held-out RMSE of single representations at default and tuned model settings, outer fold 0 only. One row is one dataset, model, and representation. Lower is better. RMSE is in eV for QM9, pK_i units for hERG K_i, $\log_{10}$ efflux ratio for Caco-2, and percentage points of yield for Buchwald–Hartwig; LogD has no unit.

Dataset	Model	Representation	Default	Tuned
QM9	DNN	Mordred	0.400	0.389
QM9	DNN	Mol2vec	0.4867	0.4867
QM9	LightGBM	Mordred	0.412	0.403
QM9	LightGBM	Avalon	0.459	0.455
hERG K_i	SVM	MHG-GNN	0.640	0.618
hERG K_i	SVM	ECFP4	0.675	0.640
hERG K_i	DNN	MHG-GNN	0.638	0.628
hERG K_i	DNN	GROVER	0.5891	0.5891
Caco-2 efflux	DNN	MHG-GNN	0.276	0.282
Caco-2 efflux	DNN	GROVER	0.266	0.261
Caco-2 efflux	SVM	PDV	0.276	0.267
Caco-2 efflux	SVM	MoLFormer	0.287	0.282
LogD	DNN	MHG-GNN	0.431	0.428
LogD	DNN	Mordred	0.417	0.409
LogD	SVM	MHG-GNN	0.475	0.421
LogD	SVM	MoLFormer	0.495	0.474
Buchwald–Hartwig	RF	Avalon (difference)	18.9	19.9
Buchwald–Hartwig	RF	CGR fingerprint	18.0	20.1
Buchwald–Hartwig	LightGBM	CGR fingerprint	19.5	19.3
Buchwald–Hartwig	LightGBM	Avalon (difference)	20.22	20.21

Additional file 21: Representation best in hindsight, and how many representations the hybrid draws on

One row is one dataset and downstream model. Table S.23 covers small molecules and reactions, and Table S.24 covers antibodies.

Dataset	Model	Best in hindsight	Its score	Hybrid	Effective number	Top share
RMSE ↓						
QM9	DNN	Mordred	0.362	0.316	3.86	0.47
	LightGBM	Mordred	0.372	0.346	1.27	**0.88**
HK_i	SVM	MHG-GNN	0.588	0.559	**7.17**	0.19
	DNN	MHG-GNN	0.593	0.569	6.74	0.20
Caco-2	DNN	PDV	**0.298**	**0.277**	5.59	0.26
	SVM	PDV	0.300	0.286	6.08	0.22
LogD	DNN	MHG-GNN	0.447	0.409	6.35	0.26
	SVM	MHG-GNN	0.493	0.430	6.46	0.22
RLM	SVM	PDV	0.521	0.482	6.10	0.25
	DNN	PDV	0.533	0.497	5.96	0.25
MDR1	SVM	PDV	0.468	0.437	5.49	0.29
	DNN	PDV	0.473	0.444	5.79	0.26
BH	RF	Difference-Avalon	14.2	15.2	1.56	0.75
	LightGBM	CGR-Morgan	15.2	15.6	1.78	0.64
AUROC ↑						
HB	LightGBM	MHG-GNN	0.860	0.867	2.28	0.56
	XGBoost	Mordred	0.855	0.864	2.77	0.49

Table S.23: Hybrid against the single representation best on the held-out molecules: small molecules and reactions. One row is one dataset and downstream model. Scores are held-out means over five outer folds in the dataset's own metric. The last two columns come from shuffling each representation's features in the hybrid on the held-out molecules: the effective number of representations behind its score, and the share of the score drop carried by the representation that carries most. Dataset short names as in Table 5.8.

Dataset	Model	Best in hindsight	Its score	Hybrid	Share built here	Effective number
Spearman ↑						
VEGF-E	SVM	ESM3-CDR	0.725	**0.773**		
	DNN	ESM3-CDR	0.724	0.735		
VEGF-B	RF	SaProt	0.445	0.511	0.10	4.28
	SVM	ProtT5	0.490	0.521	0.07	5.34
Nb-Tm	DNN	SaProt	0.694	0.676	0.13	2.81
	SVM	SaProt	0.6901	0.6899	0.17	3.01
EGFR	SVM	AbRoBERTa-CDR	0.357	0.314	0.21	3.53
	DNN	AbRoBERTa	0.290	0.216	0.17	2.79
Ang2	GP	SaProt	0.601	0.505	0.11	2.28
	SVM	AntiBERTy	0.577	0.522	0.09	3.77
$\Delta\Delta G$	SVM	Mutation-Features	0.203	0.218	**0.83**	2.30
	DNN	ESM2-WT-Delta	0.222	0.354	**0.81**	1.87
SMAC	DNN	ESM2-Early	0.313	0.268	0.00	2.62
	SVM	ProtT5	0.394	0.348	0.00	1.96
HIC-R	GP	ESM2-Concat	0.600	0.551	0.01	3.14
	DNN	ESM2-Concat	0.531	0.529	0.04	4.32
AC-SINS	LightGBM	IgT5	0.472	0.524	0.02	2.45
	GP	AbLang2	0.542	0.555	0.02	3.25
Titer	DNN	ProtT5	0.372	0.367	0.08	3.24
	RF	IgT5	0.362	0.271	0.10	1.73
PR-CHO	GP	ProtT5	0.510	0.442	0.14	3.80
	DNN	ESMC	0.431	0.413	0.09	3.60
AUROC ↑						
Hum	XGBoost	IgT5	0.963	0.955	0.05	1.41
	RF	AntiBERTy-CDR	0.967	0.961	0.15	1.95

Table S.24: Hybrid against the single representation best on the held-out antibodies. One row is one dataset and downstream model. Scores are held-out means over five outer folds, in the dataset's own metric. The last two columns come from shuffling each representation's features in the hybrid on the held-out antibodies: the share of the score drop carried by representations built for this thesis (Table 5.11), and the effective number of representations behind the score. Dataset short names as in Table 5.8.

Additional file 22: Search's hybrid against the top representation alone and an even split, at every budget

Table S.25: Search's hybrid against the budget filled from the top-ranked representation alone and against an even split over the same representations. One row is one dataset, downstream model, and budget (number of features). Held-out means over five outer folds, in the dataset's own metric. Dataset short names as in Table 5.8.

Dataset	Model	Budget	Top representation	Search's hybrid	Even split
RMSE ↓					
QM9	DNN	500	0.357	**0.314**	**0.328**
		1,000	0.364	0.315	0.326
		2,000	0.361	0.315	0.323
	LightGBM	500	0.372	0.347	0.346
		1,000	0.374	0.347	0.346
		2,000	0.372	0.3455	0.3447
HK_i	SVM	250	0.599	0.564	0.562
		500	0.601	0.559	0.557
		1,000	0.603	0.555	0.557
		2,000	0.609	0.562	0.559
	DNN	250	0.574	0.569	0.571
		500	0.582	0.572	0.564
		1,000	0.591	0.580	0.570
		2,000	0.589	0.574	0.564
Caco-2	DNN	500	0.307	0.2827	0.2829
		1,000	0.303	0.279	0.280
		2,000	0.303	0.275	0.273
	SVM	500	0.302	0.285	0.283
		1,000	0.304	0.286	0.285
		2,000	0.310	0.2854	0.2847
LogD	DNN	500	0.455	0.4165	0.4175
		1,000	0.460	0.410	0.411
		2,000	0.461	0.409	0.406
	SVM	500	0.499	0.438	0.440
		1,000	0.503	0.430	0.432
		2,000	0.504	0.4313	0.4312
RLM	SVM	500	0.547	0.4876	0.4881
		1,000	0.553	0.48346	0.48347
		2,000	0.561	0.481	0.480
	DNN	500	0.560	0.506	0.511
		1,000	0.557	0.500	0.504
		2,000	0.553	0.4956	0.4964
MDR1	SVM	500	0.474	0.438	0.441
		1,000	0.488	0.436	0.435
		2,000	0.488	0.436	0.437
	DNN	500	0.495	0.452	0.460
		1,000	0.496	0.440	0.443
		2,000	0.498	0.441	0.438

Dataset	Model	Budget	Top representation	Search's hybrid	Even split
BH	RF	500	15.5	15.4	14.9
		1,000	16.3	14.94	14.85
		2,000	15.6	16.0	16.1
	LightGBM	500	16.2	15.5	16.1
		1,000	15.9	15.70	15.67
		2,000	16.5	16.3	16.2
AUROC ↑					
HB	LightGBM	100	0.849	0.858	0.862
		250	0.854	0.865	0.863
		500	0.856	0.866	0.868
		1,000	0.856	0.866	0.865
		2,000	0.859	0.867	0.866
	XGBoost	100	0.848	0.863	0.856
		250	0.850	0.859	0.864
		500	0.854	0.864	0.863
		1,000	0.853	0.866	0.864
		2,000	0.855	0.859	0.860
Hum	XGBoost	10	0.894	0.941	0.936
		25	0.919	0.952	0.948
		50	0.929	0.949	0.948
		100	0.942	0.95623	0.95621
		250	0.930	0.956	0.962
		500	0.935	0.959	0.956
		1,000	0.938	0.961	0.960
		2,000	0.940	0.961	0.958
	RF	10	0.932	0.952	0.944
		25	0.943	0.960	0.954
		50	0.954	0.96235	0.96239
		100	0.958	0.961	0.968
		250	0.923	0.964	0.968
		500	0.900	0.966	0.963
		1,000	0.913	0.963	0.964
		2,000	0.956	0.961	0.964
Spearman ↑					
VEGF-E	SVM	250	0.661	0.762	0.763
		500	0.664	0.772	0.767
		1,000	0.670	0.776	0.778
		2,000	0.671	0.773	0.779
	DNN	250	0.658	0.732	0.735
		500	0.683	0.744	0.745
		1,000	0.660	0.742	0.739
		2,000	0.630	0.735	0.743
VEGF-B	RF	250	0.365	0.5119	0.5121
		500	0.389	0.517	0.514
		1,000	0.382	0.509	0.511
	SVM	250	0.371	0.516	0.504
		500	0.386	0.514	0.500
		1,000	0.407	0.526	0.506

Dataset	Model	Budget	Top representation	Search's hybrid	Even split
Nb-Tm	DNN	50	0.553	**0.624**	**0.572**
		100	0.536	0.638	0.632
		250	0.568	0.698	0.679
		500	0.588	0.659	0.664
		1,000	0.597	0.669	0.645
		2,000	0.615	0.662	0.650
	SVM	50	0.534	0.643	0.622
		100	0.511	0.618	0.652
		250	0.539	0.687	0.674
		500	0.541	0.6809	0.6812
		1,000	0.559	0.682	0.661
		2,000	0.549	0.649	0.665
EGFR	SVM	250	0.254	0.321	0.357
		500	0.250	0.338	0.347
		1,000	0.260	0.311	0.337
		2,000	0.265	0.310	0.325
	DNN	250	0.222	0.247	0.251
		500	0.145	0.198	0.234
		1,000	0.140	0.209	0.263
		2,000	0.145	0.113	0.237
Ang2	GP	25	0.348	0.506	0.490
		50	0.344	0.450	0.507
		100	0.374	0.463	0.522
	SVM	25	0.397	0.549	0.477
		50	0.396	0.452	0.503
		100	0.408	0.479	0.535
$\Delta\Delta G$	SVM	100	0.125	0.240	0.358
		250	0.084	0.318	0.347
		500	0.117	0.326	0.373
	DNN	100	0.026	0.342	0.267
		250	0.127	0.2674	0.2670
		500	0.132	0.339	0.343
SMAC	DNN	500	0.259	0.308	0.255
		1,000	0.188	0.216	0.231
		2,000	0.219	0.299	0.295
	SVM	500	0.282	0.346	0.296
		1,000	0.344	0.349	0.321
		2,000	0.324	0.367	0.374
HIC-R	GP	250	0.431	0.552	0.499
		500	0.485	0.550	0.537
		1,000	0.428	0.542	0.522
		2,000	0.394	0.568	0.566
	DNN	250	0.421	0.447	0.446
		500	0.429	0.504	0.496
		1,000	0.429	0.519	0.501
		2,000	0.382	0.501	0.510
AC-SINS	LightGBM	50	0.323	0.521	0.516
		100	0.321	0.481	0.530

Dataset	Model	Budget	Top representation	Search's hybrid	Even split
		250	0.294	0.502	0.547
		500	0.328	0.523	0.518
		1,000	0.296	0.557	0.503
		2,000	0.353	0.572	0.497
	GP	50	0.336	0.521	0.491
		100	0.416	0.542	0.515
		250	0.392	0.552	0.540
		500	0.416	0.553	0.529
		1,000	0.400	0.543	0.550
		2,000	0.373	0.557	0.552
Titer	DNN	250	0.230	0.296	0.292
		500	0.266	0.315	0.346
		1,000	0.298	0.349	0.341
		2,000	0.292	0.333	0.322
	RF	250	0.247	0.292	0.381
		500	0.276	0.381	0.280
		1,000	0.293	0.291	0.298
		2,000	0.260	0.288	0.317
PR-CHO	GP	250	0.294	0.471	0.483
		500	0.302	0.483	0.456
		1,000	0.323	0.480	0.487
		2,000	0.353	0.458	0.495
	DNN	250	0.304	0.406	0.448
		500	0.319	0.410	0.401
		1,000	0.297	0.404	0.410
		2,000	0.359	0.439	0.425

Additional file 23: Hybrid against PCA: three experiments that disagree

Chapter 5 compares the hybrid with principal component analysis (PCA) of all representations joined, at the budget each fold's search picked. The ensemble experiment repeats this comparison on the same folds and agrees on LogD and Caco-2. A third experiment disagrees on both, with the DNN hybrid ahead on every fold, 0.409 against 0.477 RMSE on LogD at 1,257 features. That experiment sets the width as a fixed share of the dataset's molecules, rescales each principal component before the model, and does not cap held-out features at the training range. The two experiments therefore differ in three ways at once, and we cannot say which causes the disagreement. All their DNN results for LogD and Caco-2 predate the DNN correction.

Additional file 24: Hybrid against other feature selections at the same number of features

Table S.26: Hybrid against recursive feature elimination, randomly chosen features, and mRMR, each over all single representations joined, at the same number of features (budget). One row is one dataset, downstream model, and budget. Held-out means over five outer folds, in the dataset's own metric. mRMR ran only at budgets up to 100 features. Dataset short names as in Table 5.8.

Dataset	Model	Budget	Hybrid	Recursive elimination	Random	mRMR
RMSE ↓						
QM9	DNN	500	0.315	0.340	0.375	
		1,000	0.315	0.338	0.347	
		2,000	0.315	0.346	0.331	
	LightGBM	500	0.347	0.372	0.485	
		1,000	0.347	0.370	0.420	
		2,000	0.345	0.370	0.405	
HK_i	SVM	250	0.564	0.577	0.600	
		500	0.562	0.574	0.591	
		1,000	0.558	0.568	0.580	
		2,000	0.565	0.572	0.581	
	DNN	250	0.571	0.589	0.587	
		500	0.575	0.587	0.566	
		1,000	**0.611**	**0.579**	**0.570**	
		2,000	0.620	0.596	0.581	
Caco-2	DNN	500	0.285	0.287	0.288	
		1,000	0.276	0.290	0.287	
		2,000	0.278	0.282	0.283	
	SVM	500	0.285	0.289	0.300	
		1,000	0.286	0.290	0.299	
		2,000	0.285	0.291	0.299	
LogD	DNN	500	0.423	0.430	0.458	
		1,000	0.410	0.415	0.431	
		2,000	0.410	0.411	0.416	
	SVM	500	0.439	0.458	0.495	
		1,000	0.430	0.445	0.475	
		2,000	0.432	0.444	0.458	
MDR1	SVM	500	0.438	0.446	0.469	
		1,000	0.435	0.447	0.459	
		2,000	0.437	0.452	0.455	
	DNN	500	0.450	0.460	0.473	
		1,000	0.441	0.454	0.462	
		2,000	0.438	0.448	0.458	
BH	RF	500	15.3	15.1	17.0	
		1,000	15.0	15.2	16.4	
		2,000	16.0	15.4	16.1	
	LightGBM	500	15.5	15.4	16.4	
		1,000	15.7	15.3	15.6	

Dataset	Model	Budget	Hybrid	Recursive elimination	Random	mRMR
		2,000	16.3	15.3	14.9	
Spearman ↑						
VEGF-E	SVM	250	0.761	0.753	0.737	
		500	0.771	0.768	0.756	
		1,000	0.7743	0.7742	0.763	
		2,000	0.772	0.773	0.767	
	DNN	250	0.732	0.726	0.698	
		500	0.743	0.736	0.728	
		1,000	0.739	0.750	0.726	
		2,000	0.739	0.761	0.741	
Nb-Tm	DNN	50	0.620	0.649	0.459	0.595
		100	0.628	0.632	0.600	0.631
		250	0.699	0.676	0.597	
		500	0.651	0.669	0.641	
		1,000	0.671	0.669	0.618	
		2,000	0.663	0.684	0.655	
	SVM	50	0.645	0.652	0.478	0.615
		100	0.610	0.653	0.614	0.629
		250	0.686	0.681	0.623	
		500	0.6787	0.6791	0.639	
		1,000	0.679	0.683	0.636	
		2,000	0.647	0.699	0.648	
EGFR	SVM	250	0.325	0.295	0.329	
		500	0.341	0.287	0.335	
		1,000	0.313	0.302	0.361	
		2,000	0.312	0.322	0.362	
	DNN	250	0.250	0.229	0.177	
		500	0.190	0.180	0.204	
		1,000	0.208	0.245	0.151	
		2,000	0.106	0.186	0.151	
Ang2	GP	25	0.503	0.493	0.345	0.526
		50	0.443	0.524	0.486	0.513
		100	0.456	0.530	0.408	0.523
	SVM	25	0.567	0.481	0.385	0.570
		50	0.450	0.516	0.517	0.603
		100	0.467	0.581	0.501	0.565
$\Delta\Delta G$	SVM	100	0.239	0.360	0.027	0.079
		250	0.314	0.376	0.040	
		500	0.325	0.334	0.218	
	DNN	100	0.329	0.289	-0.029	0.043
		250	0.276	0.295	0.036	
		500	0.349	0.289	0.198	
SMAC	DNN	500	0.309	0.301	0.172	
		1,000	0.214	0.285	0.216	
		2,000	0.297	0.231	0.254	
	SVM	500	0.340	0.345	0.363	
		1,000	0.342	0.365	0.358	
		2,000	0.359	0.377	0.370	

Dataset	Model	Budget	Hybrid	Recursive elimination	Random	mRMR
AC-SINS	LightGBM	50	0.521	0.484	0.376	0.448
		100	0.481	0.490	0.473	0.453
		250	0.502	0.508	0.507	
		500	0.523	0.470	0.511	
		1,000	0.557	0.500	0.457	
		2,000	0.572	0.533	0.447	
	GP	50	0.517	0.524	0.386	0.460
		100	0.535	0.484	0.437	0.543
		250	0.556	0.502	0.461	
		500	0.553	0.518	0.507	
		1,000	0.546	0.538	0.496	
		2,000	0.554	0.538	0.492	
Titer	DNN	250	0.291	0.385	0.187	
		500	0.321	0.356	0.318	
		1,000	0.338	0.298	0.293	
		2,000	0.333	0.344	0.272	
	RF	250	0.294	0.305	0.333	
		500	0.380	0.3185	0.3188	
		1,000	0.291	0.284	0.282	
		2,000	0.308	0.297	0.294	
PR-CHO	GP	250	0.468	0.482	0.394	
		500	0.482	0.497	0.423	
		1,000	0.482	0.518	0.435	
		2,000	0.460	0.536	0.447	
	DNN	250	0.419	0.458	0.252	
		500	0.437	0.451	0.402	
		1,000	0.414	0.419	0.372	
		2,000	0.420	0.461	0.371	

Additional file 25: Budget and number of representations each fold picked

		Budget					Representations				
Dataset	Model	1	2	3	4	5	1	2	3	4	5
Small molecules and reactions											
QM9	DNN	2,000	1,000	1,000	1,000	1,000	10	10	3	10	10
	LightGBM	1,000	2,000	1,000	500	1,000	3	3	3	3	3
HK_i	SVM	1,000	250	250	500	500	10	10	10	10	10
	DNN	2,000	250	2,000	250	2,000	10	10	10	9	10
Caco-2	DNN	2,000	2,000	2,000	1,000	1,000	10	10	10	10	10
	SVM	500	500	500	500	1,000	10	10	10	10	10
LogD	DNN	2,000	2,000	2,000	2,000	2,000	10	10	10	5	10
	SVM	1,000	1,000	1,000	1,000	2,000	10	10	10	10	10
RLM	SVM	2,000	2,000	2,000	2,000	1,000	10	10	10	10	10
	DNN	2,000	2,000	2,000	1,000	2,000	10	10	10	10	10
MDR1	SVM	2,000	1,000	1,000	2,000	500	10	10	10	10	10
	DNN	2,000	2,000	2,000	1,000	2,000	10	10	10	10	10
BH	RF	500	1,000	500	500	1,000	2	2	2	2	2
	LightGBM	1,000	1,000	500	1,000	2,000	2	2	2	2	2
HB	LightGBM	2,000	2,000	2,000	1,000	2,000	3	3	10	3	3
	XGBoost	1,000	500	2,000	2,000	1,000	10	3	10	3	3
Antibodies											
VEGF-E	SVM	2,000	2,000	2,000	2,000	2,000	10	9	10	9	10
	DNN	2,000	2,000	2,000	2,000	2,000	10	10	10	10	10
Nb-Tm	DNN	1,000	1,000	1,000	1,000	250	10	3	5	10	5
	SVM	250	250	250	500	500	10	2	9	10	5
EGFR	SVM	2,000	2,000	2,000	2,000	250	3	10	10	5	10
	DNN	2,000	500	1,000	2,000	250	3	10	9	10	10
Ang2	GP	100	100	100	100	25	3	10	5	3	10
	SVM	100	100	100	100	25	10	10	3	5	10
$\Delta\Delta G$	SVM	100	500	100	100	100	3	5	8	8	8
	DNN	250	100	100	100	250	3	8	3	3	8
SMAC	DNN	500	500	2,000	2,000	2,000	10	10	4	5	10
	SVM	500	1,000	500	2,000	2,000	5	5	3	7	3
HIC-R	GP	250	500	500	250	500	10	5	5	9	5
	DNN	1,000	500	1,000	500	1,000	10	10	10	10	10
AC-SINS	LightGBM	2,000	2,000	100	500	500	10	5	10	10	5
	GP	250	250	1,000	500	2,000	2	10	5	10	10
Titer	DNN	500	1,000	500	2,000	500	5	10	2	10	2
	RF	250	2,000	250	1,000	2,000	2	10	2	10	5
PR-CHO	GP	2,000	2,000	500	2,000	1,000	10	5	10	10	5
	DNN	2,000	250	500	2,000	500	5	10	10	10	10
Hum	XGBoost	250	1,000	500	50	500	2	2	5	4	3
	RF	250	100	1,000	1,000	250	3	3	10	10	2

Table S.27: Budget (number of features) and number of single representations with at least one feature in the hybrid that each outer fold's search picked on its training samples. One row is one dataset and downstream model; columns 1 to 5 are the outer folds. Dataset short names as in Table 5.8.

Additional file 26: Budget picked at each setting of the SVM and the DNN

Dataset	Model	Setting	1	2	3	4	5
HK_i	SVM	$C = 0.01$	250	250	250	250	250
		$C = 0.1$	250	250	250	250	250
		$C = 1$	1,000	250	250	500	500
		$C = 10$	2,000	500	500	250	250
Caco-2	SVM	$C = 0.01$	500	500	500	500	500
		$C = 0.1$	500	500	500	500	500
		$C = 1$	500	500	500	500	1,000
		$C = 10$	1,000	1,000	500	1,000	1,000
LogD	DNN	16 units	2,000	2,000	2,000	2,000	2,000
		64 units	1,000	2,000	2,000	2,000	2,000
		128–64 units	2,000	2,000	2,000	2,000	2,000
		256–128–64 units	2,000	2,000	2,000	2,000	2,000
	SVM	$C = 0.01$	500	500	500	500	500
		$C = 0.1$	500	500	500	500	500
		$C = 1$	1,000	1,000	1,000	1,000	2,000
		$C = 10$	1,000	2,000	1,000	2,000	2,000
RLM	SVM	$C = 0.01$	500	500	500		
		$C = 0.1$	500	500	500		
		$C = 1$	2,000	2,000	2,000		
		$C = 10$	2,000	2,000	2,000		
MDR1	SVM	$C = 0.01$	500	500	500	500	500
		$C = 0.1$	500	500	500	500	500
		$C = 1$	2,000	1,000	1,000	2,000	500
		$C = 10$	2,000	1,000	1,000	2,000	1,000

Table S.28: Budget (number of features) picked on each outer fold's training samples at each regularization setting C of the SVM, and at each hidden-layer width of the DNN on LogD. One row is one dataset, model, and setting; columns 1 to 5 are the outer folds. Rat microsomal clearance has three folds. Dataset short names as in Table 5.8.

Additional file 27: Internal evaluator matched to the downstream model, or from another family

Table S.29: Held-out score and number of features of the hybrid when candidate hybrids are scored by the downstream model itself (matched), by a model of a near family, or by a model of a far family. One row is one dataset and downstream model; the evaluator columns name the near and far model. Held-out means over five outer folds (three for QM9 and MDR1–MDCK efflux), in the dataset's own metric; features are means over folds. Dataset short names as in Table 5.8.

		Matched		Near family			Far family		
Dataset	Model	Score	Features	Evaluator	Score	Features	Evaluator	Score	Features
RMSE ↓									
QM9	DNN	0.339	1,333	Small DNN	0.337	1,667	*k*NN	0.343	1,333
	LightGBM	0.346	1,015	RF	0.347	959	SVM	0.363	1,941
HK_i	SVM	0.5587	450	GP	0.5589	624	LightGBM	0.557	549
	DNN	0.582	1,421	Small DNN	0.5778	1,235	*k*NN	0.5775	1,048
Caco-2	DNN	0.286	1,964	Small DNN	0.2840	1,769	*k*NN	0.2843	1,360
	SVM	0.2848	600	GP	0.28527	600	LightGBM	0.28531	600
LogD	DNN	0.412	2,000	Small DNN	0.413	2,000	*k*NN	0.411	1,987
	SVM	0.433	1,400	GP	0.438	1,932	LightGBM	0.431	1,200
MDR1	SVM	0.4332	1,318	GP	0.4335	1,318	LightGBM	0.4333	1,318
	DNN	0.434	1,916	Small DNN	0.438	1,916	*k*NN	0.431	1,904
BH	RF	14.5	526	LightGBM	14.6	653	SVM	14.4	698
	LightGBM	15.3	580	RF	16.2	813	SVM	15.7	581
AUROC ↑									
HB	LightGBM	0.8630	1,368	RF	0.866	1,143	SVM	0.8629	1,536
	XGBoost	0.8613	1,075	LightGBM	0.863	909	SVM	0.8609	1,423
Hum	XGBoost	0.956	713	LightGBM	0.958	690	SVM	0.962	640
	RF	0.966	357	LightGBM	0.9632	490	SVM	0.9628	514
Spearman ↑									
Nb-Tm	DNN	0.644	550	Small DNN	0.667	1,097	*k*NN	0.645	571

		Matched		Near family			Far family		
Dataset	Model	Score	Features	Evaluator	Score	Features	Evaluator	Score	Features
	SVM	0.673	770	GP	0.654	720	LightGBM	0.697	428
EGFR	SVM	**0.353**	1,059	GP	0.324	1,086	LightGBM	**0.299**	1,486
	DNN	0.259	878	Small DNN	0.230	645	*k*NN	0.162	946
Ang2	GP	0.482	70	SVM	0.480	65	LightGBM	0.475	65
	SVM	0.470	75	GP	0.550	70	LightGBM	0.466	75
$\Delta\Delta G$	SVM	0.245	259	GP	0.174	159	LightGBM	0.183	100
	DNN	0.201	254	Small DNN	0.281	202	*k*NN	0.197	202
SMAC	DNN	0.344	818	Small DNN	0.327	668	*k*NN	0.341	700
	SVM	0.381	1,235	GP	0.350	897	LightGBM	0.363	1,604
HIC-R	GP	0.570	507	SVM	0.559	668	LightGBM	0.555	750
	DNN	0.487	900	Small DNN	0.453	1,010	*k*NN	0.470	750
AC-SINS	LightGBM	0.512	420	RF	0.571	1,025	SVM	0.487	560
	GP			SVM	0.564	400	LightGBM	0.544	800
Titer	DNN	0.336	796	Small DNN	0.318	997	*k*NN	0.330	689
	RF	0.305	900	LightGBM	0.250	1,100	SVM	0.264	1,000
PR-CHO	GP	0.479	1,500	SVM	0.478	750	LightGBM	0.464	1,174
	DNN	0.465	1,000	Small DNN	0.475	1,700	*k*NN	0.435	650

Additional file 28: A new embedding's features against random features in their place

Table S.30: Hybrid with a new embedding among its single representations, against the same hybrid with the embedding's features replaced by as many Gaussian random features, with no new search. One row is one dataset and embedding; the two pairs of columns are the dataset's two downstream models, named in the line above its rows. Held-out means over five outer folds (three for MDR1–MDCK efflux, hERG blocking, and antibody humanness), in the dataset's own metric. LogD and hERG blocking ran on 2,000 molecules. Equal scores: the hybrid took no features from the embedding. Dataset short names as in Table 5.8.

Dataset	Embedding	With its features	Random in their place	With its features	Random in their place
RMSE ↓					
HK_i		SVM		DNN	
	ChemBERT	0.560	0.574	0.587	0.605
	ChemFM	0.555	0.575	0.567	0.590
	GROVER	0.560	0.576	0.587	0.595
	MHG-GNN	0.560	0.583	0.587	0.592
	SMITED	0.560	0.572	0.587	0.598
	Uni-Mol2	0.559	0.560	0.574	0.583
Caco-2		DNN		SVM	
	ChemFM	0.274	0.279	0.287	0.290
	GROVER	0.279	0.287	0.286	0.297
	MHG-GNN	0.279	0.295	0.286	0.293
	MoLFormer	0.279	0.280	0.286	0.293
	SMITED	0.279	0.288	0.286	0.294
	Uni-Mol2	0.275	0.277	0.286	0.287
LogD		DNN		SVM	
	ChemBERT	0.490	0.506	0.517	0.529
	ChemFM	0.490	0.507	0.521	0.527
	GraphMVP	0.490	0.525	0.517	0.532
	GROVER	0.490	0.534	0.517	0.537
	MHG-GNN	0.490	0.530	0.517	0.542
	MoLFormer	0.490	0.508	0.517	0.543
	SMITED	0.490	0.517	0.517	0.539
	Uni-Mol2	**0.4908**	**0.4907**	0.5178	0.5182
MDR1		SVM		DNN	
	GROVER	0.433	0.441	0.441	0.443
	MHG-GNN	0.433	0.442	0.441	0.446
	SMITED	0.433	0.445	0.441	0.447
BH		RF		LightGBM	
	Difference-MHG-GNN	15.2	17.3	15.6	15.4
	Reactant-GROVER	15.2	15.2	15.6	15.6
	Reactant-Mol2Vec	15.2	15.2	15.6	15.6
	ReactionT5	15.2	15.5	15.6	15.8

Dataset	Embedding	With its features	Random in their place	With its features	Random in their place
	RXNFP	15.2	15.2	15.6	15.6
AUROC ↑					
HB		LightGBM		XGBoost	
	GROVER	0.829	0.831	0.822	0.818
	MHG-GNN	0.829	0.808	0.822	0.804
	Mol2Vec	0.829	0.823	0.822	0.825
	MoLFormer	0.829	0.831	0.822	0.818
Hum		XGBoost		RF	
	AbLang2	0.956	0.962	0.960	0.961
	AbRoBERTa	0.956	0.954	0.960	0.961
	AntiBERTy-CDR	0.956	0.953	0.960	0.964
	ESM2-Early	0.956	0.956	0.960	0.960
	ESM2-Midlayers	0.956	0.956	0.960	0.960
	ESM2-Per-CDR	0.956	0.956	0.960	0.964
	ESMC	0.9561	0.9555	0.960	0.960
	IgT5	0.956	0.951	0.960	0.963
	ProtT5	0.956	0.956	0.960	0.960
Spearman ↑					
EGFR		SVM		DNN	
	AbLang2	0.298	0.291	0.224	0.214
	AbRoBERTa	0.298	0.284	0.224	0.270
	AbRoBERTa-CDR	0.298	0.295	0.224	0.228
	ESM2-Concat	0.298	0.254	0.224	0.253
	ESM2-Early	0.298	0.291	0.224	0.209
	ESM3	0.298	0.295	0.224	0.206
	ESMC	0.298	0.316	0.224	0.218
	SaProt-CDR	0.298	0.302	0.2239	0.2235
$\Delta\Delta G$		SVM		DNN	
	AbCDR-ESMC	0.328	0.323	0.400	0.370
	AntiBERTy	0.213	0.184	0.346	0.308
	AntiBERTy-CDR	0.213	0.201	0.346	0.338
	CurrAb	0.242	0.243	0.259	0.262
	ESM2-Concat	0.213	0.206	0.346	0.299
	ESM2-Layer24	0.213	0.223	0.346	0.332
	ESM2-WT-Delta	0.213	0.169	0.346	0.127
	IgBert	0.261	0.258	0.232	0.272
	ISM	0.226	0.246	0.329	0.308
	MINT	0.250	0.221	0.316	0.310
	p-IgGen	0.240	0.206	0.331	0.310
	ProtT5-CDR	0.213	0.174	0.346	0.314
SMAC		DNN		SVM	
	AbCDR-ESMC	0.310	0.330	0.350	0.357
	AbLang2	0.265	0.288	0.345	0.334
	AntiBERTy	0.265	0.270	0.345	0.345
	CurrAb	0.260	0.229	0.365	0.370

Dataset	Embedding	With its features	Random in their place	With its features	Random in their place
	ESM2-Concat	0.265	0.335	0.345	0.351
	ESM2-Early	0.265	0.253	0.345	0.351
	ESM2-Entropy-Top20	0.265	0.284	0.345	0.345
	ESM3	0.265	0.287	0.345	0.337
	ESMC	0.265	0.268	0.345	0.345
	IgBert	0.304	0.326	0.374	0.381
	IgT5	0.265	0.326	0.345	0.370
	ISM	0.213	0.231	0.3479	0.3475
	MINT	0.210	0.203	0.339	0.349
	p-IgGen	0.337	0.323	0.381	0.386
	ProtT5	0.265	0.206	0.345	0.330
	SaProt	0.265	0.370	0.345	0.353
Titer		DNN		RF	
	AbCDR-ESMC	0.338	0.268	0.307	0.271
	AbLang2	0.368	0.323	0.271	0.195
	CurrAb	0.283	0.294	0.256	0.279
	ESM2	0.368	0.329	0.271	0.294
	ESM2-Concat	0.368	0.354	0.271	0.264
	ESM2-Entropy-Top20	0.368	0.387	0.271	0.290
	ESM2-Midlayers	0.368	0.378	0.271	0.285
	ESM3	0.368	0.382	0.271	0.280
	ESMC	0.368	0.364	0.271	0.294
	IgBert	0.317	0.320	0.290	0.291
	IgT5	0.368	0.332	0.271	0.234
	ISM	0.311	0.311	0.285	0.257
	MINT	0.359	0.360	0.295	0.299
	p-IgGen	0.311	0.311	0.369	0.362
	ProtT5	0.368	0.374	0.271	0.266
	SaProt	0.368	0.366	0.271	0.305
PR-CHO		GP		DNN	
	AbLang2	0.443	0.428	0.413	0.436
	ESM2-Concat	0.443	0.424	0.413	0.431
	ESM2-Entropy-Top20	0.443	0.447	0.413	0.421
	ESM2-Per-Region	0.443	0.415	0.413	0.429
	ESMC	0.443	0.439	0.413	0.416
	IgT5	0.443	0.434	0.413	0.378
	ProtT5	0.443	0.411	0.413	0.420
	SaProt	0.443	0.454	0.413	0.427

Additional file 29: Coverage and width of 90% conformal intervals

		Best single representation		Hybrid	
Dataset	Model	Coverage	Width	Coverage	Width
Small molecules and reactions					
QM9	DNN	0.893	1.153	0.891	0.999
	LightGBM	0.901	1.221	0.899	1.117
HK_i	SVM	0.906	2.010	0.904	1.883
	DNN	0.924	2.181	0.906	1.890
Caco-2	DNN	0.903	1.005	0.889	0.867
	SVM	0.891	0.948	0.886	0.876
LogD	DNN	0.905	1.527	0.909	1.362
	SVM	0.905	**1.652**	0.898	**1.403**
RLM	SVM	0.908	1.773	0.904	1.628
	DNN	0.905	1.802	0.902	1.659
MDR1	SVM	0.904	1.627	0.896	1.483
	DNN	0.916	1.703	0.893	1.498
BH	RF	0.878	48.0	0.880	53.4
	LightGBM	0.869	50.0	0.887	58.0
Antibodies					
VEGF-E	SVM	0.902	1.294	0.890	1.116
	DNN	0.893	1.233	0.880	1.094
VEGF-B	RF	0.903	1.667	0.892	1.514
	SVM	0.903	1.666	0.890	1.555
Nb-Tm	DNN	0.872	24.469	0.851	23.308
	SVM	0.873	23.912	0.833	21.126
EGFR	SVM	0.862	0.811	0.854	0.766
	DNN	0.864	0.878	0.839	0.799
Ang2	GP	0.833	0.609	0.736	0.500
	SVM	**0.804**	0.599	**0.720**	0.492
$\Delta\Delta G$	SVM	0.884	4.835	0.828	4.522
	DNN	0.888	5.565	0.844	4.703
SMAC	DNN	0.890	2.489	0.886	1.873
	SVM	0.938	1.660	0.938	1.571
HIC-R	GP	0.900	0.810	0.895	0.797
	DNN	0.917	0.955	0.895	0.813
AC-SINS	LightGBM	0.921	31.184	0.958	28.894
	GP	0.946	32.460	0.909	27.139
Titer	DNN	0.917	389.977	0.852	328.471
	RF	0.925	376.215	0.930	360.947
PR-CHO	GP	0.901	0.448	0.882	0.426
	DNN	0.908	0.486	0.881	0.448

Table S.31: Coverage and mean width of split-conformal intervals at 90%, for the best single representation and the hybrid. One row is one dataset and downstream model. Coverage is the share of held-out samples inside their interval (target 0.900); width is in the label's units, lower is narrower. Means over five outer folds. Dataset short names as in Table 5.8.

Additional file 30: Three further experiments with separate sites

The site models' predictions are combined in one of two ways. Under distance weighting, each site's vote on a molecule is weighted by how close the molecule is to that site's training molecules. The distance is measured to the 8 nearest molecules in a random 20% sample of that site's training molecules, on the features the site model uses. Distances are Manhattan by default, whereas the collaborator's code uses Euclidean. Under score weighting, each site has one weight, its own cross-validated score. Only predictions on shared molecules cross a site boundary. The published process then trains a student model on the most confident combined labels.

In the first of three further experiments, labels are made two-class: an efflux ratio of at least 2 on Caco-2, and a LogD of at least 3. Five sites are formed by k-means clusters of the molecules. Up to 30% of each site's training labels are flipped, at different rates on different sites. At each flip level, selection at the sites is compared with the ensembles and the student model of the published process. It is compared as well with one model on the pooled molecules, and with the same selection made on all molecules pooled. A version in which each site first removes the molecules it flags as noisy is not reported, because some sites were then left with too few molecules, groups, or classes to score a representation.

In the second experiment, each of five sites holds the same molecules as a different representation. One hybrid over all sites is compared with the best single site and with the published ensembles. In the third, the molecules are split into k-means clusters and features are selected per cluster. These features are used either for one model on all molecules or for a separate model per cluster. The same experiment asks whether the clusters agree on which representations matter, against random subsets of the same size. The first and third experiments report MCC on the two-class labels, and the second reports RMSE.

When each of five sites holds a different representation, one hybrid over all sites beats the best single site on both datasets with both models (Table S.32). It also beats the collaborator's two ensembles, which combine the sites' predictions, except on Caco-2 with the DNN, where both ensembles have the lower RMSE.

Only Caco-2 has finished this experiment, with three seeds for the DNN and two for the SVM. There, with two-class labels and up to 30% of each site's labels flipped, selecting features at each site beats the collaborator's ensembles and student model at every flip level (Table S.33). With the DNN it stays well ahead of the better ensemble from no flips to 30% flipped. It beats the same selection made with all molecules pooled only with the SVM at 20% flips, so keeping the data apart still costs something. With

Dataset	Model	Best single site	Hybrid over all sites	Ensemble	
				Distance	Score
Caco-2	DNN	0.306	**0.284**	**0.273**	0.272
	SVM	0.3004	0.290	0.298	0.2999
LogD	DNN	0.451	0.427	0.433	0.428
	SVM	0.507	0.447	0.495	0.490

Table S.32: Five sites that each hold a different representation of the same molecules. One row is one dataset and downstream model. Columns: the best single site; one hybrid over all sites; the collaborator's ensembles of the sites' predictions, weighted by distance and by site score. Held-out RMSE, lower is better, mean over three seeds of the five-fold mean.

the DNN at 30% flips the two are almost equal.

Model	Flipped	Selection		Pooled model	Ensemble		Student
		Pooled	Per site		Distance	Score	
DNN	0%	0.520	**0.509**	0.443	**0.390**	0.344	0.391
	5%	0.508	0.494	0.424	0.403	0.354	0.378
	10%	0.494	0.490	0.434	0.383	0.334	0.376
	15%	0.502	0.477	0.419	0.392	0.343	0.367
	20%	0.481	0.478	0.410	0.369	0.334	0.347
	30%	**0.4324**	**0.4323**	0.382	0.336	0.298	0.320
SVM	0%	0.537	0.515	0.479	0.440	0.399	0.414
	5%	0.531	0.516	0.475	0.429	0.377	0.407
	10%	0.526	0.501	0.468	0.427	0.382	0.408
	15%	0.506	0.495	0.465	0.415	0.385	0.393
	20%	0.507	0.510	0.452	0.417	0.380	0.404
	30%	0.447	0.433	0.423	0.374	0.347	0.363

Table S.33: Selection at separate sites when up to the given share of each site's two-class Caco-2 training labels is flipped, at different rates on different sites. One row is one downstream model and flip level. Columns: the same selection with all molecules pooled; selection at each site; one model on the pooled molecules; the collaborator's ensembles weighted by distance and by site score; their student model. Held-out MCC, higher is better, five folds, mean over three seeds for the DNN and two for the SVM.

We selected features separately in each of five k-means clusters of Caco-2 and fitted one model on all molecules. This matches one global selection, while fitting a separate model per cluster is worse (Table S.34). The clusters agree less on which representations matter than random parts of the same size do, measured by mean rank correlation.

	MCC			Agreement	
Model	Global	Per cluster, one model	Model per cluster	Clusters	Random parts
DNN	**0.532**	**0.529**	0.486	0.585	0.654
SVM	0.535	0.539	0.499	0.525	0.618

Table S.34: Features selected per k-means cluster of Caco-2 (five clusters, two-class labels). One row is one downstream model. MCC columns (held-out, higher is better): one global selection; features selected per cluster with one model on all molecules; one model per cluster. Agreement columns: mean rank correlation of the representations' importance between clusters, and between random parts of the same size. Means over three seeds of five folds.

S.5 Similarity in an adaptive hybrid representation

Additional file 31: Prediction from nearest neighbors by setting and by fold

One row is one dataset and feature space. It supports Section 6.3.2.

Table S.35: Prediction of each held-out molecule from its nearest training molecules, in the kNN model's best single representation and in the kNN hybrid. One row is one dataset and feature space, in the dataset's own metric (arrow gives the direction). By setting: mean over five outer folds and over the other settings. Fold columns: Manhattan distance, 5 neighbors weighted by the inverse of their rank, unweighted. Buchwald–Hartwig RMSE is in percentage points of yield.

	Unweighted, by setting						All settings		Setting fixed before the run, by fold				
	cosine	Manhattan	Minkowski	5 neighbors	10 neighbors	20 neighbors	Un-weighted	Weighted	1	2	3	4	5
Small molecules													
QM9, RMSE ↓													
Single	0.651	0.659		0.664	0.650	0.652	0.655	0.577	0.819	0.641	0.608	0.627	0.651
Hybrid	0.421	0.423		0.422	0.418	0.427	0.422	0.468	0.436	0.410	0.407	0.434	0.415
Caco-2, RMSE ↓													
Single	0.332	0.333	0.335	0.338	0.331	0.330	0.333	0.330	0.321	0.316	0.339	0.363	0.355
Hybrid	0.322	0.320	0.322	0.327	0.319	0.318	0.3214	0.3210	0.312	0.295	0.341	0.332	0.343
MDR1–MDCK efflux, RMSE ↓													
Single	0.530	0.531	0.531	0.546	0.526	0.520	0.531	0.517	0.585	0.522	0.571	0.518	0.575
Hybrid	0.495	0.500	0.503	0.513	0.496	0.490	0.4996	0.4997	0.543	0.497	0.536	0.490	0.535
hERG K_i, RMSE ↓													
Single	0.607	0.619	0.618	0.619	0.609	0.616	0.615	0.604	0.709	0.563	0.651	0.594	0.623
Hybrid	0.597	0.582	0.585	0.591	0.584	0.589	0.588	0.586	0.597	0.498	0.597	0.578	0.641
LogD, RMSE ↓													
Single	0.661	0.674		0.668	0.662	0.673	0.668	**0.604**	0.675	0.687	0.658	0.700	0.662
Hybrid	0.578	0.580		0.582	0.576	0.580	0.5790	**0.5790**	0.592	0.595	0.563	0.610	0.556
hERG blocking, AUROC ↑													
Single	0.815	0.818		0.813	0.819	0.818	0.817	0.830	0.816	0.795	0.825	0.799	0.836

	Unweighted, by setting						All settings		Setting fixed before the run, by fold				
	cosine	Manhattan	Minkowski	5 neighbors	10 neighbors	20 neighbors	Un-weighted	Weighted	1	2	3	4	5
Hybrid	0.840	0.840		0.835	0.841	0.843	0.840	0.843	0.818	0.838	0.840	0.827	0.844
Antibodies													
EGFR binding, Spearman ↑													
Single	0.251	0.196	0.196	0.206	0.215	0.222	0.214	0.229	0.018	0.060	0.422	0.322	0.058
Hybrid	0.244	0.225	0.224	0.222	0.235	0.237	0.231	0.243	0.330	0.111	0.409	0.262	−0.049
Nanobody thermostability, Spearman ↑													
Single	0.517	0.510	0.509	0.493	0.506	0.537	0.512	0.568	0.524	0.550	0.457	0.548	0.423
Hybrid	0.605	0.580	0.582	0.565	0.590	0.611	0.589	0.572	0.667	0.610	0.418	0.530	0.505
AC-SINS self-association, Spearman ↑													
Single	0.445	0.414	0.397	0.399	0.428	0.429	0.419	0.437	0.416	0.398	0.277	0.339	0.526
Hybrid	0.450	0.424	0.418	0.398	0.444	0.451	0.431	0.444	0.479	0.531	0.442	**−0.155**	0.645
SMAC colloidal stability, Spearman ↑													
Single	0.158	0.154	0.143	0.124	0.157	0.174	0.152	0.177	−0.052	0.028	0.246	0.384	0.028
Hybrid	0.265	0.226	0.173	0.219	0.214	0.231	0.221	0.227	0.214	0.358	0.124	0.545	−0.032
CHO polyreactivity, Spearman ↑													
Single	0.266	0.264	0.261	0.214	0.263	0.314	0.263	0.317	0.089	0.319	0.239	0.244	0.139
Hybrid	0.364	0.383	0.386	0.364	0.374	0.394	0.377	0.390	0.397	0.402	0.411	0.254	0.331
Expression titer, Spearman ↑													
Single	0.345	0.311	0.290	0.289	0.334	0.322	0.315	0.277	0.222	0.354	0.349	0.236	0.100
Hybrid	0.281	0.267	0.248	0.242	0.271	0.282	0.265	0.247	0.270	0.158	0.431	0.329	0.123
Binding free energy ($\Delta\Delta G$), Spearman ↑													
Single	0.148	0.152	0.153	0.140	0.146	0.166	0.151	0.144	0.127	0.158	0.008	0.189	0.229
Hybrid	0.116	0.113	0.043	0.071	0.092	0.110	0.091	0.116	0.339	**−0.171**	−0.168	0.260	0.190
Reactions													
Buchwald–Hartwig yield, RMSE ↓													
Single	14.3	14.4	14.1	14.3	14.2	14.3	14.3	14.1	18.1	16.9	14.7	9.0	13.6
Hybrid	17.2	15.9	15.4	16.4	16.0	16.1	16.2	16.4	19.8	16.5	13.6	14.8	16.1

Additional file 32: Label gap to the nearest neighbors by setting and by fold

One row is one dataset and feature space. It supports Section 6.3.2.

Table S.36: Label gap between each held-out molecule and its nearest training molecules (lower is better), in the kNN model's best single representation and in the kNN hybrid, unweighted. One row is one dataset and feature space. The gap is in the dataset's label units; on hERG blocking it is one minus the weighted share of neighbors in the molecule's class. By setting: mean over five outer folds and over the other settings. Fold columns: Manhattan distance, 5 neighbors weighted by the inverse of their rank. Buchwald–Hartwig is in percentage points of yield.

	By setting						Setting fixed before the run, by fold				
	cosine	Manhattan	Minkowski	5 neighbors	10 neighbors	20 neighbors	1	2	3	4	5
Small molecules											
QM9											
Single	0.682	0.696		0.649	0.688	0.731	0.741	0.627	0.601	0.618	0.627
Hybrid	0.436	0.421		0.404	0.427	0.453	0.408	0.377	0.382	0.402	0.394
Caco-2											
Single	0.311	0.311	0.308	0.300	0.310	0.319	0.289	0.300	0.300	0.316	0.311
Hybrid	0.302	0.298	0.296	0.290	0.299	0.307	0.279	0.270	0.301	0.300	0.300
MDR1–MDCK efflux											
Single	0.532	0.535	0.532	0.522	0.533	0.544	0.554	0.505	0.516	0.505	0.537
Hybrid	0.492	0.485	0.484	0.476	0.487	0.497	0.482	0.475	0.467	0.459	0.489
hERG K_i											
Single	0.573	0.580	0.562	0.543	0.568	0.603	0.568	0.493	0.564	0.580	0.553
Hybrid	0.557	0.549	0.547	0.524	0.550	0.579	0.499	0.451	0.538	0.549	0.560
LogD											
Single	0.665	0.676		0.630	0.670	0.711	0.624	0.663	0.632	0.658	0.630
Hybrid	0.589	0.590		0.557	0.589	0.622	0.550	0.584	0.548	0.581	0.548
hERG blocking											
Single	0.313	0.309		0.296	0.310	0.327	0.297	0.314	0.280	0.301	0.282
Hybrid	0.288	0.292		0.275	0.290	0.305	0.291	0.275	0.268	0.279	0.265
Antibodies											

	By setting						Setting fixed before the run, by fold				
	cosine	Manhattan	Minkowski	5 neighbors	10 neighbors	20 neighbors	1	2	3	4	5
EGFR binding											
Single	0.273	0.2885	0.2878	0.282	0.283	0.285	0.307	0.325	0.251	0.261	0.300
Hybrid	0.272	0.2894	0.2876	0.284	0.282	0.283	0.264	0.357	0.244	0.275	0.331
Nanobody thermostability											
Single	8.975	9.084	9.071	8.753	9.050	9.326	7.861	9.598	8.210	8.982	8.745
Hybrid	8.363	8.179	8.235	8.137	8.243	8.397	7.192	8.577	7.752	8.587	7.956
AC-SINS self-association											
Single	7.517	7.693	7.783	7.521	7.674	7.796	6.568	8.711	8.997	7.060	6.166
Hybrid	7.256	6.869	6.903	6.884	6.984	7.161	5.463	7.451	7.855	6.406	6.315
SMAC colloidal stability											
Single	**0.421**	**0.423**	0.429	0.4126	0.425	0.436	0.347	0.363	0.502	0.347	0.448
Hybrid	**0.478**	**0.395**	0.404	0.4131	0.427	0.437	0.233	0.344	0.503	0.368	0.450
CHO polyreactivity											
Single	0.157	0.158	0.159	0.159	0.158	0.158	0.159	0.149	0.160	0.153	0.169
Hybrid	0.149	0.149	0.149	0.146	0.149	0.151	0.130	0.146	0.161	0.148	0.144
Expression titer											
Single	120.372	125.559	126.552	123.650	124.064	124.770	127.385	126.048	128.136	112.123	138.232
Hybrid	120.315	117.830	118.107	117.470	118.896	119.886	122.145	110.696	130.607	98.581	116.706
Binding free energy ($\Delta\Delta G$)											
Single	1.533	1.484	1.469	1.466	1.502	1.519	1.170	1.423	1.654	1.723	1.345
Hybrid	1.627	1.560	1.581	1.584	1.586	1.599	1.017	2.044	1.960	1.549	1.097
Reactions											
Buchwald–Hartwig yield											
Single	14.0	14.0	13.2	12.9	13.7	14.7	14.7	13.7	13.2	11.2	13.2
Hybrid	15.4	14.9	14.4	14.2	14.8	15.8	16.2	13.8	11.9	13.6	15.2

Additional file 33: Label information in the features a hybrid keeps

One row is one dataset. It supports Section 6.3.4.

Dataset	Kept features	Principal components	Random subset
Small molecules			
QM9	2.68	1.93	2.02
Caco-2	0.79	0.91	0.73
MDR1–MDCK efflux	0.35	0.40	0.30
hERG K_i	2.10	2.11	1.77
LogD	**1.63**	**1.68**	1.53
hERG blocking	0.44	0.46	0.49
Antibodies			
EGFR binding	0.52	0.79	0.51
Nanobody thermostability	2.77	4.36	2.73
AC-SINS self-association	0.05	0.11	0.08
SMAC colloidal stability	0.00	0.00	0.01
CHO polyreactivity	0.11	0.31	0.07
Expression titer	0.17	0.39	0.22
Binding free energy ($\Delta\Delta G$)	0.63	0.72	0.55
Reactions			
Buchwald–Hartwig yield	7.24	5.98	5.97

Table S.37: Label information in the features a hybrid keeps from each representation, against principal components and a random subset of the same number. One row is one dataset. Values are the median, over folds, models, and representations, of a mutual-information estimate summed over at most 50 components of the held-out molecules, in nats, minus its mean under three label permutations (higher is more information).

S.6 Detecting and handling label noise

Additional file 34: How often each group of noise-handling methods improved the held-out score

The table counts fold 0 only: six of its experiments also have results on folds 1 and 2.

Table S.38: Conditions where a method improved the best single representation, the hybrid, both (and beat its control, where one exists) or neither. One condition is one dataset, noise condition, model and method. "As delivered" means no noise added; "added damage" means the main damaged level of each noise kind. "Hybrid ahead after" means the hybrid scores better than the single representation after the method. Outer fold 0.

Group	Labels	Conditions	Both improved	Both, hybrid ahead after	Both, single ahead after	Both, tie after	Single only	Hybrid only	Neither	No score
All 14 datasets; QM9 with no noise added left out										
Noise-robust training	As delivered	226	97	76	21	0	38	54	37	0
	Added damage	1086	536	437	99	0	197	166	187	0
	All	1312	633	513	120	0	235	220	224	0
Mitigation, selected by a detector	As delivered	497	178	140	38	0	88	65	156	10
	Added damage	1753	943	761	181	1	173	238	359	40
	All	2250	1121	901	219	1	261	303	515	50
Mitigation, selected by a known flag	As delivered	106	21	21	0	0	17	1	39	28
	Added damage	494	237	185	52	0	67	68	117	5
	All	600	258	206	52	0	84	69	156	33
Mitigation, every molecule	As delivered	8	2	2	0	0	1	0	5	0
	Added damage	40	24	20	4	0	2	0	14	0
	All	48	26	22	4	0	3	0	19	0
Mitigation, all three	As delivered	611	201	163	38	0	106	66	200	38

Group	Labels	Conditions	Both improved	Both, hybrid ahead after	Both, single ahead after	Both, tie after	Single only	Hybrid only	Neither	No score
	Added damage	2287	1204	966	237	1	242	306	490	45
	All	2898	1405	1129	275	1	348	372	690	83
All four groups	As delivered	837	298	239	59	0	144	120	237	38
	Added damage	3373	1740	1403	336	1	439	472	677	45
	All	4210	2038	1642	395	1	583	592	914	83
QM9, no noise added (computed labels)										
Noise-robust training	As delivered	26	12	12	0	0	2	3	9	0
Mitigation, selected by a detector	As delivered	55	18	18	0	0	3	13	21	0
Mitigation, selected by a known flag	As delivered	10	0	0	0	0	0	0	6	4
Mitigation, every molecule	As delivered	2	0	0	0	0	0	0	2	0
Mitigation, all three	As delivered	67	18	18	0	0	3	13	29	4

Additional file 35: How often each broad noise mitigation method improved the held-out score

Table S.39: One row is one method. "Datasets" counts datasets with a score; a condition is one dataset, noise condition and model. The other columns count conditions where the method improved both the best single representation and the hybrid (and beat its control, where one exists), only one of them, or neither. QM9 with no noise added is left out. Outer fold 0.

Method	Datasets	Conditions	Both improved	Single only	Hybrid only	Neither	No score
Selected by a detector							
Neighbor weights, importance-scaled features	13	130	78	15	18	19	0
Neighbor weights, unscaled features	13	130	71	12	23	24	0
Move toward prediction; Forest residual	7	92	29	5	13	20	25
Score-scaled shrink; Forest residual	7	92	31	5	11	20	25
Score down-weight; Forest residual or Neighbor-label gap	7	62	36	5	7	14	0
Forest residual or Neighbor-label gap; Huber-style weight	7	62	36	4	6	16	0
Neighbor-mean smoothing; Forest residual or Neighbor-label gap	7	62	38	5	6	13	0
Close-pair weights	11	42	27	3	5	7	0
Top-share down-weight; LightGBM residual	4	40	24	3	2	11	0
Delete flagged; Forest residual, labels as delivered only	11	22	5	3	2	12	0
Neighbor-agreement weights	12	22	12	4	2	4	0
Cliff put-back	5	40	18	2	7	13	0
Selected by a known flag							
Scaffold-family offset	5	28	10	2	1	15	0
Scaffold-family offset, families of 3 or more	5	28	11	6	3	8	0
Scaffold-family offset, random intercept	5	28	15	5	3	5	0
Source offset	8	76	29	16	13	18	0
Source offset, then Source weights	8	76	40	11	11	14	0
Source weights, from error against number of sources	8	76	32	17	15	12	0

Method	Datasets	Conditions	Both improved	Single only	Hybrid only	Neither	No score
Source weights, by spread within and between sources	8	76	37	12	13	14	0
Count and spread weights, by count	1	4	2	0	0	2	0
Count and spread weights, inverse error variance, pooled spread	1	4	2	1	0	1	0
Shrink by spread	1	4	2	1	0	1	0
Every molecule							
Huber blend	4	48	26	3	0	19	0

Table S.39 counts outer fold 0 only, though three of its experiments also have folds 1 and 2.

Additional file 36: Model-uncertainty detectors and detectors from outside cheminformatics

Dataset	Damage	First run		Second run		
		GP spread and residual	Cliff-aware gap	LightGBM residual	Nearest-neighbor Shapley value	Cleanlab score
QM9	Clipped 25%[a]	0.162	0.208	0.153	0.394	0.159
	Outliers[a]	0.409	0.419	0.388	0.339	0.374
	Shift in 10%[a]			0.715	0.388	0.772
Caco-2	Clipped 25%	0.275	0.270			
	Outliers[a]	0.345	0.335	0.308	0.322	0.308
	Shift in 10%	0.223	0.250			
	Shift in 10%, model error	0.149	0.159			
HK_i	Clipped 25%	0.227	0.267	0.233	0.383	0.235
	Outliers	0.363	0.360	0.356	0.325	0.367
	Shift in 10%[a]	0.233	0.310	0.232	0.203	0.303
LogD	Clipped 25%	0.212	0.184	0.174	0.331	0.180
	Outliers[a]	0.408	0.385	0.384	0.364	0.407
	Shift in 10%[a]	0.376	0.444	0.419	0.278	0.515
HB	Uniform flips	0.345	0.465			
	One-way flips	0.273	0.378			
	Class-dependent flips	0.346	0.422			
	Class-frequency flips	0.250	0.360			
	Instance flips	0.341	0.458			
	Confusion-matrix flips	0.353	0.447			
BH	Clipped 25%	0.270	0.212	0.195	**0.391**	0.205
	Outliers	0.354	0.381	0.331	0.340	0.380
	Shift in 10%	0.240	0.691	0.296	0.351	**0.829**
Nb-Tm	Outliers[a]			0.434	0.418	0.405
$\Delta\Delta G$	Clipped 25%			0.286	0.450	0.298
	Outliers			0.374	0.462	0.447
	Shift in 10%			0.160	0.185	0.256
Titer	Clipped 25%			0.330	0.381	0.312
	Outliers			0.348	0.386	0.405
SMAC	Clipped 25%			0.490	**0.640**	
	Outliers			0.266	0.310	
	Shift in 10%			0.168	0.296	

Table S.40: Model-uncertainty detectors and detectors from outside cheminformatics. The first two columns and the last three come from two separate runs. GP SPREAD AND RESIDUAL: the Gaussian process's predicted spread plus its residual. One row is one dataset and damage kind at its main level. Average precision of the damaged molecules, mean over configurations and settings; higher is better. Chance is in Table 7.16. Outer fold 0. [a]Second run: fewer passes than the other rows. Dataset short names as in Table 5.8.

Additional file 37: Three ways to raise clipped labels, every model and configuration

Model	Configuration	Before clipping	Clipped	Constant lift	Truncated-normal mean	Censored EM
LogD						
DNN	First single	0.431	0.635	0.499	0.531	0.479
DNN	Second single	0.418	0.633	0.485	0.523	0.487
DNN	First hybrid	0.406	0.605	0.460	0.507	0.461
DNN	Second hybrid	0.402	0.622	0.480	0.520	0.476
SVM	First single	0.475	0.637	0.522	0.553	0.506
SVM	Second single	0.495	0.649	0.518	0.560	0.530
SVM	First hybrid	0.421	0.598	0.472	0.503	0.461
SVM	Second hybrid	0.422	0.599	0.474	0.505	0.463
QM9						
DNN	First single	0.396	0.575	0.483	0.498	0.464
DNN	Second single	0.495	0.594	0.521	0.530	0.525
DNN	First hybrid	0.351	0.520	0.450	0.444	0.441
DNN	Second hybrid	0.344	0.524	0.449	0.450	0.446
LightGBM	First single	0.412	0.553	0.484	0.492	0.460
LightGBM	Second single	0.459	0.569	0.504	0.516	0.487
LightGBM	First hybrid	0.369	0.524	0.450	0.461	0.430
LightGBM	Second hybrid	0.372	0.527	0.457	0.466	0.427
hERG K_i						
DNN	First single	0.638	0.742	0.677	0.702	0.680
DNN	Second single	0.589	0.749	**0.666**	0.709	0.704
DNN	First hybrid	0.608	0.731	0.668	0.693	0.664
DNN	Second hybrid	0.612	0.736	0.675	0.685	0.660
SVM	First single	0.640	0.760	0.702	0.724	0.710
SVM	Second single	0.675	0.774	0.726	0.741	0.735
SVM	First hybrid	0.621	0.746	0.684	0.710	0.697
SVM	Second hybrid	0.628	0.750	0.687	0.712	0.700
Caco-2						
DNN	First single	0.276	0.320	0.295	0.308	0.300
DNN	Second single	0.267	0.313	0.281	0.293	0.280
DNN	First hybrid	0.246	0.304	0.275	0.288	0.274
DNN	Second hybrid	0.237	0.303	0.279	0.288	0.281
SVM	First single	0.276	0.311	0.289	0.297	0.291
SVM	Second single	0.287	0.320	0.299	0.309	0.304
SVM	First hybrid	0.262	0.307	0.283	0.292	0.286
SVM	Second hybrid	0.261	0.306	0.282	0.291	0.285
MDR1–MDCK efflux						
DNN	First single	0.483	0.539	0.484	0.507	0.491
DNN	Second single	0.506	0.546	0.509	0.519	0.504
DNN	First hybrid	0.456	0.511	0.461	0.483	0.466
DNN	Second hybrid	0.47886	0.506	0.453	0.47892	0.466
SVM	First single	0.479	0.541	0.487	0.512	0.497

Model	Configuration	Before clipping	Clipped	Constant lift	Truncated-normal mean	Censored EM
SVM	Second single	0.487	0.536	0.488	0.511	0.501
SVM	First hybrid	0.445	0.512	0.452	0.480	0.469
SVM	Second hybrid	0.444	0.512	0.452	0.480	0.469

Table S.41: Three ways to raise clipped training labels on small molecules, with 25% of training labels clipped. One row is one model and configuration. Held-out RMSE in the label's units, lower is better. Outer fold 0.

Model	Configuration	Before clipping	Clipped	Constant lift	Truncated-normal mean	Censored EM
Buchwald–Hartwig, RMSE in yield points (lower is better)						
LightGBM	First single	19.49	19.51	19.61	19.4	19.58
LightGBM	Second single	20.2	19.9	20.1	19.5	19.3
LightGBM	First hybrid	20.4	20.2	20.5	19.50	19.49
LightGBM	Second hybrid	19.8	20.3	20.7	19.6	19.9
RF	First single	18.9	19.6	18.9	18.8	18.5
RF	Second single	18.0	19.1	18.1	18.3	18.0
RF	First hybrid	17.6	18.7	18.0	18.1	**17.6**
RF	Second hybrid	17.2	18.9	17.9	18.1	17.6
Nanobody thermostability, Spearman (higher is better)						
DNN	First single	0.770	0.633	0.759	0.657	0.694
DNN	Second single	0.548	0.441	0.515	0.427	0.460
DNN	First hybrid	0.749	0.610	**0.7164**	0.7156	0.699
DNN	Second hybrid	0.670	0.626	0.612	0.642	0.623
SVM	First single	0.752	0.654	0.749	0.721	0.743
SVM	Second single	0.581	0.480	0.652	0.539	0.580
SVM	First hybrid	0.710	0.628	0.703	0.690	0.714
SVM	Second hybrid	0.665	0.577	0.654	0.650	0.668
SMAC colloidal stability, Spearman (higher is better)						
DNN	First single	0.065	0.166	0.150	0.186	0.135
DNN	Second single	0.097	0.339	0.285	0.323	0.322
DNN	First hybrid	0.162	0.1469	0.141	0.1467	0.129
DNN	Second hybrid	0.012	0.198	0.186	0.207	0.179
SVM	First single	0.274	0.329	0.321	0.325	0.328
SVM	Second single	0.294	0.298	0.278	0.284	0.286
SVM	First hybrid	0.293	0.361	0.358	0.365	0.360
SVM	Second hybrid	0.316	0.332	0.264	0.310	0.313
$\Delta\Delta G$, *Spearman (higher is better)*						
DNN	First single	0.341	0.469	0.422	0.434	0.424
DNN	Second single	0.380	0.296	0.319	0.322	0.313
DNN	First hybrid	0.305	0.351	0.302	0.341	0.346
DNN	Second hybrid	0.428	0.4313	0.437	0.4315	0.434
SVM	First single	**0.119**	**0.248**	0.086	0.162	0.150
SVM	Second single	0.095	0.133	0.115	0.120	0.112
SVM	First hybrid	0.002	0.051	0.016	0.031	0.018

Model	Configuration	Before clipping	Clipped	Constant lift	Truncated-normal mean	Censored EM
SVM	Second hybrid	−0.012	0.001	−0.047	−0.044	−0.039
Expression titer, Spearman (higher is better)						
DNN	First single	0.380	0.409	0.4108	0.4107	0.416
DNN	Second single	−0.033	0.097	0.093	0.086	0.083
DNN	First hybrid	0.348	0.3687	0.396	0.367	0.3694
DNN	Second hybrid	0.222	0.374	0.236	0.381	0.382
RF	First single	0.309	0.315	0.242	0.365	0.408
RF	Second single	0.345	0.301	0.393	0.356	0.360
RF	First hybrid	0.283	0.271	0.292	0.307	0.356
RF	Second hybrid	0.253	0.283	0.278	0.156	0.213
AC-SINS self-association, Spearman (higher is better)						
LightGBM	First single	0.437	0.461	0.377	0.456	0.435
LightGBM	Second single	0.468	0.467	0.356	0.4361	0.4363
LightGBM	First hybrid	0.476	0.464	0.424	0.364	0.407
LightGBM	Second hybrid	0.519	0.453	0.442	0.451	0.474
GP	First single	0.522	0.496	0.490	0.499	0.506
GP	Second single	0.429	0.420	0.392	0.421	0.419
GP	First hybrid	0.605	0.609	0.596	0.608	0.612
GP	Second hybrid	0.551	0.590	0.557	0.585	0.594

Table S.42: Three ways to raise clipped training labels on Buchwald–Hartwig and five antibody datasets, with 25% of training labels clipped. One row is one model and configuration. Buchwald–Hartwig RMSE is in percentage points of yield. Outer fold 0.

Additional file 38: How often each broad noise-robust training method improved the held-out score

Table S.43: One row is one method. "Datasets" counts datasets with a score; a condition is one dataset, noise condition and model. The other columns count conditions where the method improved both the best single representation and the hybrid (and beat its control, where one exists), only one of them, or neither. QM9 with no noise added is left out. Outer fold 0.

Method	Datasets	Conditions	Both improved	Single only	Hybrid only	Neither	No score
Several datasets							
Heteroscedastic GP	11	111	59	22	19	11	0
Evidential network	11	111	51	23	17	20	0
Huber loss, DNN	3	19	15	3	0	1	0
Each model's robust loss	10	26	9	9	6	2	0
Two-learner selection	9	58	16	9	8	25	0
Repeat measurements, hERG IC_{50} only							
One row per measurement	1	2	2	0	0	0	0

Method	Datasets	Conditions	Both improved	Single only	Hybrid only	Neither	No score
Log-count feature	1	4	1	0	3	0	0
NGBoost with the log-count feature	1	2	2	0	0	0	0
Interval model	1	2	1	0	1	0	0

S.7 Datasets and curation

This section gives, for every dataset, its source, its label, and each curation step in the order the steps run. Chapter 5 and Table 5.7 give the labels, metrics, fold groupings, and splitting rules of its 20 datasets. Table S.44 gives the sample counts before and after curation, obtained by running the curation code on the downloaded files.

S.7.1 Small molecules

Three steps are shared by several small-molecule datasets. The first is structure standardization with RDKit [16]: a general cleanup, keeping the largest fragment, neutralizing charges, and selecting the canonical tautomer, in that order. Molecules that then contain a metal atom are rejected. Standardization is applied to hERG K_i, LogD, and Caco-2; the Biogen sets, QM9, and the hERG blocker sets are only checked to parse in RDKit.

The second removes duplicate molecules by InChIKey [17], keeping the first occurrence, so that one molecule cannot sit in both a training and a test fold. It is applied to hERG K_i, LogD, Caco-2, the two Biogen sets, and QM9, and to the hERG blocker sets only across sites (below). The third is fold grouping by Bemis–Murcko scaffold [18] (Chapter 5).

hERG K_i. hERG records (ChEMBL target CHEMBL240) were retrieved from ChEMBL [19], keeping binding assays (assay type B) with standard type K_i, a pChEMBL value, an exact relation ("="), and no curator data-validity comment. The label is the pChEMBL value, $\mathrm{p}K_i = -\log_{10} K_i$ with K_i in molar units. Repeat measurements of one compound are merged into their median, and compounds whose repeats have an SD above 1.0 $\mathrm{p}K_i$ unit are removed, following Landrum and Riniker [20]. This gives 1,415 compounds, 226 of them measured more than once and none removed by the standard-deviation rule. After standardization, InChIKey deduplication removes 3, leaving 1,412 compounds with $\mathrm{p}K_i$ from 4.07 to 9.41. Because repeats

Dataset	Released	Used	Steps that remove samples
Small molecules			
hERG K_i	1,815	1,412	no pChEMBL value (3); not a binding assay (75); repeats merged into one median per compound (322); InChIKey duplicates (3)
LogD	5,326	5,028	no value (287); repeats merged after standardization (11)
Caco-2 efflux	5,326	2,158	no value (3,165); repeats merged after standardization (3)
Rat liver microsome clearance	3,521	3,053	no value (467); InChIKey duplicates (1)
MDR1–MDCK efflux	3,521	2,641	no value (879); InChIKey duplicates (1)
QM9	130,831	9,978	not parsable by RDKit (1,403); random draw of 10,000; InChIKey duplicates (22)
hERG blockers, training set	7,772	7,606	also in the hERG K_i set (166)
hERG blockers, test set	4,383	4,213	also in the hERG K_i set (170)
hERG IC_{50}	12,010	12,008	missing or non-positive value (2)
Chemical reactions			
Buchwald–Hartwig yield	3,955	3,955	none
Buchwald–Hartwig plates	4,599	4,312	no aryl halide (287)
Antibodies			
Koenig expression	4,275	4,275	none
Nanobody thermostability	672	672	none
Humanness	564	564	none
Trastuzumab–HER2 binding	524,346	5,000	stratified random draw of 5,000
Nucleotide sequences			
H3K4me3	18,244	18,244	none (training and test files merged)
TATA promoter	5,274	5,274	none (training and test files merged)

Table S.44: Sample counts of each dataset before and after curation. Released is the number of rows in the downloaded files, and Used the number left after curation. The last column lists the steps that remove samples, with the number removed where it applies.

are merged on ChEMBL's own SMILES before standardization, two ChEMBL entries that standardize to the same molecule are not merged by median; the InChIKey step keeps the first of them. The file read by the experiments was pulled on 1 February 2026; the ChEMBL release was not recorded at the time. Re-running the same query in September 2026 against ChEMBL release 37 gives a byte-identical file of 1,415 compounds.

LogD and Caco-2 efflux. Both come from the training split of the OpenADMET–ExpansionRx challenge data on Hugging Face [21]. For each endpoint, rows without a value are dropped, structures are standardized, and rows that standardize to the same SMILES are merged into their median label. The unitless efflux ratio is transformed to $\log_{10}$. Labels are clipped at the 1st and 99th percentiles of each dataset, -1.0 and 4.5 for LogD and -0.319 and 1.679 for the $\log_{10}$ efflux ratio. This sets 116 LogD and 45 Caco-2 labels to a limit before any noise is added in Chapter 7. InChIKey deduplication removes nothing.

Rat liver microsome clearance and MDR1–MDCK efflux. These are two endpoints of the Biogen ADME set of 3,521 compounds [22], from Polaris as `biogen/adme-fang-v1` [23]: intrinsic clearance $\mathrm{CL}_{\mathrm{int}}$ in mL/min/kg and an efflux ratio, both released and used as $\log_{10}$ values. Rows without a value, or whose SMILES RDKit cannot parse, are dropped, and InChIKey deduplication removes one molecule from each. There is no standardization, median merge, or clipping.

QM9. QM9 [24, 25] was read through PyTorch Geometric [26], which distributes 130,831 molecules. Of these, 129,428 parse in RDKit. From these, 10,000 molecules were drawn once at random with seed 42, so the subsample is a random draw of molecules, not a random split. InChIKey deduplication removes 22, leaving 9,978. The label is the HOMO–LUMO gap in eV, from 1.88 to 11.37 eV in this sample.

hERG blockers held at separate sites. The separate-site experiments of Chapter 5 use two further hERG sets from Lhasa Limited [27, 28]: a training set of 7,772 molecules (3,630 blockers, 46.7%) and a test set of 4,383 molecules (1,084 blockers). Each label is Lhasa's own binary blocker call, made under different rules in the two sets. Both are proprietary (Section S.8). The hERG K_i set acts as a third site, with a compound counted as a blocker when $\mathrm{p}K_i \geq 6.0$, giving 467 blockers and 945 non-blockers. Molecules are matched by InChIKey across the sites, in the order hERG K_i,

training set, test set, keeping the first copy. This removes 166 training and 170 test molecules, all of them also in the hERG K_i set; the two Lhasa sets share none.

hERG IC_{50}. This set is used only in Chapter 7, where its repeat measurements and curator flags serve as real evidence of label error. It keeps every measurement as its own row. The query takes ChEMBL records for CHEMBL240 with standard type IC_{50} in nM and an exact relation [19], dropping records with a missing or non-positive value. The label is $pIC_{50} = 9 - \log_{10}(IC_{50}/\text{nM})$. Assay type is not filtered. Of the 12,010 records, 10,219 come from binding assays, 1,543 from toxicity assays, 217 from functional assays, and 31 from ADME assays, as ChEMBL classifies them. The 301 records with a curator data-validity comment, on 215 molecules, are kept and marked. With no standardization, median merge, or InChIKey step, the result is 12,008 measurements of 9,802 molecules. Folds are grouped by scaffold, so no molecule is split between training and test.

S.7.2 The Chapter 3 versions of LogD, Caco-2, hERG K_i, and QM9

Chapter 3 used the same sources as Chapter 5 for LogD, Caco-2, and hERG K_i but a lighter curation, which explains the different sample counts (Table S.45). Its QM9 replicates are drawn from the same 129,428 parsable molecules (Section 3.2). It also merges hERG K_i repeats a second time, on its own standardized SMILES; this merges nothing, so its set keeps 1,415 compounds.

	Chapter 3	Chapter 5
Structure standardization	largest fragment and canonical SMILES only	cleanup, largest fragment, neutralized charges, canonical tautomer, metals rejected
Label clipping (LogD, Caco-2)	none	1st and 99th percentile
InChIKey deduplication	none	first occurrence kept
Scaffold key	stereochemistry removed; each acyclic molecule its own group	stereochemistry kept; all acyclic molecules one group
Molecules: LogD, Caco-2, hERG K_i	5,039; 2,161; 1,415	5,028; 2,158; 1,412

Table S.45: Curation of LogD, Caco-2, and hERG K_i in Chapters 3 and 5. The last row gives the numbers of molecules.

S.7.3 Chemical reactions

Both reaction datasets come from the high-throughput Buchwald–Hartwig C–N coupling experiments of Ahneman et al. [29], which varied aryl halide, palladium ligand, base, and additive. The label is the yield in %, from 0 to 100. One sample is a whole reaction, written as its reactants and product.

Buchwald–Hartwig yield. This is the version released with Schwaller et al. [30], from that paper's public repository: 3,955 reactions. Each is written as aryl halide, 4-methylaniline, ligand, base, and additive going to a product, generated from the aryl halide by a C–N coupling template that succeeds on all 3,955. They fall into 22 additive groups of 178 to 180 reactions.

Buchwald–Hartwig plates. This is the data table of the public repository of Ahneman et al. [29]: 4,599 rows over three plates, with product SMILES from the file. The 287 rows without an aryl halide are dropped. The 180 remaining rows without an additive form their own group, giving 24 additive groups of 177 to 180 reactions. The 4,312 reactions kept include all 3,955 of the first set, with the same aryl halide, ligand, base, additive and yield, plus 357 more: the 180 without an additive and 177 with 5-phenyl-1,2,4-oxadiazole as the additive.

S.7.4 Antibodies

The Koenig and nanobody sets were downloaded as CSV files from the FLAb benchmark repository [31], the humanness set from the BioPhi publication repository, and the trastuzumab set from its own repository. The FLAb and BioPhi files come from each repository's main branch, with no commit pinned. Each file has been changed only once. The Koenig and nanobody files were changed in FLAb commits `c41e56e` and `005e6c1`, both on 6 August 2025, and the two humanness files in BioPhi commit `c0797f2` on 24 August 2021. The local copies of the Koenig and humanness files are identical to those commits.

The nanobody and humanness sets are grouped by sequence identity, defined for sequences a and b as

$$\mathrm{id}(a, b) = 1 - \frac{\mathrm{lev}(a, b)}{\max(|a|, |b|)}, \tag{S.2}$$

where lev is the Levenshtein (edit) distance and $|a|$ is the length in residues. Following Kenlay et al. [32], sequences are clustered by single linkage at id ≥ 0.95. The 672 nanobodies fall into 309 clusters and the 564 humanness antibodies into 461.

Koenig expression. This deep mutational scan of the anti-VEGF antibody G6 [33], taken from FLAb, has 4,275 variants with both chains (heavy chains of 120 residues, light chains of 108). Each carries exactly one mutation, 2,261 in the heavy chain and 2,014 in the light chain, at 225 distinct positions. The label is the unitless expression enrichment ratio, from 0.00098 to 3.178.

Nanobody thermostability. This is the NbThermo set [34], taken from FLAb: melting temperatures T_m from 39.0 to 98.2 °C for 672 single-domain antibodies with no light chain. The 672 rows hold 556 distinct sequences; 116 appear twice, and 106 of those pairs carry different melting temperatures, up to 8.8 °C apart. These repeats are kept, so the labels carry disagreement between measurements before any noise is added.

Humanness. This is the list of 564 therapeutic antibodies from IMGT/mAb-DB released with BioPhi [35], using the first sequence listed for each. An antibody is labeled human (1) only when its species field is exactly "Human" (198 antibodies). The other 366 are labeled non-human (0). The 564 antibodies hold 506 distinct sequences, mostly because an antibody and its drug conjugate share a sequence. One sequence is shared by three antibodies with different labels, two humanized and one human. All three sit in one identity cluster, so they never fall on both sides of a split.

Trastuzumab–HER2 binding. This is the library of trastuzumab variants in the ten-residue CDR-H3 (the third complementarity-determining region of the heavy chain), screened for HER2 binding by Chinery et al. [36]. The released training, validation, and test files are joined and re-split. All 524,346 variants have a 10-residue CDR-H3 and are kept; 32.8% are binders. From these, 5,000 variants were drawn once with seed 42, stratified so that each combination of label and number of mutated CDR-H3 residues keeps its share; the sample holds 1,639 binders (32.8%). Each 120-residue heavy chain is rebuilt by inserting the variant CDR-H3 into the trastuzumab framework, and paired with the fixed trastuzumab light chain.

S.7.5 Nucleotide sequences

Both nucleotide tasks come from the revised downstream tasks of the Nucleotide Transformer [37] on Hugging Face (`InstaDeepAI/nucleotide_transformer_downstream_tasks_revised`). The download is not pinned to a revision; the files used are those of revision `851f994`, dated 30 June 2025, the latest at the time of writing. The released

training and test files are merged, and folds are grouped by the chromosome read from each sequence's name. Nothing else is filtered. Both tasks cover chromosomes 1 to 21, X, and Y; chromosome 22 is absent.

H3K4me3. Each sample is a 1,000-base sequence, labeled by whether it carries the H3K4me3 histone mark. The merged files hold 18,244 sequences, 9,122 per class. The dataset card lists 30,000 training sequences; the count here is that of the files.

TATA promoter. Each sample is a 300-base sequence, labeled by whether it is a promoter with a TATA box. The merged files hold 5,274 sequences, 2,637 per class.

S.8 Data and code availability

The software behind this thesis includes NoiseInject, the label-noise package released with Chapter 3, and the KIRBy repository, which implements the adaptive hybrid representation and holds the experiments of Chapters 4 to 7 and the assay-dataset runs of Chapter 3.

NoiseInject is available as an open-source Python package under an MIT license [38]. Its noise conditions and metrics are described in Section 2.5.1.

- Project name: NoiseInject
- Project home page: `https://github.com/adpunt/noise_inject`
- Archived version: `https://doi.org/10.5281/zenodo.20532710`
- Version: 1.0.0
- Operating system(s): Platform independent
- Programming language: Python 3.7 or later
- Other requirements: NumPy, pandas, SciPy, and scikit-learn; see the repository's `requirements.txt`
- License: MIT
- Any restrictions to use by non-academics: None

The QM9 results of Chapter 3, with that chapter's hyperparameter tuning, variance decomposition, and figure scripts, come from a separate repository in Python and Rust, described in Chapter 3 under "QM9 Data Pipeline".

- Project name: qsar_qm_models
- Project home page: `https://github.com/adpunt/qsar_qm_models`
- Archived version: not yet archived
- Operating system(s): Linux and macOS
- Programming language: Python and Rust (2021 edition)
- Other requirements: RDKit installed through conda, to which the Rust program links; PyTorch and PyTorch Geometric
- License: GNU GPL v3.0
- Any restrictions to use by non-academics: those of the GNU GPL v3.0

KIRBy is written in Python and works on feature matrices computed beforehand, so the hybrid search applies to representations from any domain. It holds the hybrid search, loaders for the datasets in Section S.7, generators for the representations, the similarity analyses and learning-with-noisy-labels methods of Chapters 6 and 7, and every experiment script. Label noise is added through NoiseInject, so the noise definitions match Chapter 3.

- Project name: KIRBy
- Project home page: `https://github.com/adpunt/KIRBy`
- Archived version: not yet archived
- Operating system(s): Linux and macOS
- Programming language: Python 3.8 or later; several pretrained encoders need Python 3.10 or later
- Other requirements: NumPy, pandas, SciPy, scikit-learn, and RDKit installed through conda; each application area has an optional install
- License: to be set on release
- Any restrictions to use by non-academics: to be set on release

All datasets but one are public. QM9 [24] is deposited on Figshare [25]. The LogD and Caco-2 efflux data are from the OpenADMET–ExpansionRx release under CC-BY-4.0 [21]. The hERG K_i data come from ChEMBL [19], released under CC BY-SA 3.0, and can be regenerated with the protocol in Section S.7. The rat liver microsome clearance and MDR1–MDCK efflux data are the Biogen ADME set [22], on Polaris [23] under CC-BY-4.0. The reaction-yield, antibody, and nucleotide data come from the public releases cited in Section S.7. Pretrained model weights come from their original sources and are not redistributed.

The exception is Lhasa Limited's curated hERG training and test sets [39] (Section S.7), used for feature selection with data held at separate sites (Chapter 5). They are proprietary and not redistributed. The comparison with FLuID also cannot be run without Lhasa's FLuID code. Every other result in this thesis can be reproduced from public data and the code above.

Lhasa Limited supplied the curated hERG data and the FLuID code [39]. The use of generative AI is described in the declaration in the front matter.

Bibliography

[1] F. Pedregosa, G. Varoquaux, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[2] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[3] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[4] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[5] Vladimir N. Vapnik. *The Nature of Statistical Learning Theory*. Springer, New York, 1995.

[6] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. The MIT Press, Cambridge, MA, 2005.

[7] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958, 2014. URL `https://jmlr.org/papers/v15/srivastava14a.html`.

[8] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *3rd International Conference on Learning Representations*, 2015.

[9] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[10] Ron Kohavi and George H. John. Wrappers for feature subset selection. *Artificial Intelligence*, 97(1–2):273–324, 1997.

[11] Isabelle Guyon and André Elisseeff. An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182, 2003.

[12] Malkeet Singh Bahia, Omer Kaspi, Meir Touitou, Idan Binayev, Seema Dhail, Jacob Spiegel, Netaly Khazanov, Abraham Yosipof, and Hanoch Senderowitz. A comparison between 2D and 3D descriptors in QSAR modeling based on bio-active conformations. *Molecular Informatics*, 42(4):e2200186, 2023.

[13] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001.

[14] Bradley Efron, Trevor Hastie, Iain Johnstone, and Robert Tibshirani. Least angle regression. *The Annals of Statistics*, 32(2):407–499, 2004.

[15] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11(70):2079–2107, 2010.

[16] Greg Landrum. RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2023. Version 2023.09.1.

[17] Stephen R. Heller, Alan McNaught, Igor Pletnev, Stephen Stein, and Dmitrii Tchekhovskoi. InChI, the IUPAC international chemical identifier. *Journal of Cheminformatics*, 7(1):23, 2015.

[18] Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. Molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996.

[19] Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J. Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F. Mosquera, et al. The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024.

[20] Gregory A. Landrum and Sereina Riniker. Combining IC_{50} or K_i values from different sources is a source of significant noise. *Journal of Chemical Information and Modeling*, 64(5):1560–1567, 2024.

[21] Open ADMET Consortium. OpenADMET-ExpansionRx challenge dataset. https://huggingface.co/datasets/openadmet/openadmet-expansionrx-challenge-data, 2026.

[22] Cheng Fang, Ye Wang, Richard Grater, Sudarshan Kapadnis, Cheryl Black, Patrick Trapa, and Simone Sciabola. Prospective validation of machine learning algorithms for absorption, distribution, metabolism, and excretion prediction: An industrial perspective. *Journal of Chemical Information and Modeling*, 63(11): 3263–3274, 2023.

[23] Polaris contributors. polaris-hub/polaris. Zenodo, https://doi.org/10.5281/zenodo.13652587, 2025.

[24] Raghunathan Ramakrishnan, Pavlo O. Dral, Matthias Rupp, and O. Anatole von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Scientific Data*, 1(1):140022, 2014.

[25] Raghunathan Ramakrishnan, Pavlo O. Dral, Matthias Rupp, and O. Anatole von Lilienfeld. QM9 dataset. Figshare, https://doi.org/10.6084/m9.figshare.978904, 2014.

[26] Matthias Fey and Jan E. Lenssen. Fast graph representation learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

[27] Davide Bassani, Alessandro Brigo, and Andrea Andrews-Morger. Federated learning in computational toxicology: an industrial perspective on the Effiris hackathon. *Chemical Research in Toxicology*, 36(9):1503–1517, 2023.

[28] Thierry Hanser. Federated learning for molecular discovery. *Current Opinion in Structural Biology*, 79:102545, 2023.

[29] Derek T. Ahneman, Jesús G. Estrada, Shishi Lin, Spencer D. Dreher, and Abigail G. Doyle. Predicting reaction performance in C–N cross-coupling using machine learning. *Science*, 360(6385):186–190, 2018.

[30] Philippe Schwaller, Alain C. Vaucher, Teodoro Laino, and Jean-Louis Reymond. Prediction of chemical reaction yields using deep learning. *Machine Learning: Science and Technology*, 2(1):015016, 2021.

[31] Michael Chungyoun, Jeffrey Ruffolo, and Jeffrey Gray. FLAb: Benchmarking deep learning methods for antibody fitness prediction. bioRxiv, 2024.

[32] Henry Kenlay, Frédéric A. Dreyer, Aleksandr Kovaltsuk, Dom Miketa, Douglas Pires, and Charlotte M. Deane. Large scale paired antibody language models. *PLOS Computational Biology*, 20(12):e1012646, 2024.

[33] Patrick Koenig, Chingwei V. Lee, Benjamin T. Walters, Vasantharajan Janakiraman, Jeremy Stinson, Thomas W. Patapoff, and Germaine Fuh. Mutational landscape of antibody variable domains reveals a switch modulating the interdomain conformational dynamics and antigen binding. *Proceedings of the National Academy of Sciences*, 114(4):E486–E495, 2017.

[34] Mario S. Valdés-Tresanco, Mario E. Valdés-Tresanco, Esteban Molina-Abad, and Ernesto Moreno. NbThermo: a new thermostability database for nanobodies. *Database*, 2023:baad021, 2023.

[35] David Prihoda, Jad Maamary, Andrew Waight, Veronica Juan, Laurence Fayadat-Dilman, Daniel Svozil, and Danny A. Bitton. BioPhi: A platform for antibody design, humanization, and humanness evaluation based on natural antibody repertoires and deep learning. *mAbs*, 14(1):2020203, 2022.

[36] Lewis Chinery, Alissa M. Hummer, Brij Bhushan Mehta, Rahmad Akbar, Puneet Rawat, Andrei Slabodkin, Khang Le Quy, Fridtjof Lund-Johansen, Victor Greiff, Jeliazko R. Jeliazkov, et al. Baselining the buzz. Trastuzumab-HER2 affinity, and beyond. bioRxiv, 2024. Version 1; later versions retitled "Simple computational methods can outperform deep learning in designing diverse, binder-enriched antibody libraries".

[37] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, et al. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025.

[38] Adelaide Punt. NoiseInject: A framework for label noise robustness testing. Version 0.3.0, Zenodo, `https://doi.org/10.5281/zenodo.20532710`, 2026.

[39] Thierry Hanser, Ernst Ahlberg, Alexander Amberg, Lennart T. Anger, Chris Barber, Richard J. Brennan, Alessandro Brigo, Annie Delaunois, Susanne Glowienke,

Nigel Greene, et al. Data-driven federated learning in drug discovery with knowledge distillation. *Nature Machine Intelligence*, 7(3):423–436, 2025.